

Bioinorganic chemistry

Introduction

Role of metal ions in biological systems

O₂ transport in biological systems

Haemoglobin and myoglobin

Haemocyanin

Haemerythrin

Metalloenzymes

Carbonic anhydrase

Carboxypeptidase

ELEMENTAL MASS ABUNDANCE IN A 70 kg HUMAN

ELEMENTAL MASS ABUNDANCE IN A 70 kg HUMAN

Bulk elements

Oxygen	44	kg
Carbon	12.6	kg
Hydrogen	6.6	kg
Nitrogen	1.8	kg
Sulfur	0.1	kg

Macrominerals

Calcium	1700	g
Phosphorus	680	g
Potassium	250	g
Chlorine	115	g
Sodium	70	g
Magnesium	42	g

Trace elements

Iron	5000	mg
Silicon	3000	mg
Zinc	1750	mg
Rubidium	360	mg
Copper	280	mg
Strontium	280	mg
Bromine	140	mg
Tin	140	mg
Manganese	70	mg
Iodine	70	mg
Aluminum	35	mg
Lead	35	mg
Barium	21	mg
Molybdenum	14	mg
Boron	14	mg
Arsenic	~ 3	mg
Cobalt	~ 3	mg
Chromium	~ 3	mg
Nickel	~ 3	mg
Selenium	~ 2	mg
Lithium	~ 2	mg
Vanadium	~ 2	mg

Chemical elements essential to life forms can be divided into the following

(i) Bulk elements: C, H, N, O, P, S

(ii) Macrominerals and ions: Na, K, Mg, Ca, Cl, PO_4^{3-} , SO_4^{2-}

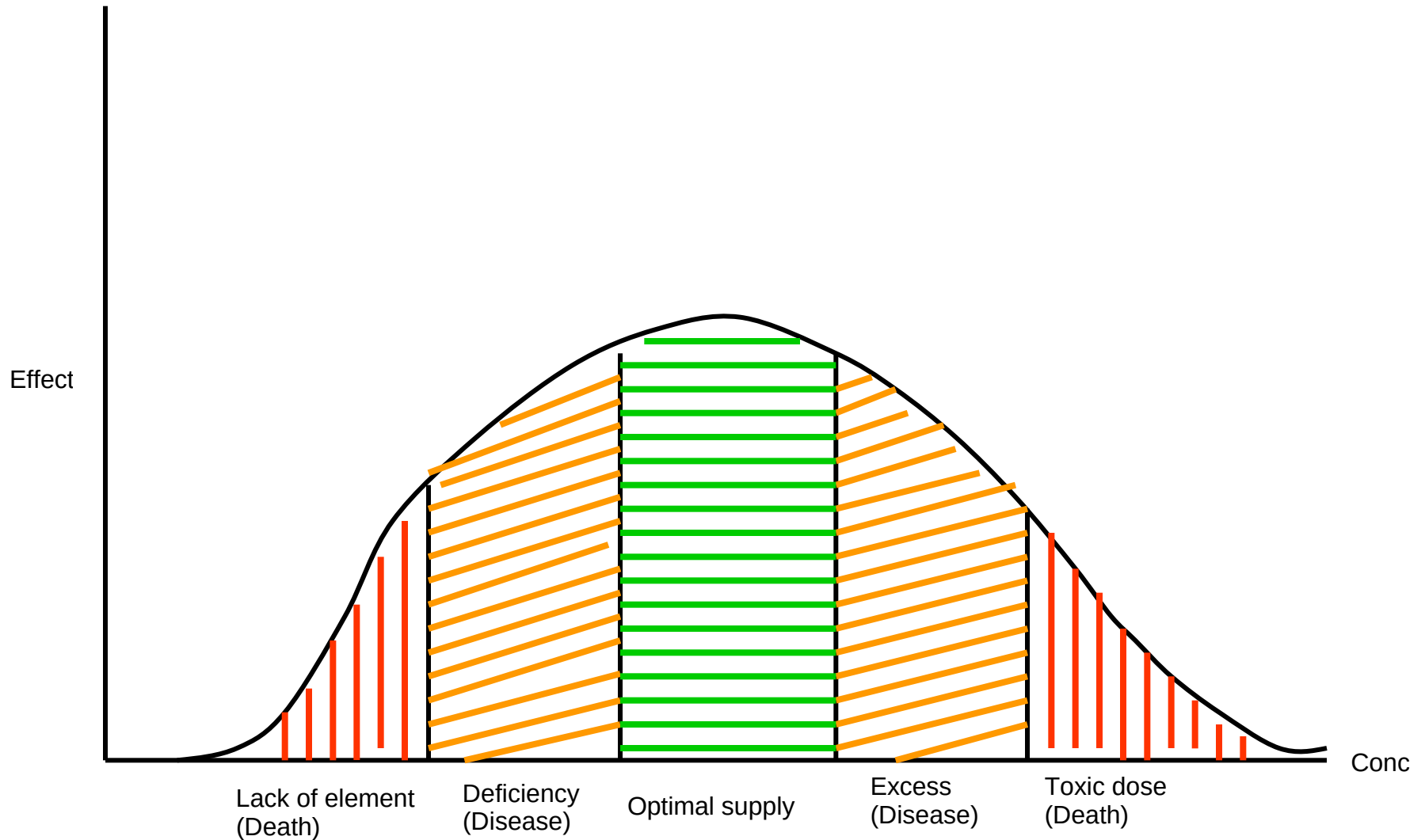
(iii) Trace elements: Fe, Zn, Cu

(iv) Ultratrace elements comprises of

(a) non-metals: F, I, Se, Si, As, B

(b) metals: Mn, Mo, Co, Cr, V, Ni, Cd, Sn, Pb, Li

Concentration and physiological effect



Role of metal ions in biological systems

Na⁺ and K⁺ Regulation of the osmotic pressure, membrane potentials, enzyme activity, signalling

Mg²⁺ Chlorophyll; anaerobic energy metabolism

V^{IV/V}, Mo^{IV/V}, W^{IV/V}, Mn^{II/III/IV}, Fe^{II/III}, Ni^{II/III}, Cu^{I/II} active centres in electron-transport (redox) enzymes, oxygenases, dismutases.

Fe and Cu: Transport of oxygen.

Fe^{III} Iron-storage proteins (ferritins).

Co Synthases and isomerases (cobalamines, e.g. vitamin-B12); methylation of inorganics.

Zn²⁺ In the active centre of hydrolases, carboanhydrase, alcohol dehydrogenase, synthases; genetic transcription (zinc fingers), stabilisation of tertiary and quartary structures of proteins; repair enzymes.

P^V (phosphate): Constituent in hydroxi- and fluorapatite ,energy metabolism (ATP), activation of organic substrate; phospholipids in cell membranes; phosphate esters (DNA, RNA,...).

Li⁺: Treatment of bipolar disorder (maniac depression) and hypertension.

Gd³⁺: Contrast agent in magnetic resonance tomography of soft tissues.

BaSO₄: Contrast agent for X-ray tomography. Sun protection.

Pt^{II}: Chemotherapy (e.g. with cisplatin *cis*-[Pt(NH₃)₂Cl₂]) of cancer (*ovaria, testes*)

Au^I: Therapy of rheumatic arthritis

Sb^{III}: Treatment of inflammatory skin pimples like acne

Bi^{III}: Treatment of gastritis

PROTEINS – CLASSES AND FUNCTIONS

dynamic

catalysis

transport

protection

muscle contraction

metabolic control

gene transcription

storage

enzymes

hemoglobin

antibodies

actin and myosin

hormones

histones

ferritin

structural

matrices for bone
and connective tissue

collagen and elastin

PROTEINS

proteins are polymers of 20 different α -amino acids, known as the common amino acids.

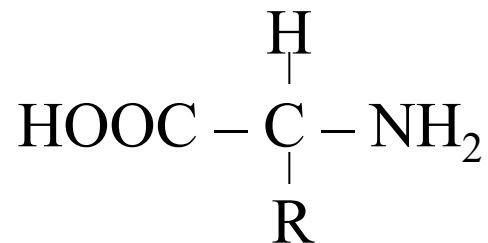
properties of 20 amino acids

α -amino group – except proline, which has an imino group

α -carboxyl group

unique R side chain and a hydrogen bound at the central α carbon

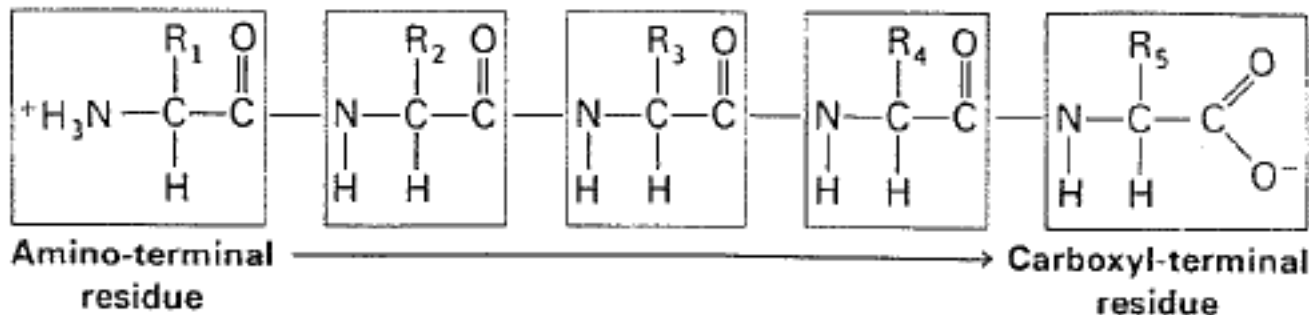
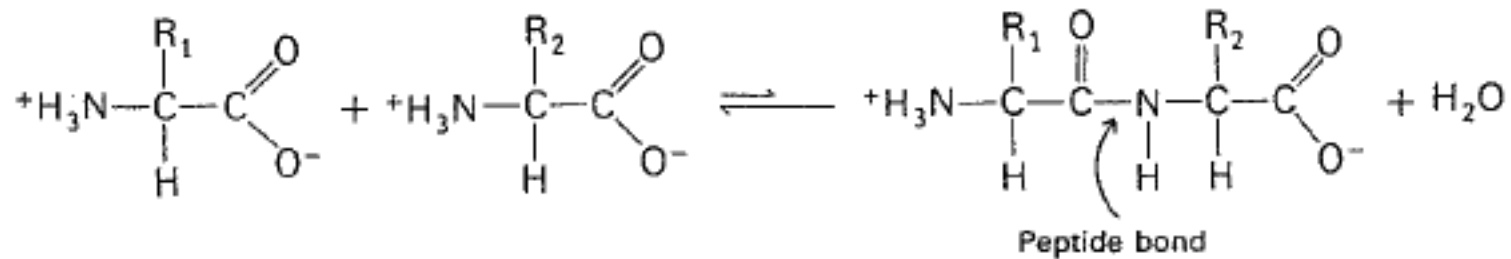
possess at least one asymmetric carbon (L form) except glycine



PROTEINS

at neutral pH, the amino and carboxyl groups are ionized, and the amino acids thus exist as zwitterions

proteins are produced by enzymatic polymerization of the 20 common amino acids, connected by peptide bonds formed by dehydration



AMINO ACIDS – ALIPHATIC

Glycine
(Gly, G) — H

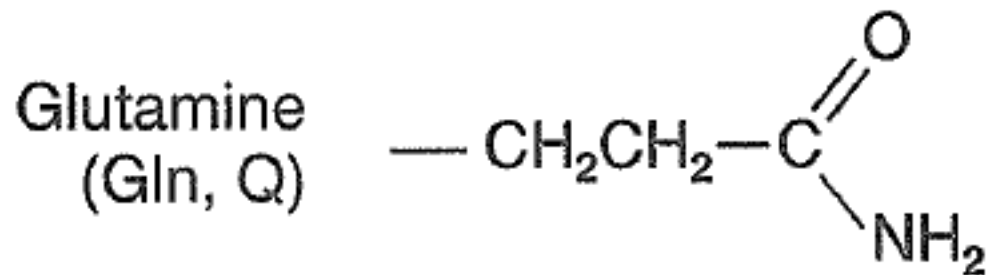
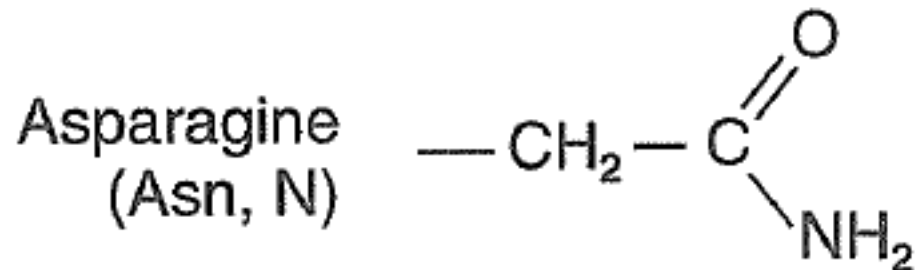
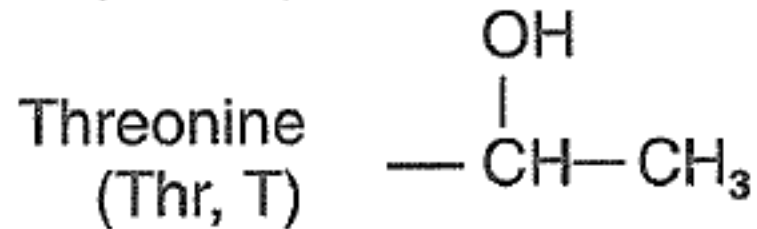
Alanine
(Ala, A) — CH₃

Valine
(Val, V) $\begin{array}{c} \text{CH}_3 \\ | \\ \text{—CH—CH}_3 \end{array}$

Leucine
(Leu, L) $\begin{array}{c} \text{CH}_3 \\ | \\ \text{—CH}_2\text{—CH—CH}_3 \end{array}$

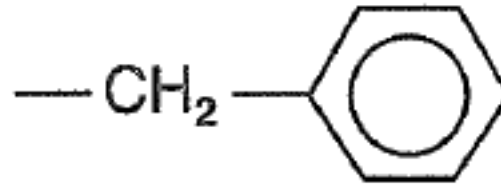
Isoleucine
(Ile, I) $\begin{array}{c} \text{CH}_3 \\ | \\ \text{—CH—CH}_2\text{—CH}_3 \end{array}$

AMINO ACIDS – POLAR

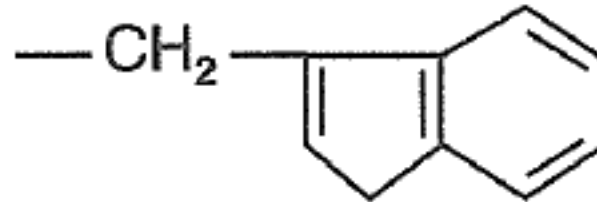


AMINO ACIDS – AROMATIC

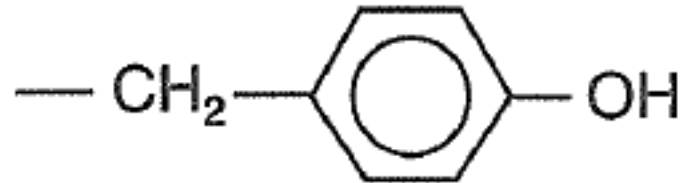
Phenylalanine
(Phe, F)



Tryptophan
(Trp, W)



Tyrosine
(Tyr, Y)



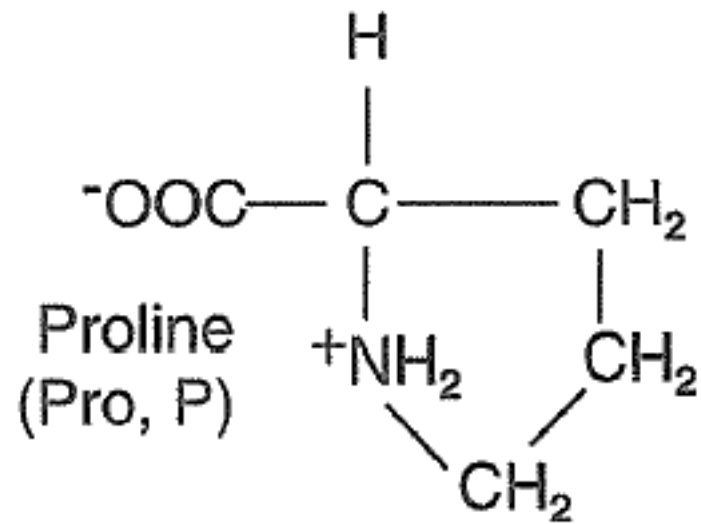
AMINO ACIDS – SULFUR OR SELENIUM

Sulfur-containing

Methionine — $\text{CH}_2\text{CH}_2\text{—S—CH}_3$
(Met, M)

Cysteine — $\text{CH}_2\text{—S—H}$
(Cys, C)

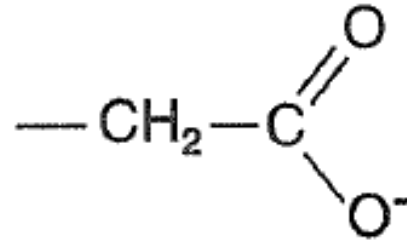
AMINO ACIDS – SECONDARY AMINE



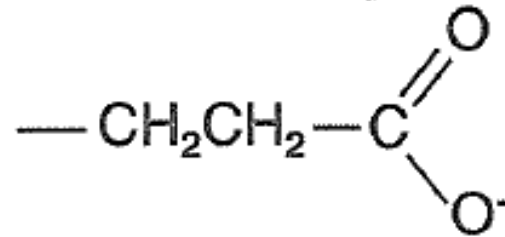
AMINO ACIDS – CHARGED

Acidic

Aspartate
(Asp, D)



Glutamate
(Glu, E)

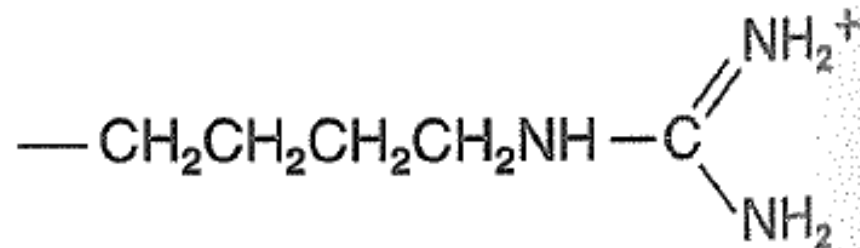


Basic

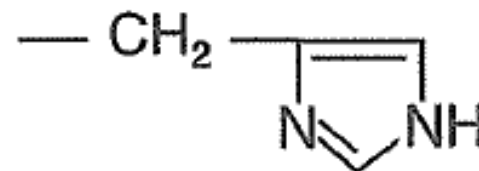
Lysine
(Lys, K)



Arginine
(Arg, R)



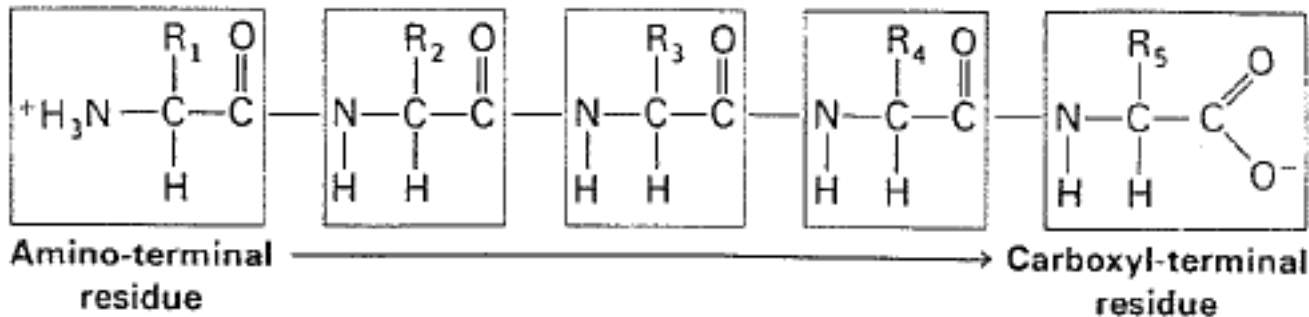
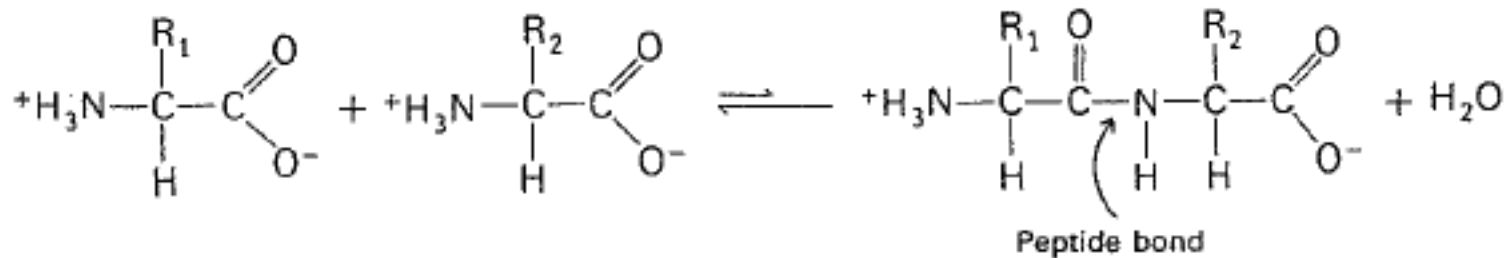
Histidine
(His, H)



PROTEINS

proteins are produced by enzymatic polymerization of the 20 common amino acids, connected by peptide bonds formed by dehydration

the specific sequence of amino acids in the polypeptide chain is called the *primary structure* of the protein.



METAL LIGATION

metal ions are bound in mononuclear or polynuclear coordination units in which amino acid side chains function as multidentate chelating ligands (protein)

common bridging ligands

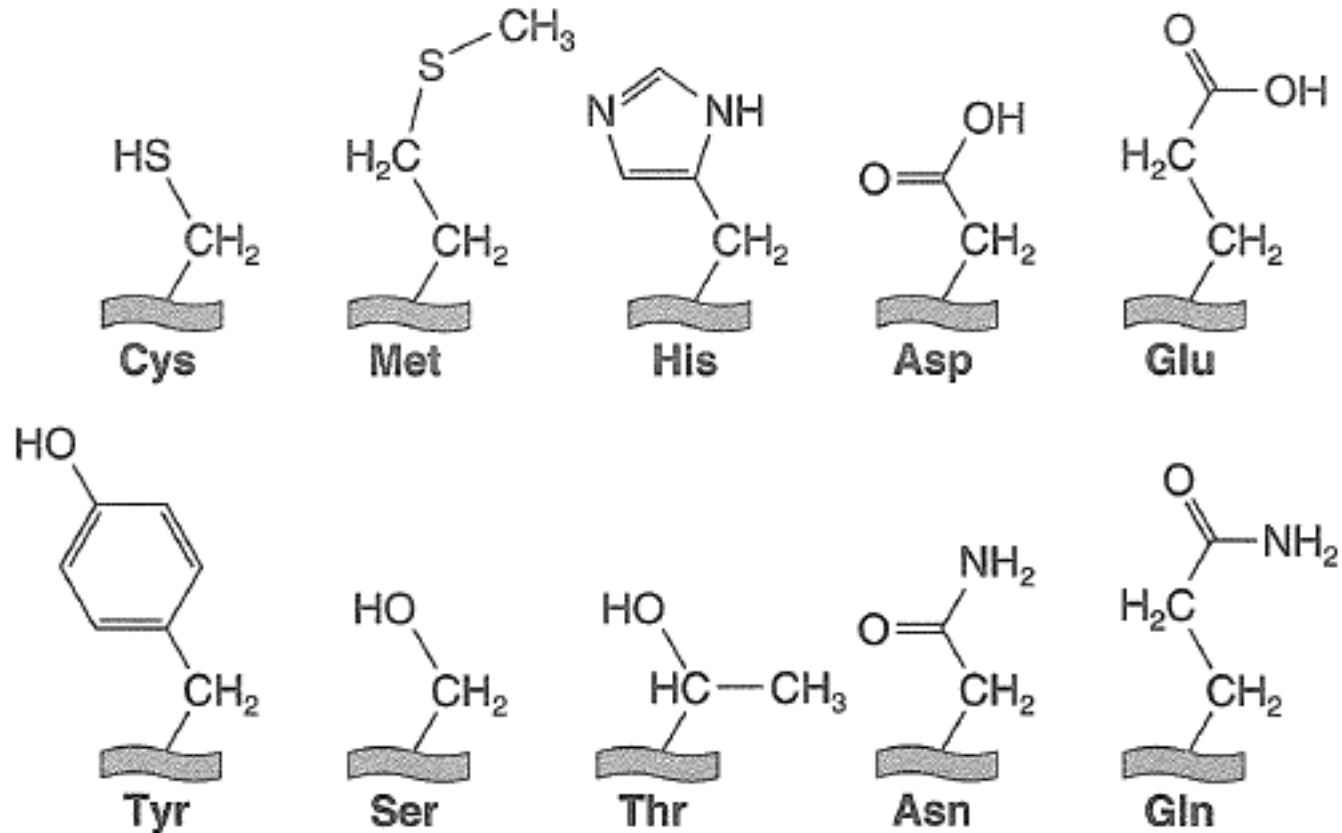
O^{2-} , OH^- , $-\text{CH}_2\text{S}^-$, S^{2-} , $-\text{CH}_2\text{CO}_2^-$, imidazole

exogenous terminal ligands are also often bound to metals

H_2O , OH^- , O^{2-} , HS^- , S^{2-}

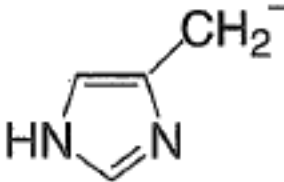
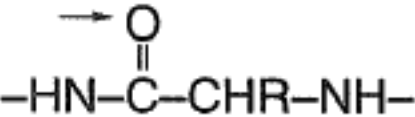
METAL LIGATION

Protein residues as ligands for metal ions



Oxygen atoms of peptide carbonyls, nitrogen atoms of deprotonated backbone amides, and lysine side chains are also available for metal coordination.

METAL LIGATION

R Group (ligand)	Amino Acid	Metals Bound ^a
$-\text{CH}_2\text{SH}^b$ $-\text{CH}_2\text{SeH}^b$	Cys SeCys	$\left\{ \begin{array}{l} \text{Fe}^{2+,3+}, \text{Ni}^{2+,3+} \\ \text{Cu}^{1+,2+}, \text{Zn}^{2+} \\ \text{Cd}^{2+}, \text{Mo}^{4+,6+} \end{array} \right.$
$-\text{CH}_2-\text{C}_6\text{H}_4-\text{OH}^b$	Tyr	$\text{Mn}^{3+}, \text{Fe}^{2+,3+}$
	His	$\text{Fe}^{2+,3+}, \text{Mn}^{2+,3+}, \text{Ni}^{2+}$ $\text{Cu}^{1+,2+}, \text{Zn}^{2+}$
$-\text{CH}_2\text{SCH}_3$	Met	$\text{Fe}^{2+,3+}, \text{Cu}^{1+,2+}$
$-(\text{CH}_2)_{1,2}\text{COOH}^b$	Asp, Glu	$\text{Mg}^{2+}, \text{Ca}^{2+}, \text{Mn}^{2+,3+}$ $\text{Fe}^{2+,3+}, \text{Ni}^{2+}, \text{Zn}^{2+}$
	any	$\text{Cu}^{2+}, \text{Zn}^{2+}, \text{Mg}^{2+}, \text{Ca}^{2+}$

The prosthetic group

It is an additional, non-amino acid component of a protein which is essential for the biological activity of the protein.

We shall be concerned with prosthetic groups containing metal centres,

e.g. **haem** is the prosthetic group in haemoglobin and myoglobin.

PROSTHETIC GROUPS

biosynthesized groups that may participate directly in catalytic processes or carry other small molecules; binding to proteins is strong

functions

- bind metal cations tightly

- may accept a substrate

- may participate in electron transfer

MACROCYCLIC LIGANDS

tetrapyrroles

most common, best known bioinorganic compounds

study of structure/function and organic synthesis of these complexes
led to several Nobel prizes

1915 – **Willstätter** (extraction of pigments, relationship between chlorophyll and heme)

1930 – **Fischer** (formula of heme and chlorophyll, first synthesis of tetrapyrroles)

1962 – **Kendrew & Perutz** (X-ray structure of hemoglobin and myoglobin)

1964 – **Crowfoot Hodgkin** (X-ray structure of vitamin B₁₂)

1965 – **Woodward** (total synthesis of vitamin B₁₂ and chlorophyll)

1988 – **Deisenhofer, Huber, & Michel** (X-ray structure of photosynthetic reaction centers containing heme and chlorophyll in bacteria)

Metalloproteins : The proteins that contain metals

Apoprotein: The form of the protein with the metal removed is called the apoportien.

The prefix apo- before a particular protein (e.g. ferritin and apoferritin) signifies the metal-free species.

The difference between a protein and the corresponding apoprotein is analogous to that between a metal complex and the corresponding free ligand

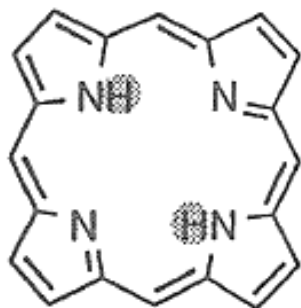
TETRAPYRROLES

partially unsaturated, tetradentate, macrocyclic ligands

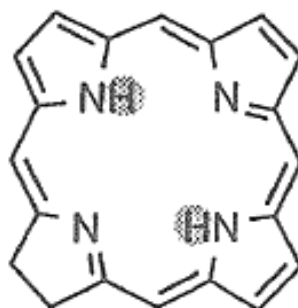
stable, rigid, planar or nearly planar ring system

deprotonated forms bind metal ions tightly and size selectively

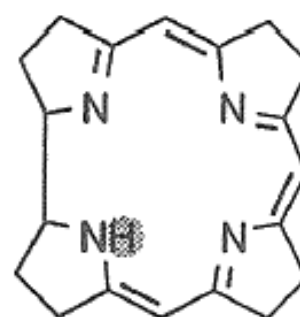
extensive conjugation leads to very intense colors (*pigments of life*) and potentially to redox activity



porphyrin
or 'porphin' (in
unsubstituted form)



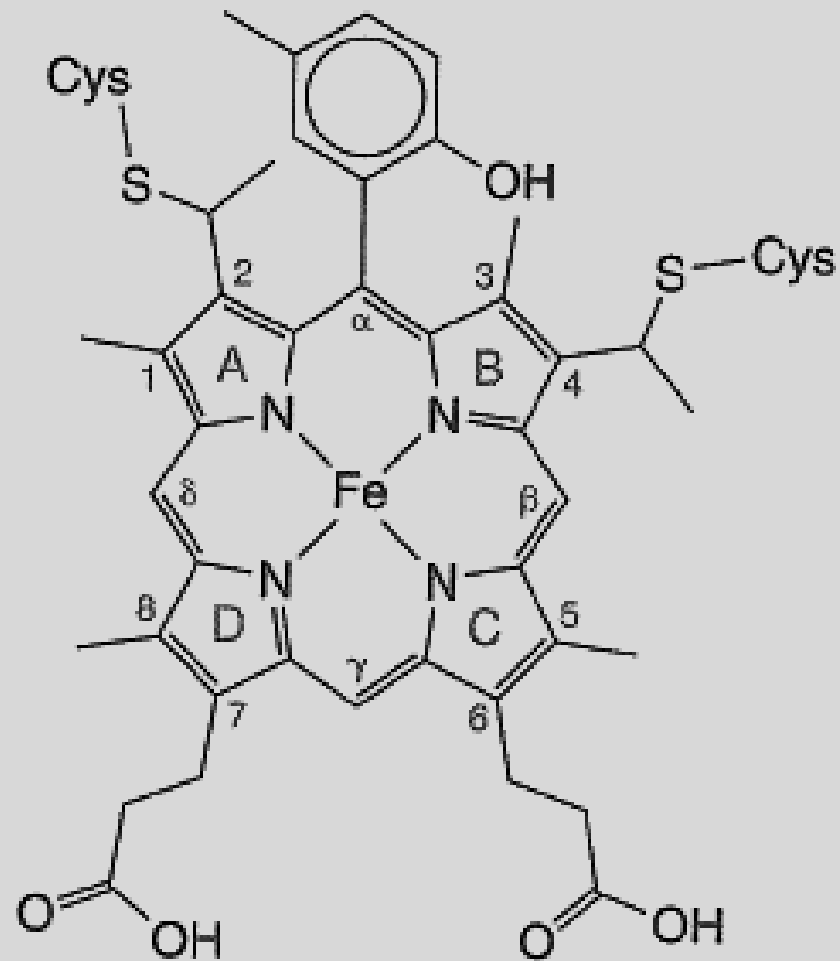
chlorin
(2,3-dihydroporphyrin)



corrin

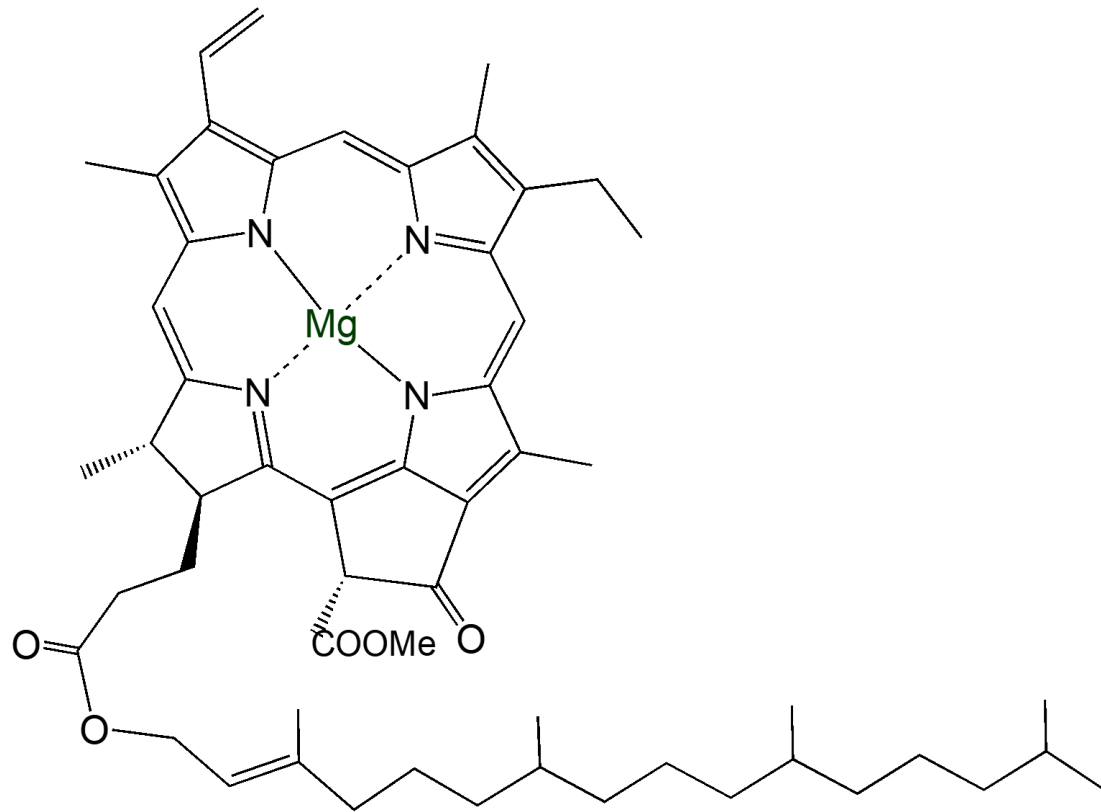
●: acidic protons which may be substituted by metal ions

PORPHYRINS



Heme P460

CHLORINS – CHLOROPHYLL *a*

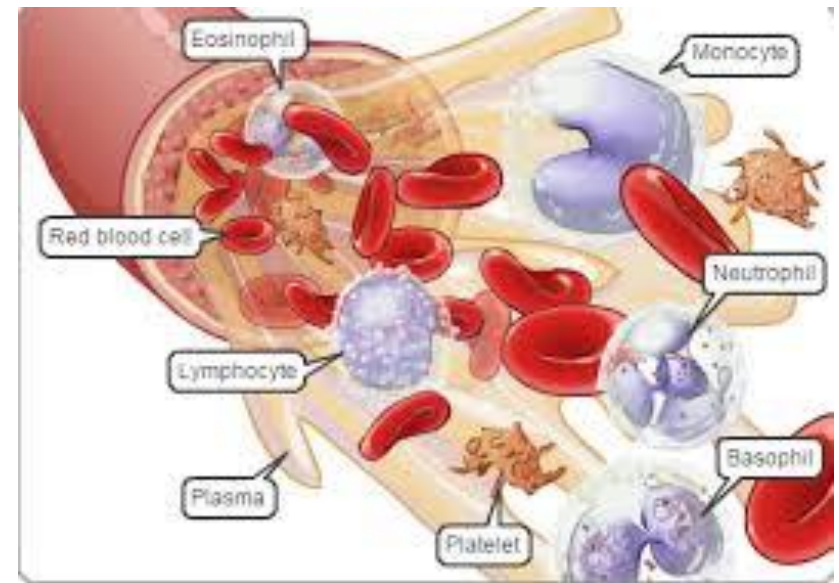
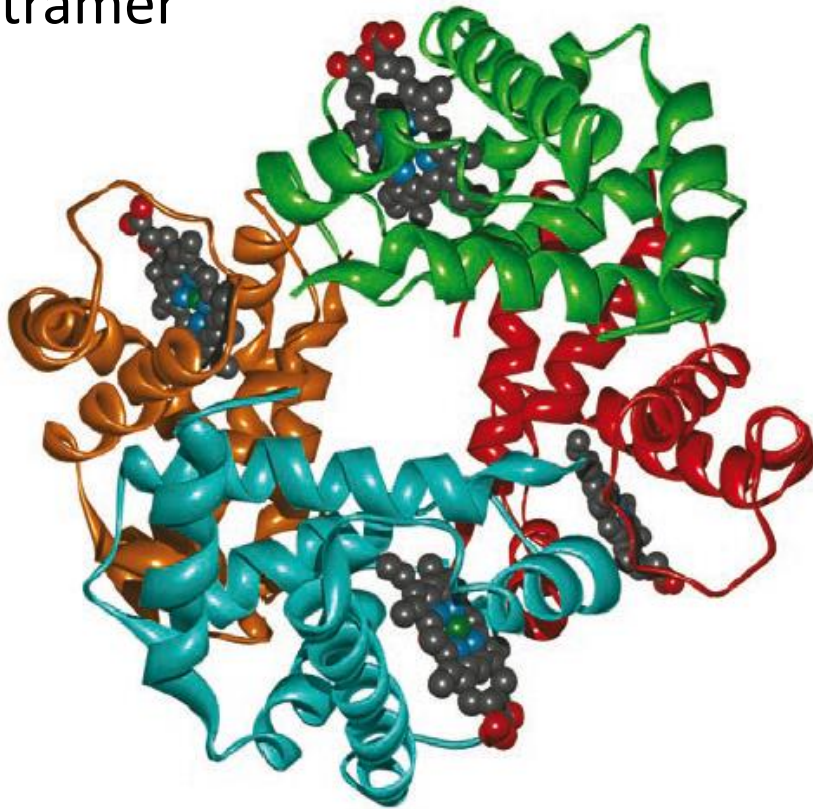


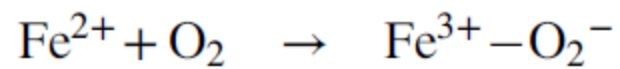
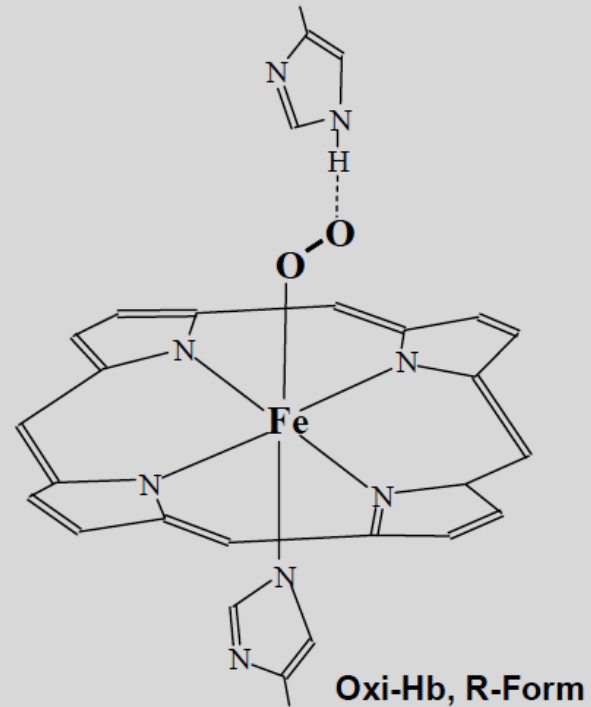
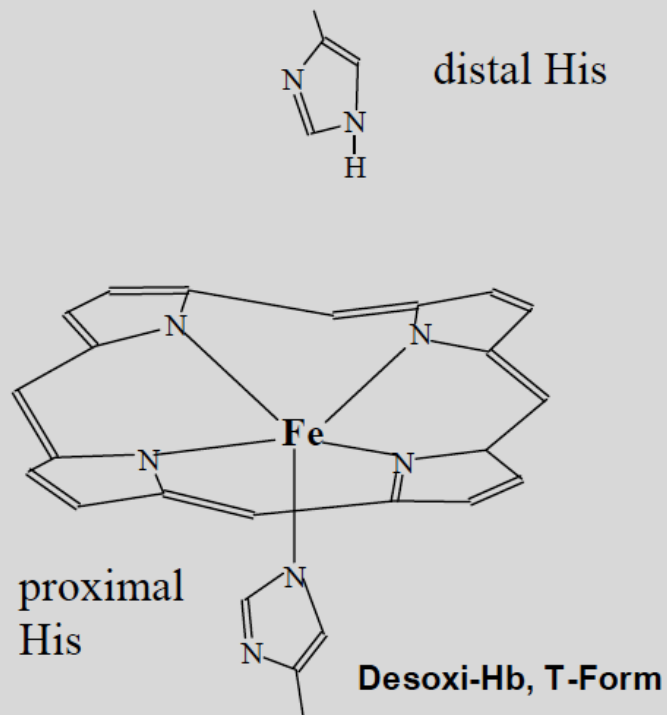
chlorophyll *a*

Haemoglobin and Myoglobin

Myoglobin has a molecular weight of 17000 and is a monomer with a protein chain consisting of 153 amino acid residues.

Haemoglobin has a molecular weight of 64500 and is a tetramer

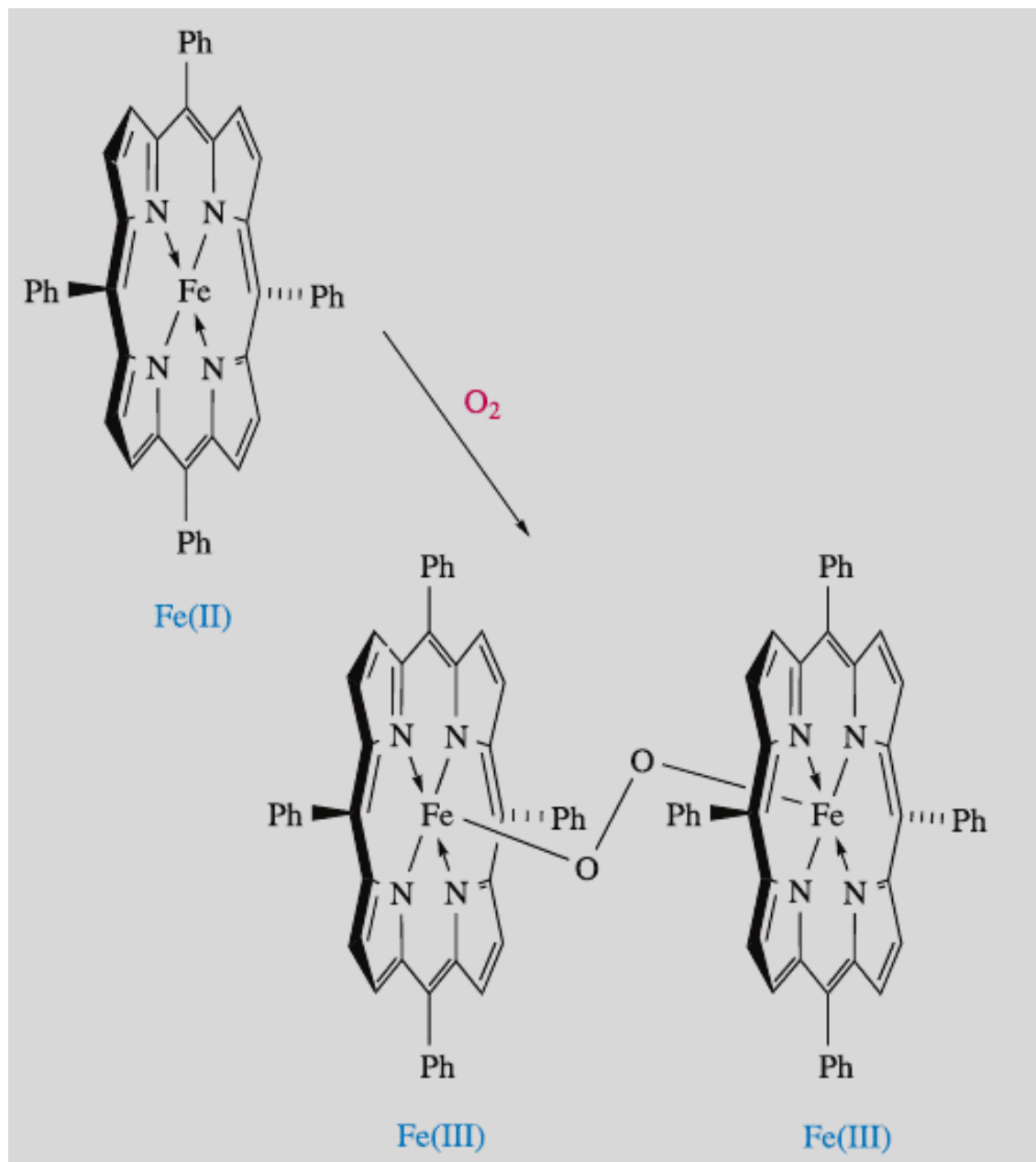




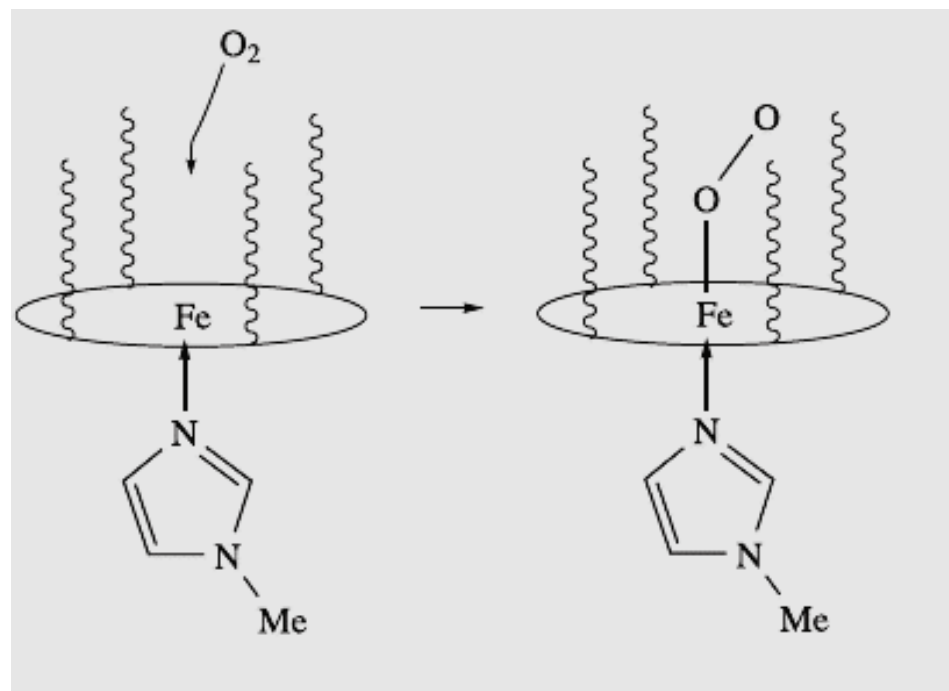
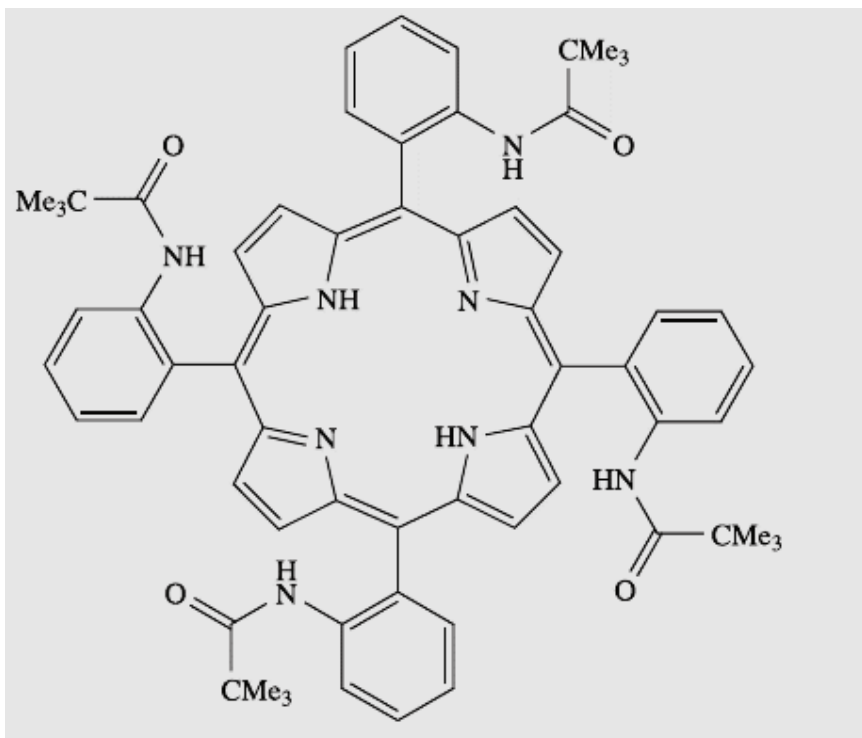
Deoxy-Mb

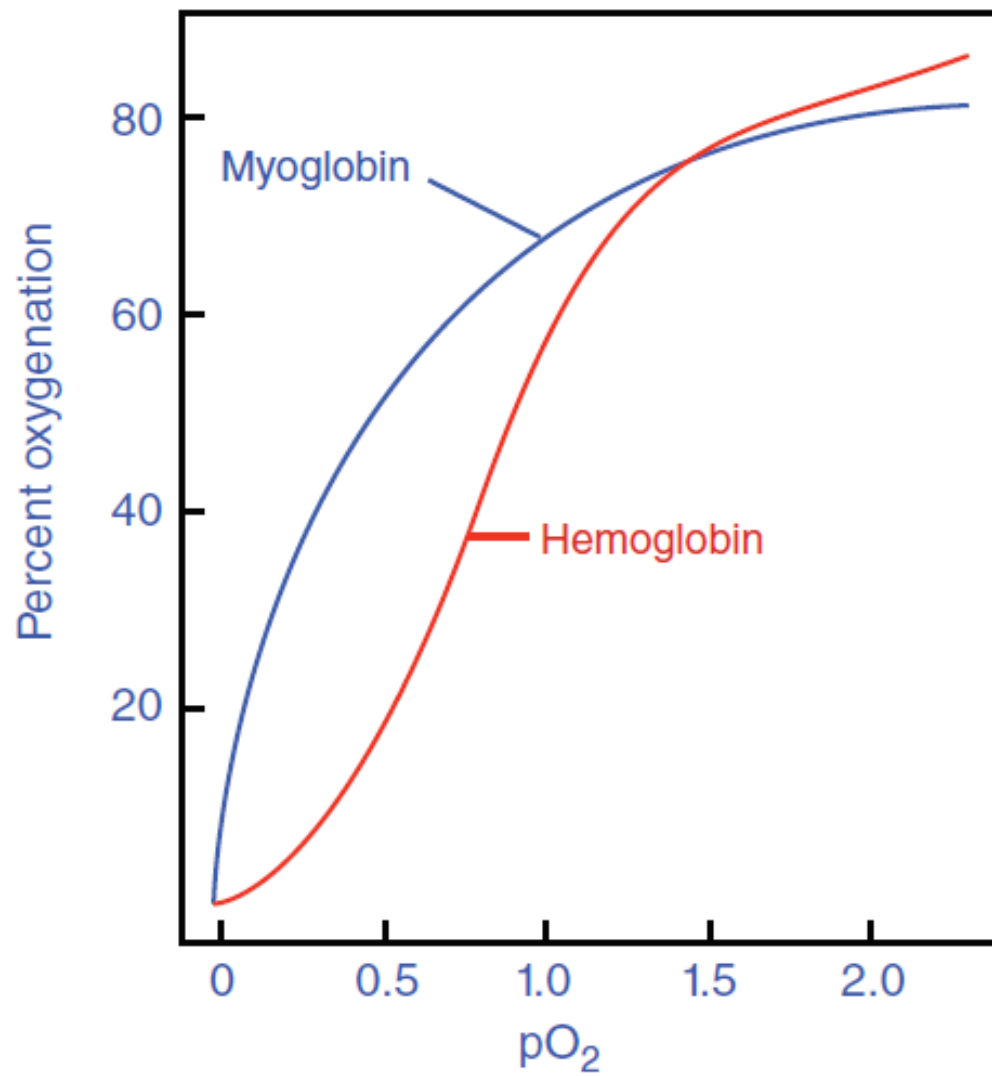
Oxy-Mb,

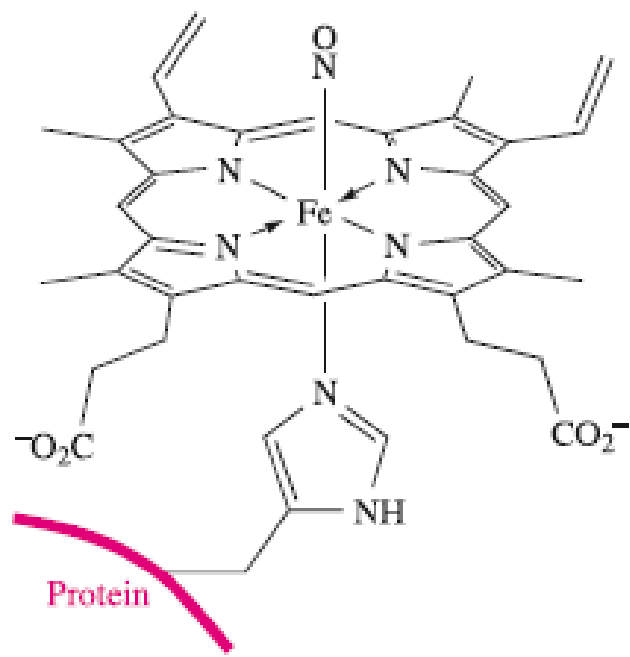
	O_2	O_2^-	O_2^{2-}
ν_{O-O} (cm^{-1})	1560	1150–1100	850–740
Bond length ($\text{\AA}$)	1.21	1.28	1.49
Bond order	2	1.5	1



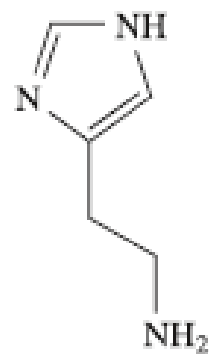
'picket-fence' porphyrin



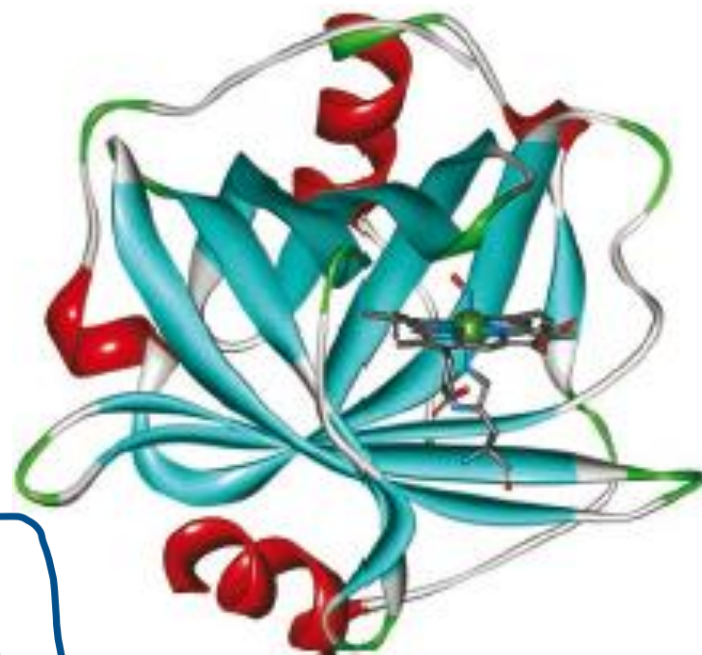




NO bound to NP1 (NP1-NO)



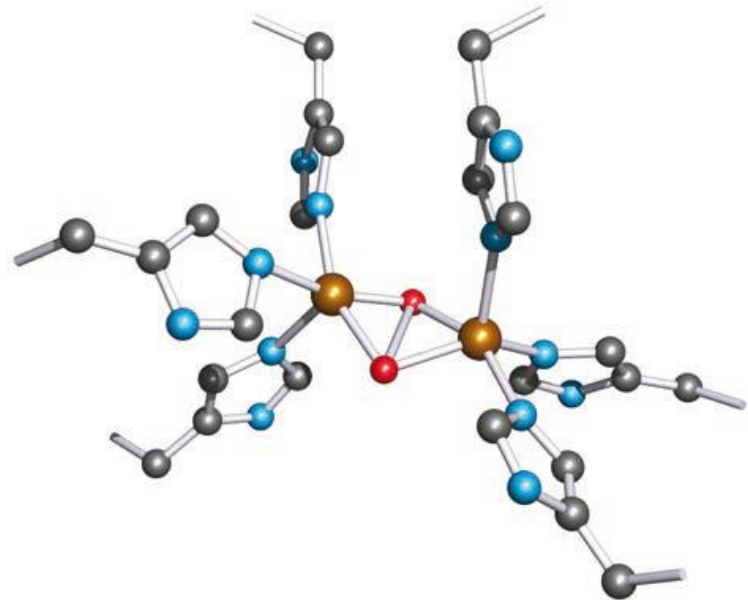
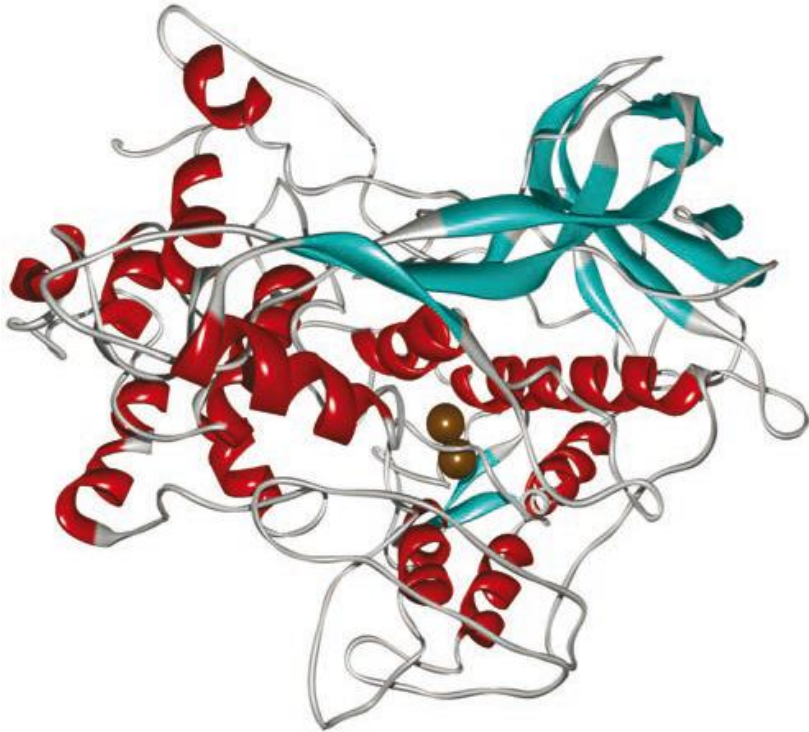
Histamine

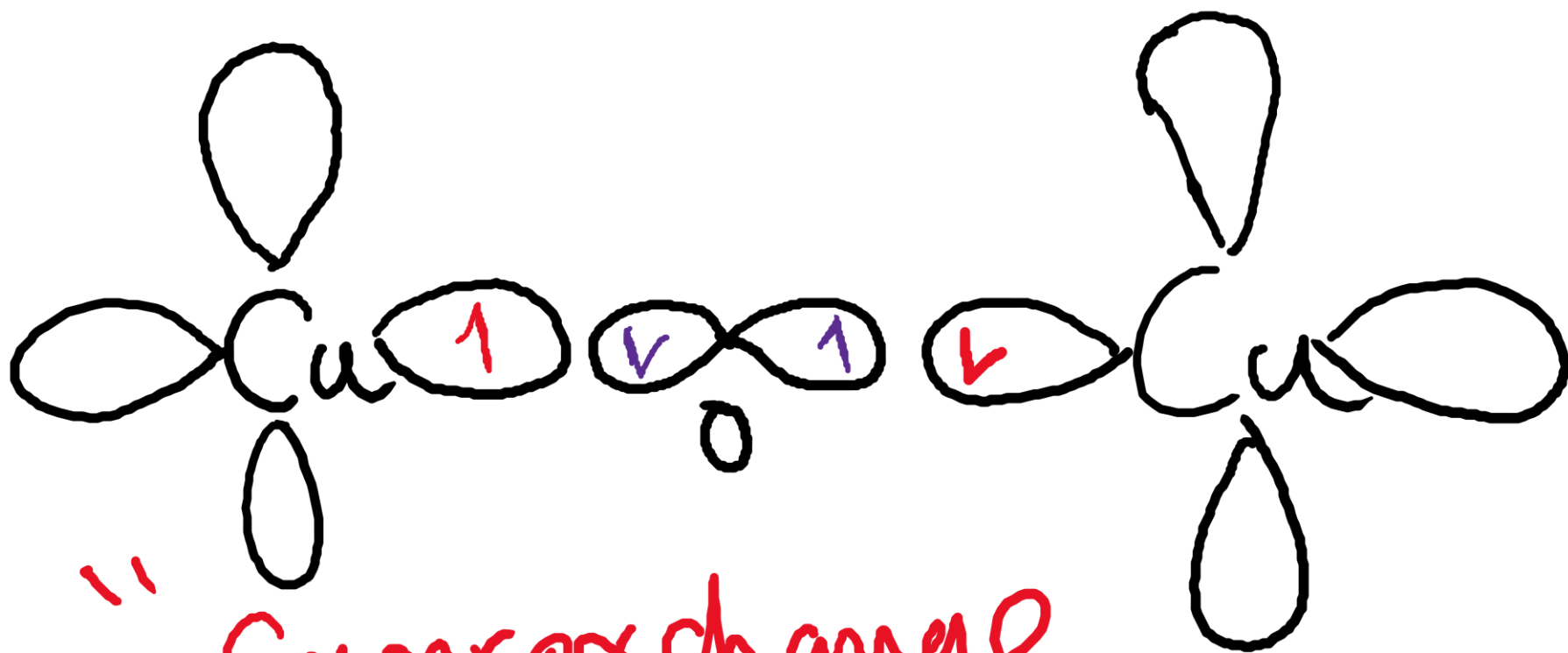


NOT FOR EXAM

Haemocyanins

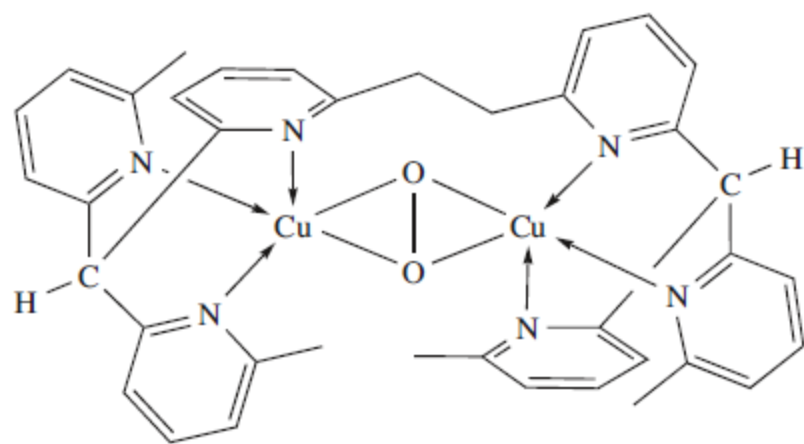
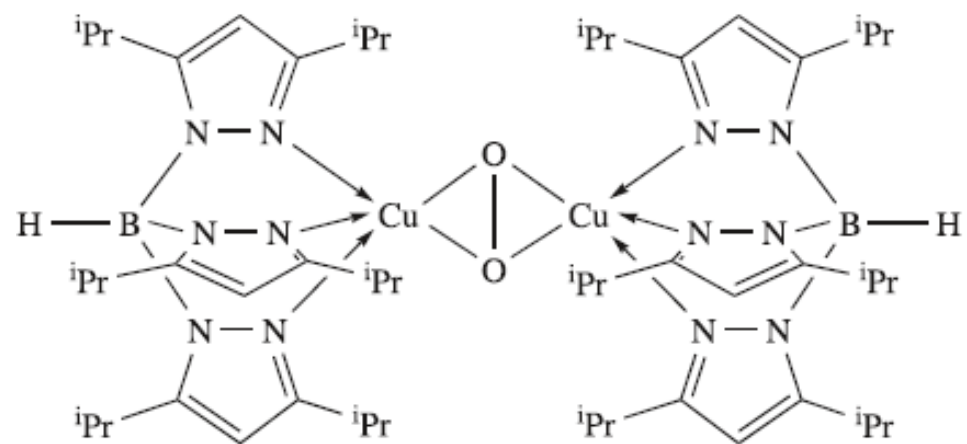
O₂-carrying copper-containing proteins in molluscs (e.g. whelks, snails, squid) and arthropods (e.g. lobsters, crabs, shrimps, horseshoe crabs, scorpions)





''

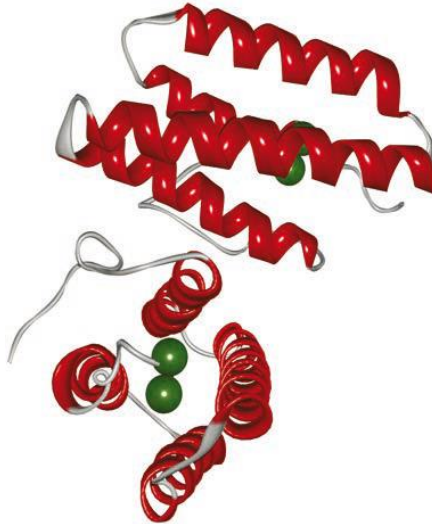
Superexchange



Haemerythrin

In marine invertebrates such as annelids (segmented earthworms), O_2 is transported by haemerythrin, a non-haem Fe-containing protein

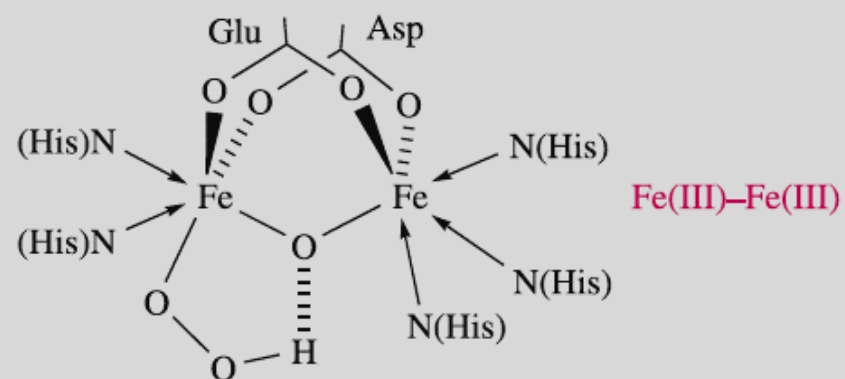
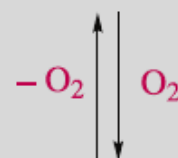
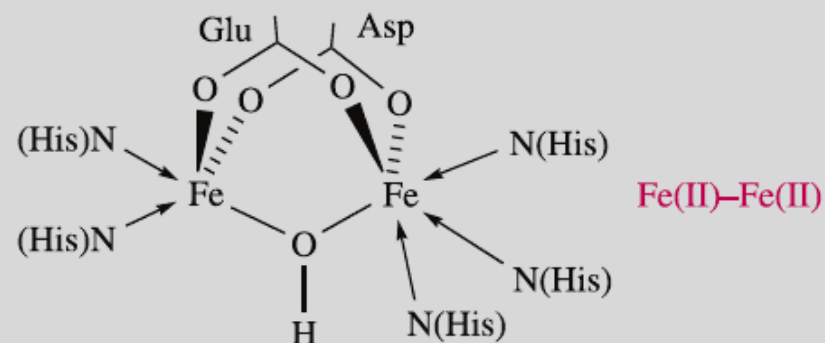
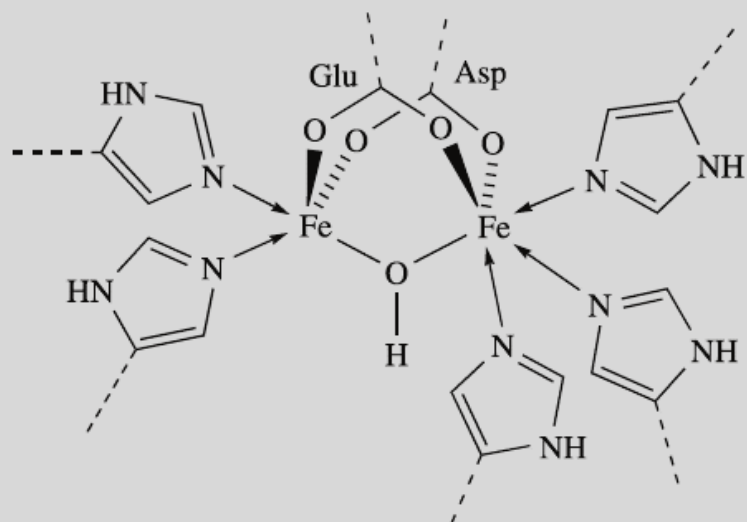
(Mr wt ~ 108 000)

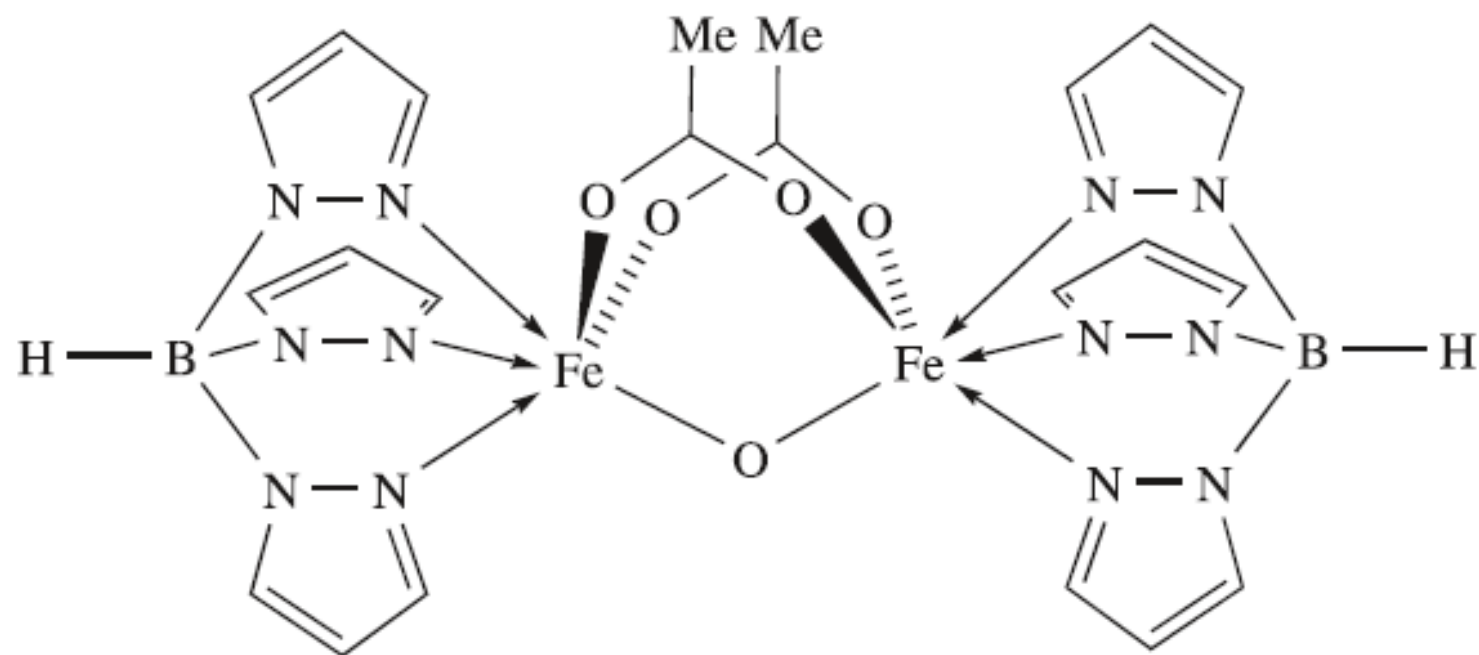


In the deoxy-form, a hydroxido-bridged $[Fe(II)]_2$ unit is present.

The two $Fe(II)$ centres in deoxyhaemerythrin are strongly antiferromagnetically coupled through the Fe-O-Fe bridge

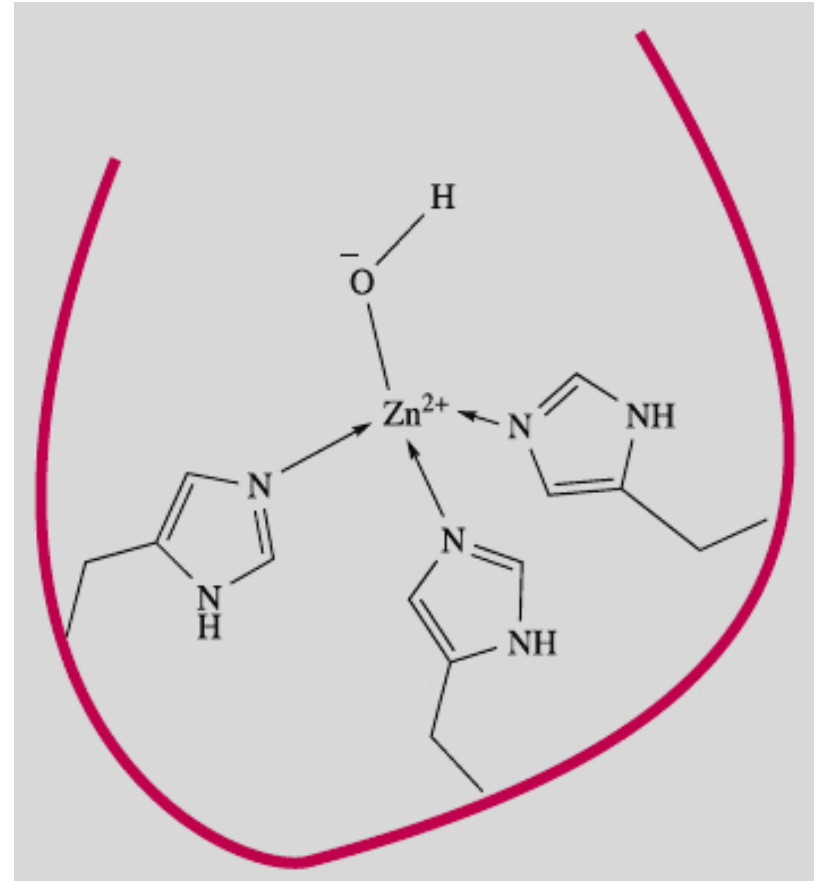
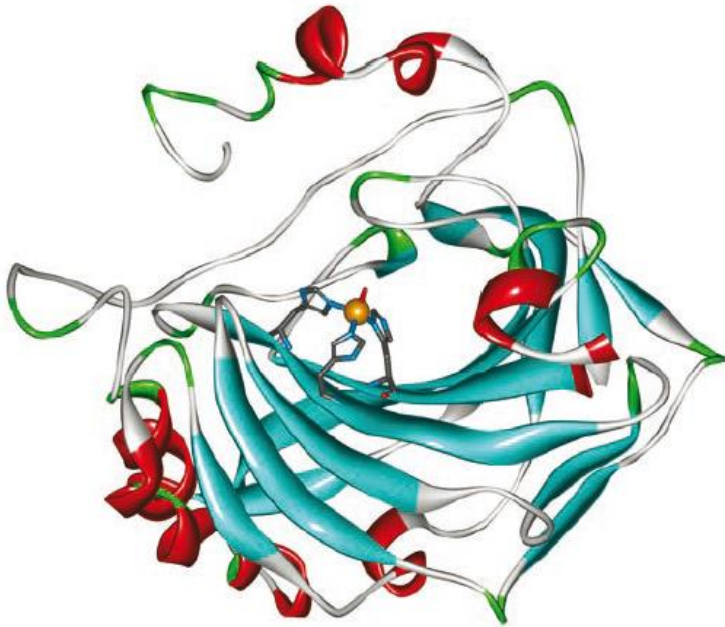
Unlike haemoglobin, haemerythrin exhibits no **cooperativity** between the sub-units during O_2 binding

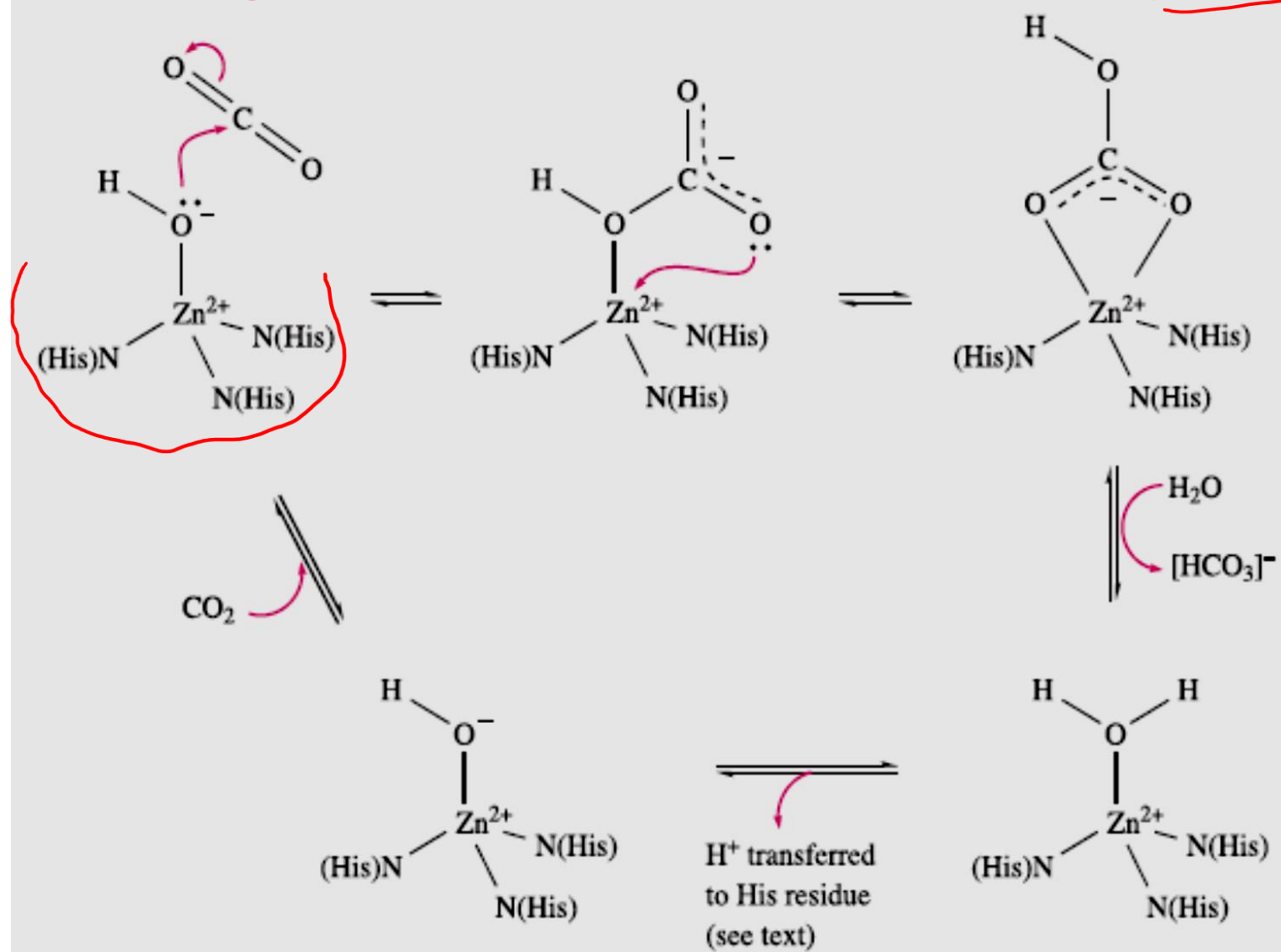


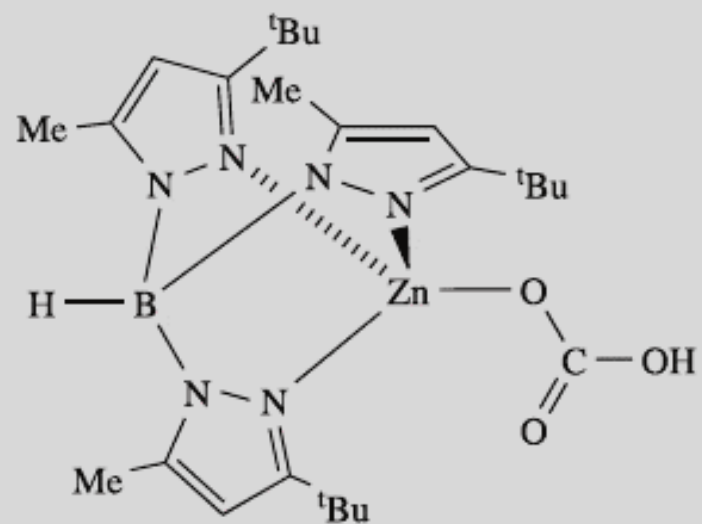
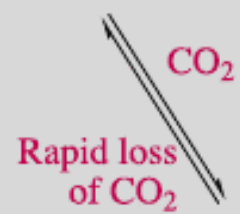
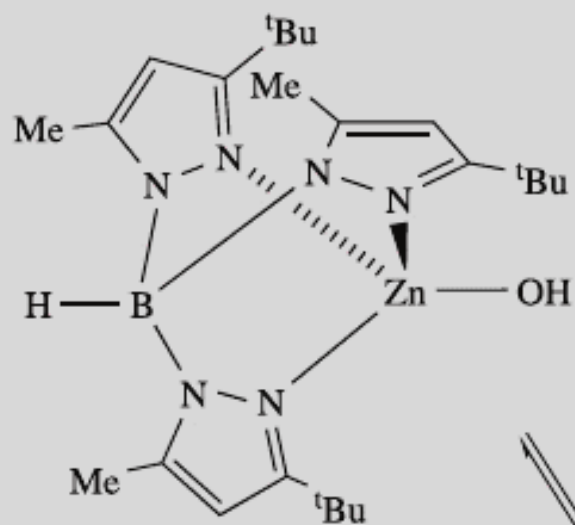


CARBONIC ANHYDRASE

The metalloprotein consists of 260 amino acids







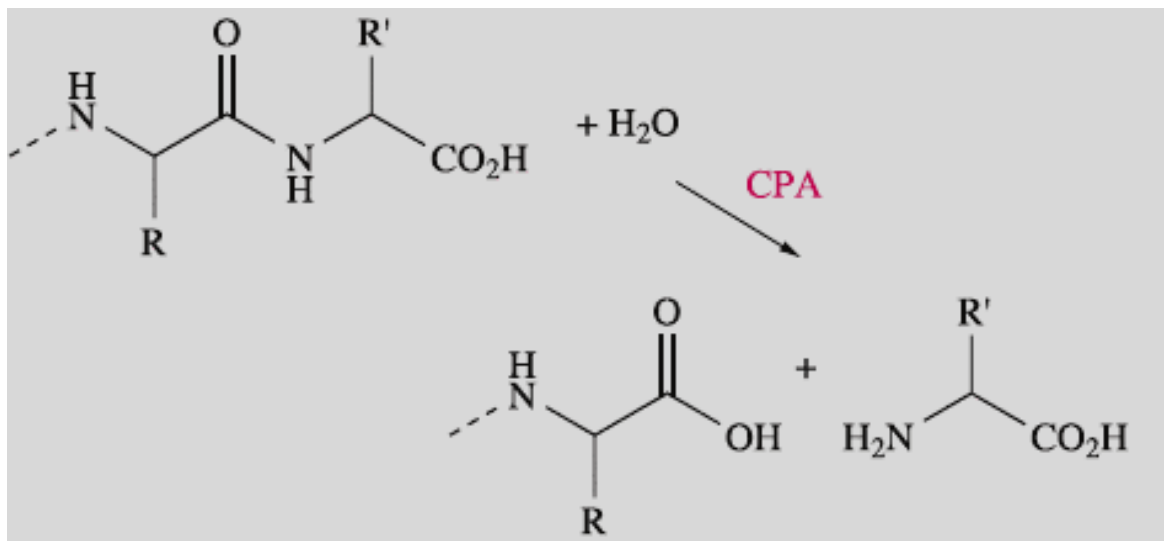
Carboxypeptidase

Metalloenzyme which catalyses the cleavage of a peptide link in a polypeptide chain

Carboxypeptidase A is monomeric (Mr 34 500)

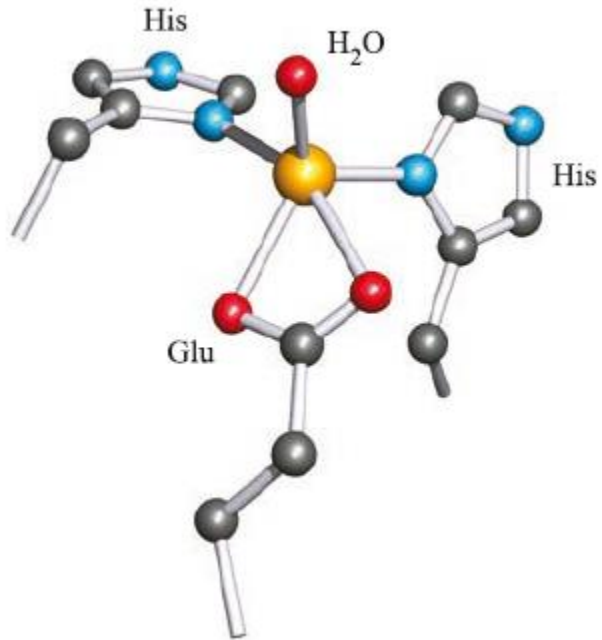
Near the surface of the protein lies a pocket in which a Zn^{2+} ion is bound to the protein backbone by one bidentate Glu and two His residues.

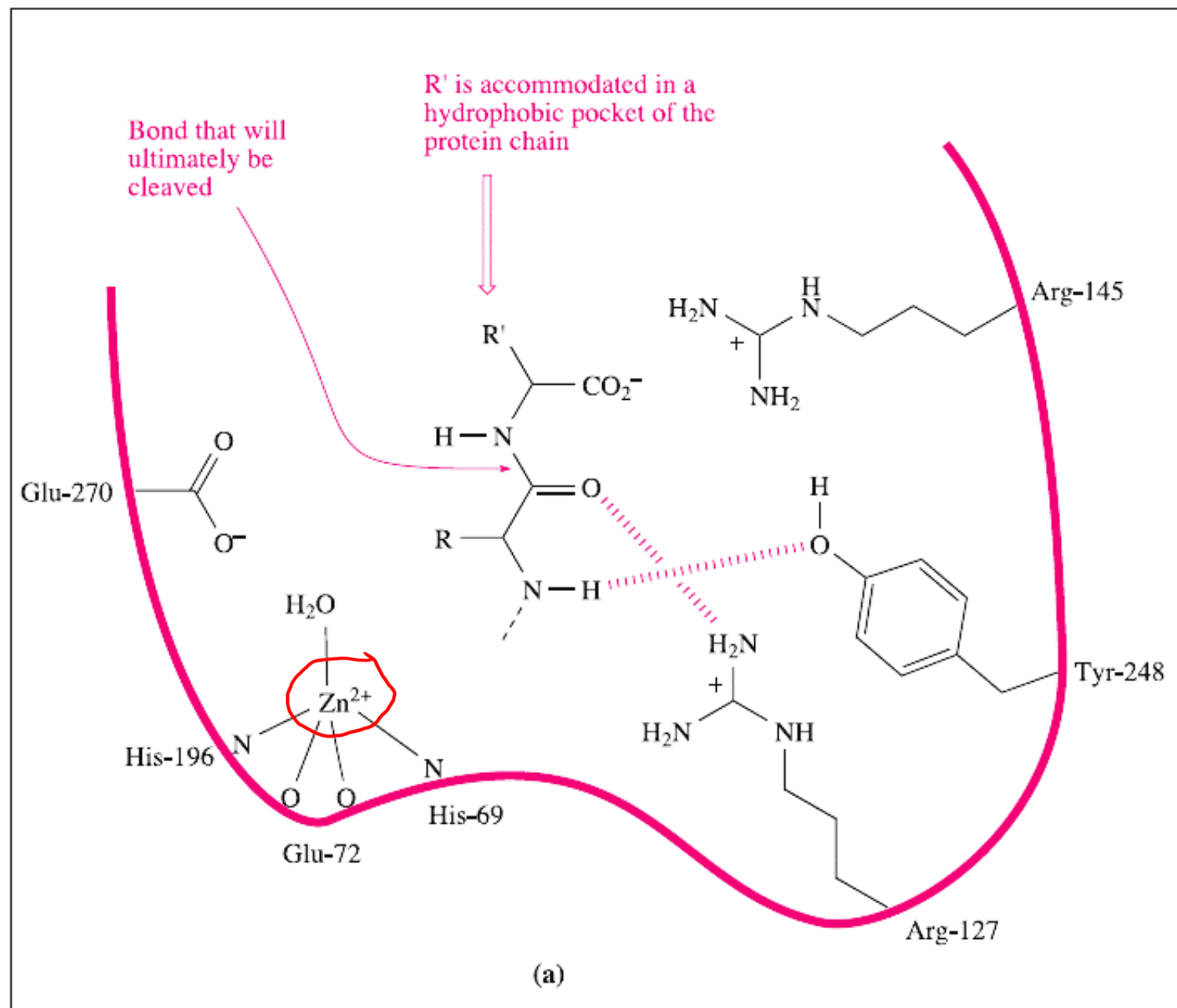
A 5-coordinate coordination sphere is completed by a water molecule



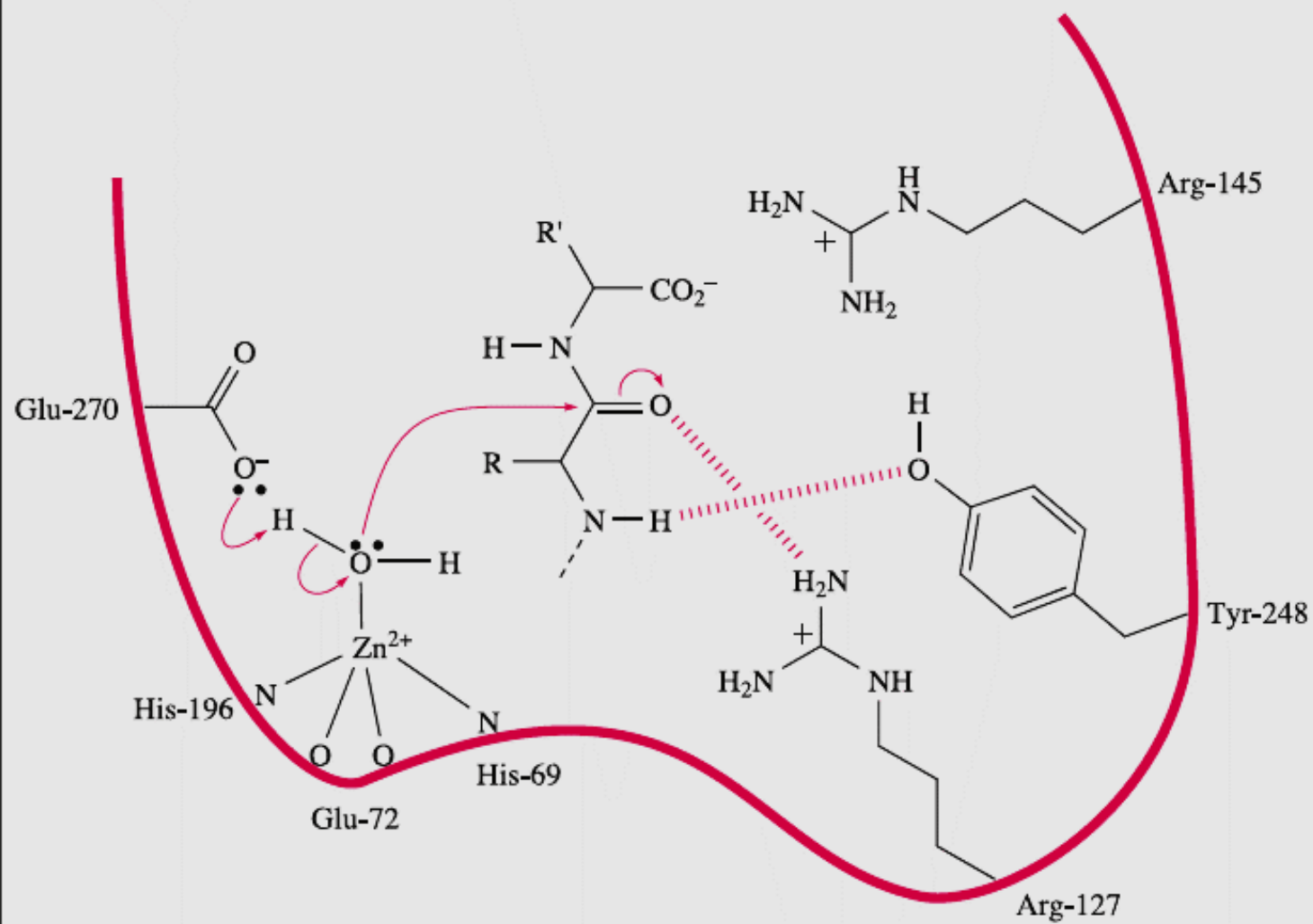
Carboxypeptidase

Active Site



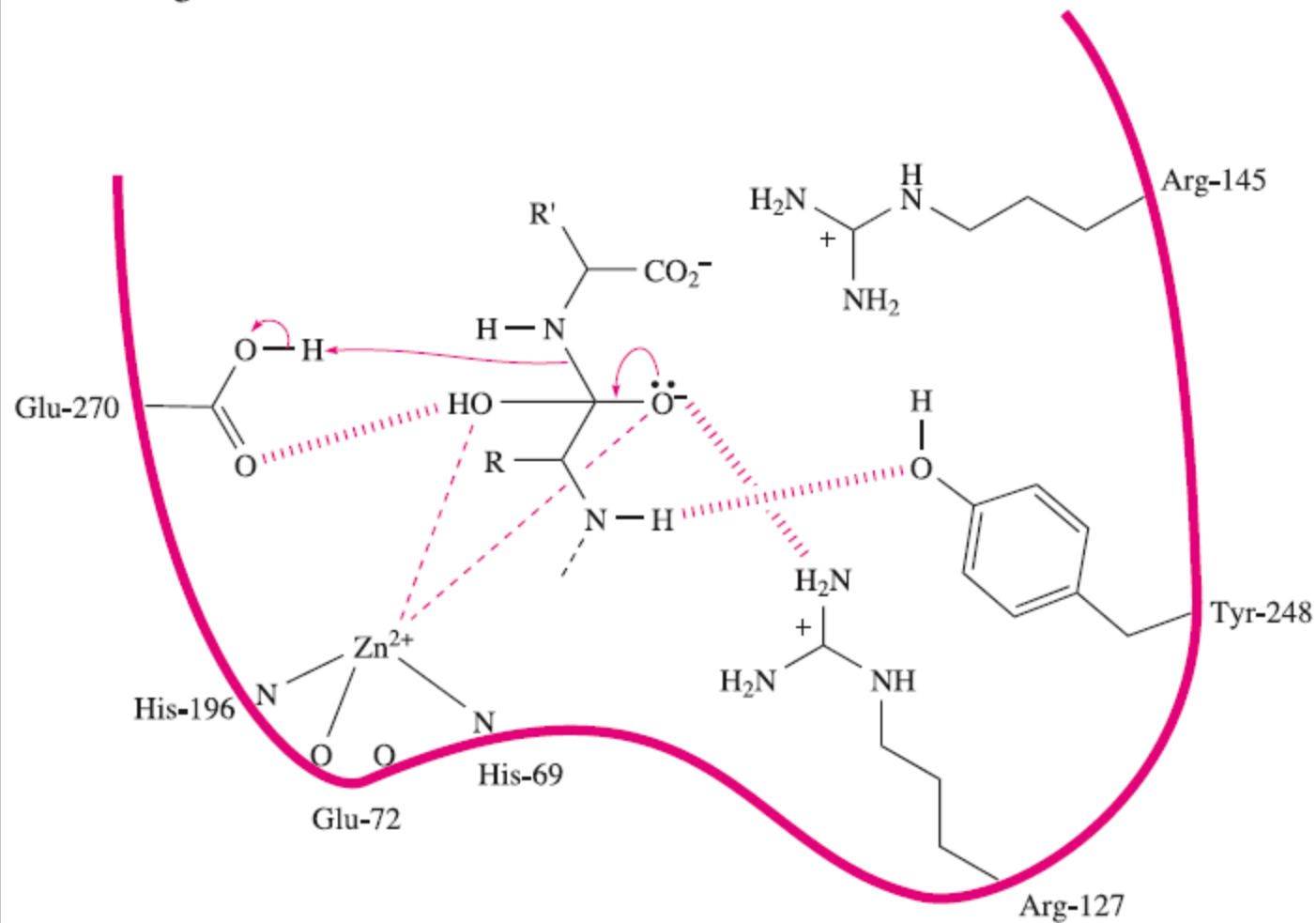


Nucleophilic attack on carbonyl C atom



(b)

Cleavage of the C–N bond



(c)

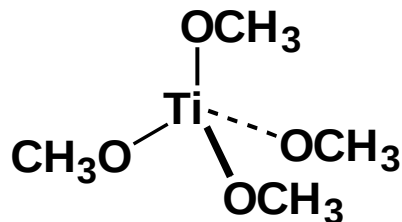
ORGANOMETALLIC CHEMISTRY

ORGANOMETALLIC CHEMISTRY

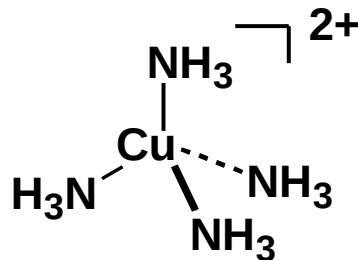
- 1. Introduction**
- 2. 16 and 18 electron rule (learning to count)**
- 3. Bonding of selected Organometallic ligands**
- 4. Characterization technique: IR**
- 5. Applications (Catalysis).**

Which of the following compounds is an organometallic compound?

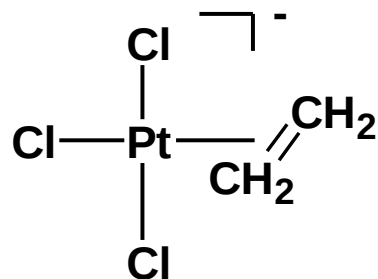
a)



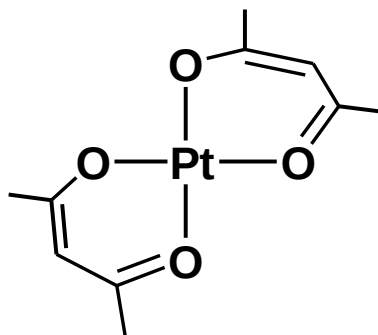
b)



c)



d)

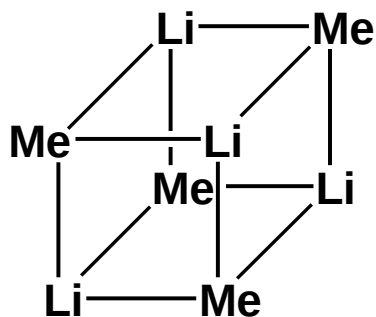


In general, metals in organometallic compounds include:

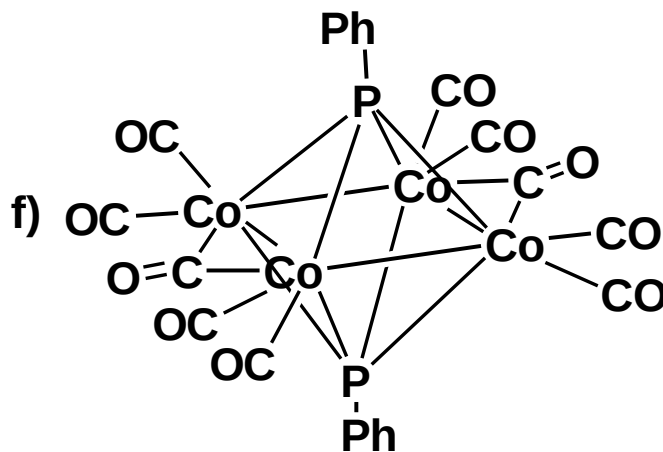
- *main group metals*
- *transition metals*
- *f-block metals*

In this course, transition metals are our main concern.

e)

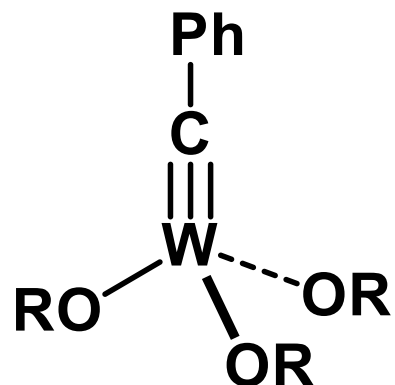
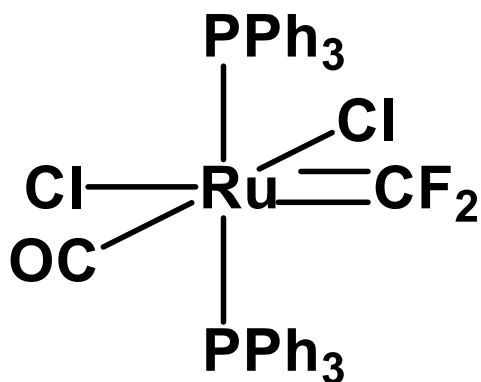
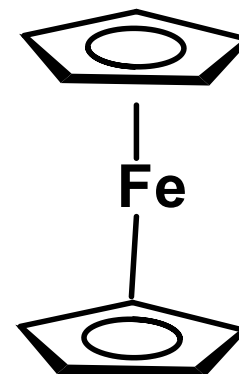
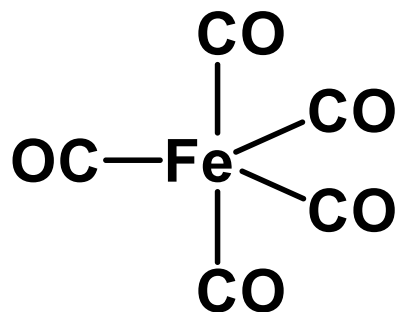
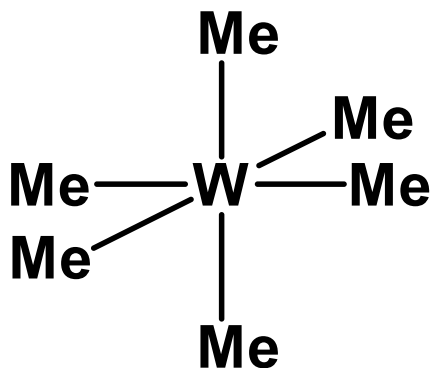


f)



Chemistry: structures, bonding and properties of molecules.

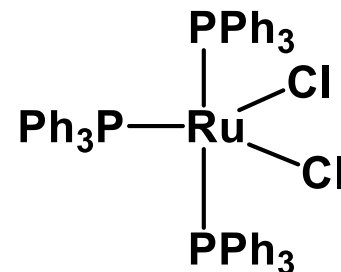
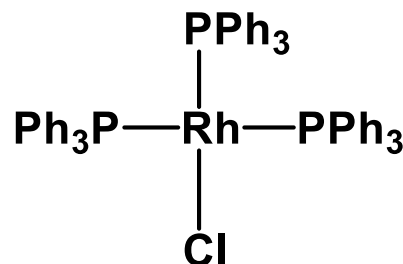
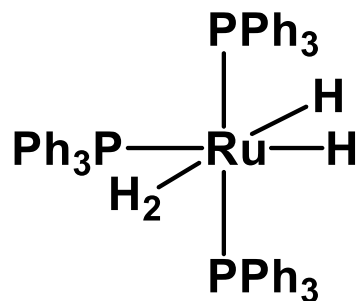
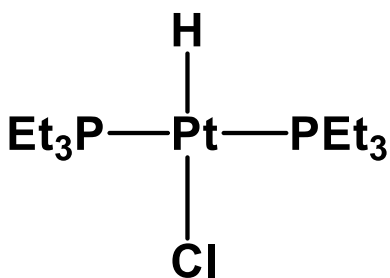
Transition metal compounds



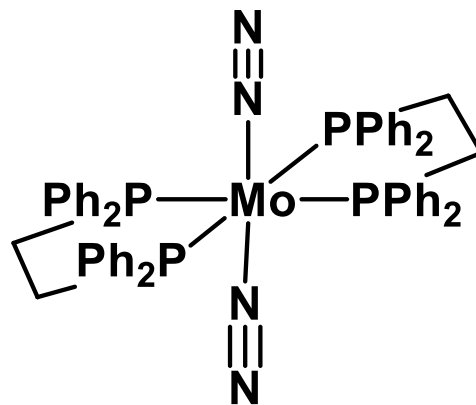
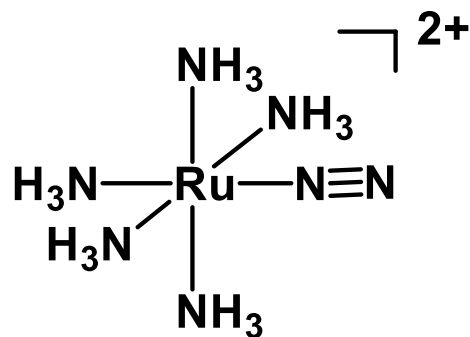
Some compounds do not contain metal-carbon bond, but they are usually included in the field of organometallic chemistry.

They include:

- Metal hydride complexes, *e.g.*
- Phosphine complexes, *e.g.*



- N₂-complexes, *e.g.*

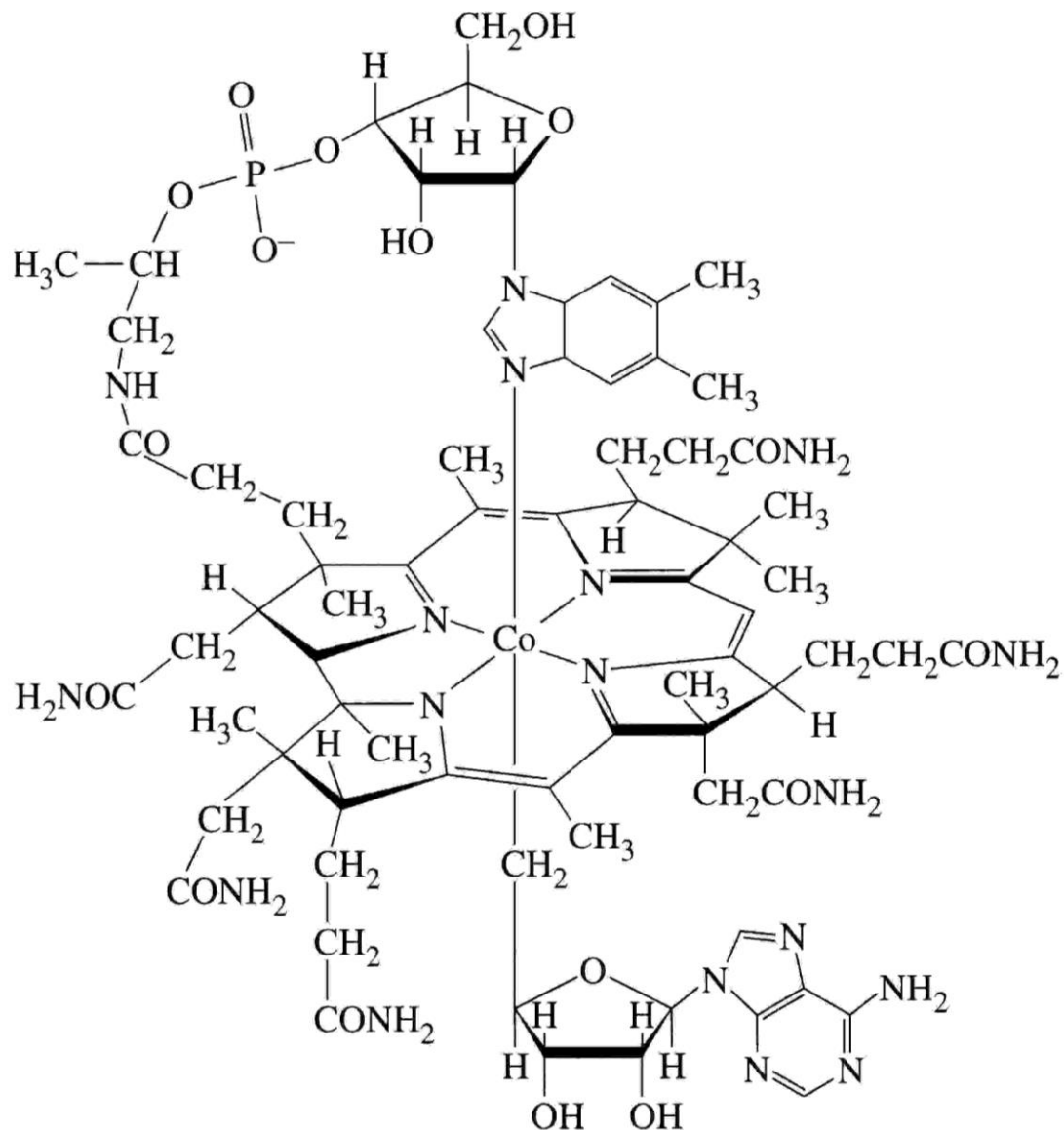


A brief history of organometallic chemistry

1) Organometallic Chemistry has really been around for millions of years

Naturally occurring
Cobalamins contain
Co—C bonds

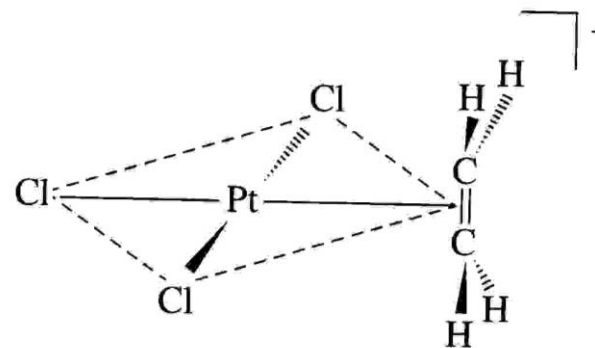
Vitamin B12



2) Zeise's Salt synthesized in 1827 = $\text{K}[\text{Pt}(\text{C}_2\text{H}_4)\text{Cl}_3] \cdot \text{H}_2\text{O}$

Confirmed to have $\text{H}_2\text{C}=\text{CH}_2$ as a ligand in 1868

Structure not fully known until 1975



3) $\text{Ni}(\text{CO})_4$ synthesized in 1890

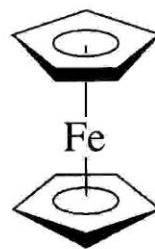
4) Grignard Reagents (XMgR) synthesized about 1900

Utility to Organic Synthesis recognized early on

5) Ferrocene synthesized in 1951

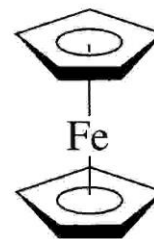
Modern Organometallic Chemistry begins with this discovery (Paulson and Miller)

1952 Fischer and Wilkinson



D_{5d}

Staggered rings



D_{5h}

Eclipsed rings

Nobel -Prize Winners related to the area:

Victor Grignard and Paul Sabatier (1912)

Grignard reagent

K. Ziegler, G. Natta (1963)

Ziegler-Natta catalyst

E. O. Fisher, G. Wilkinson (1973)

Sandwich compounds

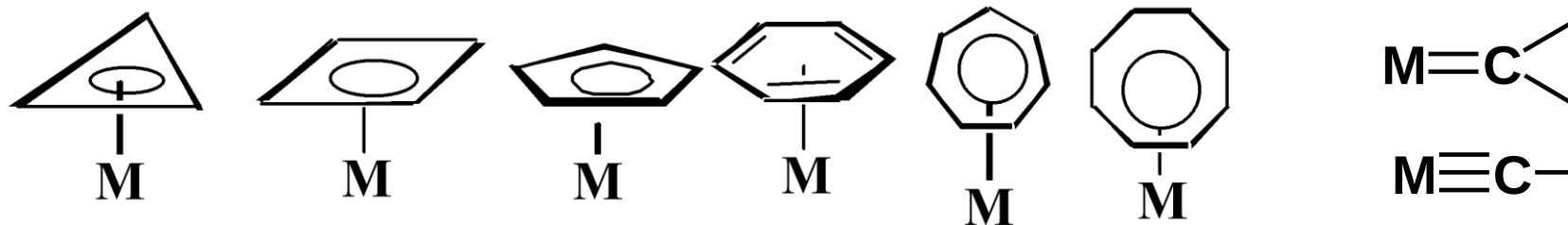
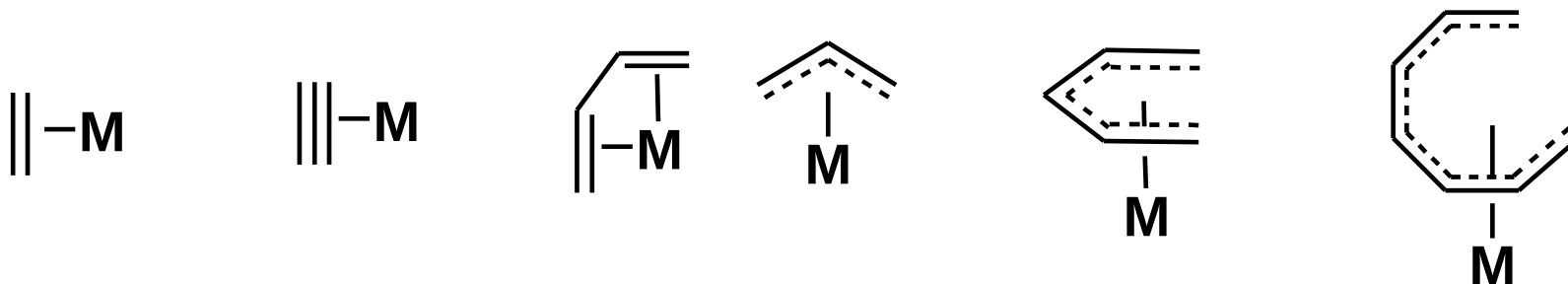
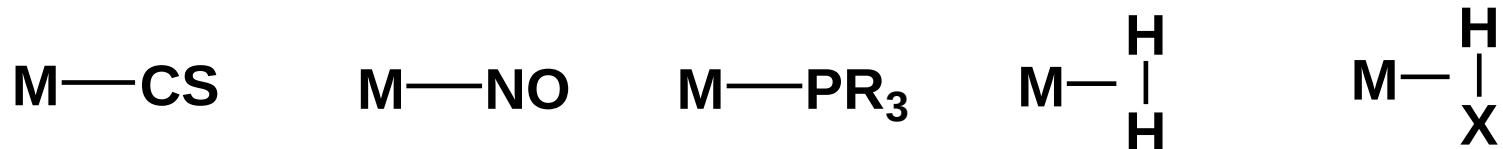
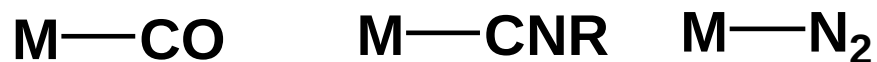
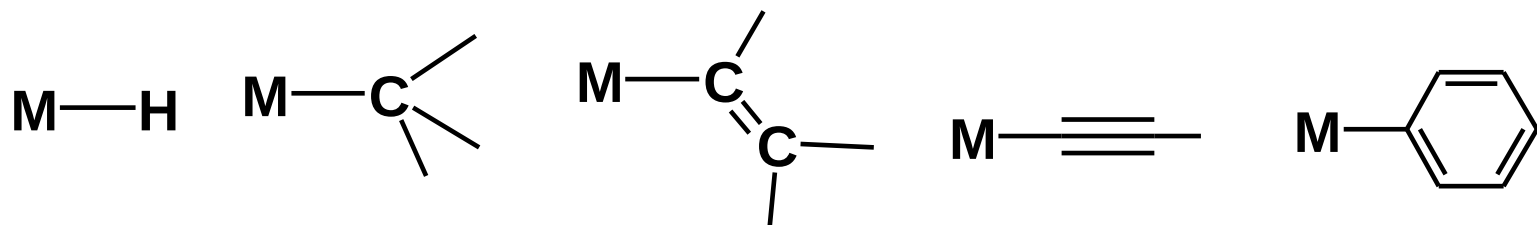
K. B. Sharpless, R. Noyori (2001)

Hydrogenation and oxidation

Yves Chauvin, Robert H. Grubbs, Richard R. Schrock (2005)

Metal-catalyzed alkene metathesis

Common organometallic ligands



Why organometallic chemistry ?

a). From practical point of view:

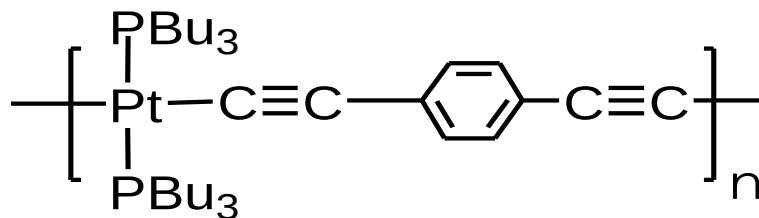
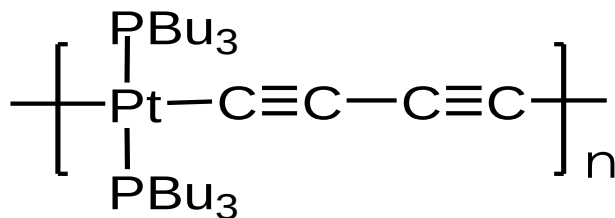
*** OMC are useful for chemical synthesis,
especially catalytic processes, *e.g.***

In production of fine chemicals

*In production of chemicals in large-scale reactions
could not be achieved traditionally*

* **Organometallic chemistry is related to material sciences.**

e.g. Organometallic Polymers



Small organometallic compounds:

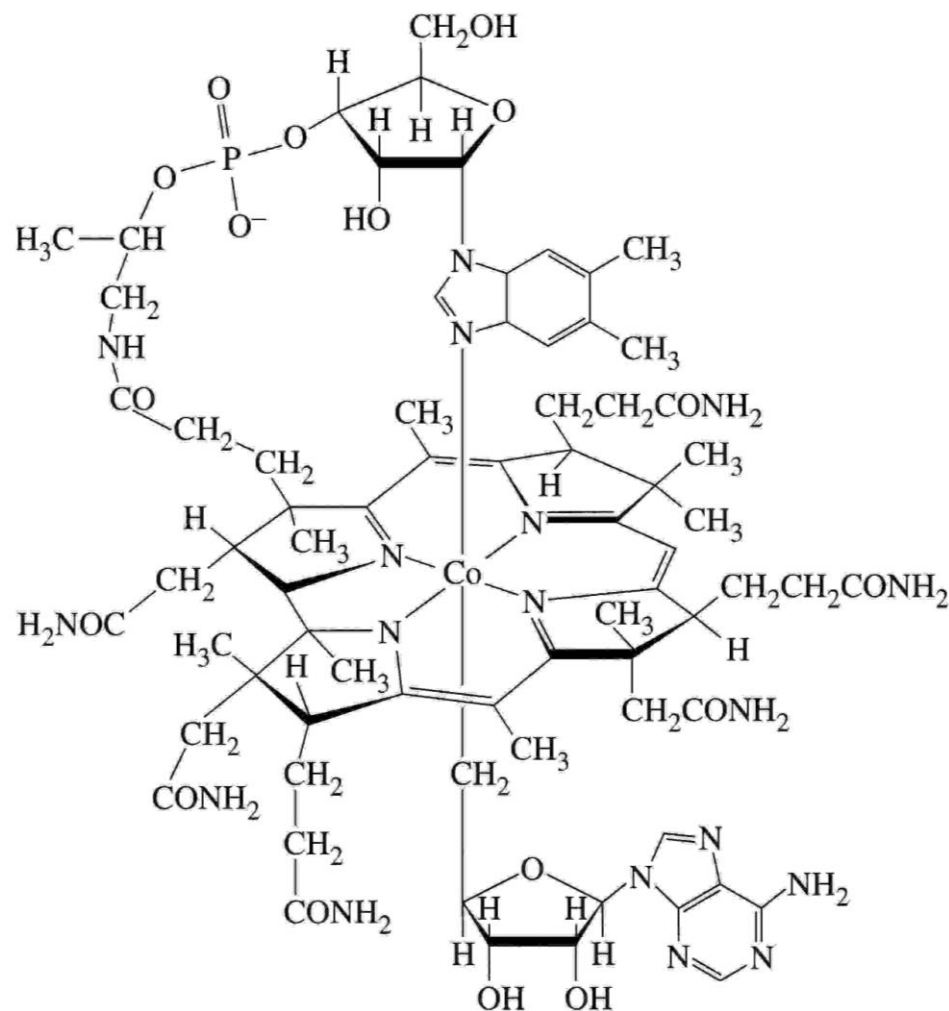
Precursors to films for coating (MOCVD)



Luminescent materials

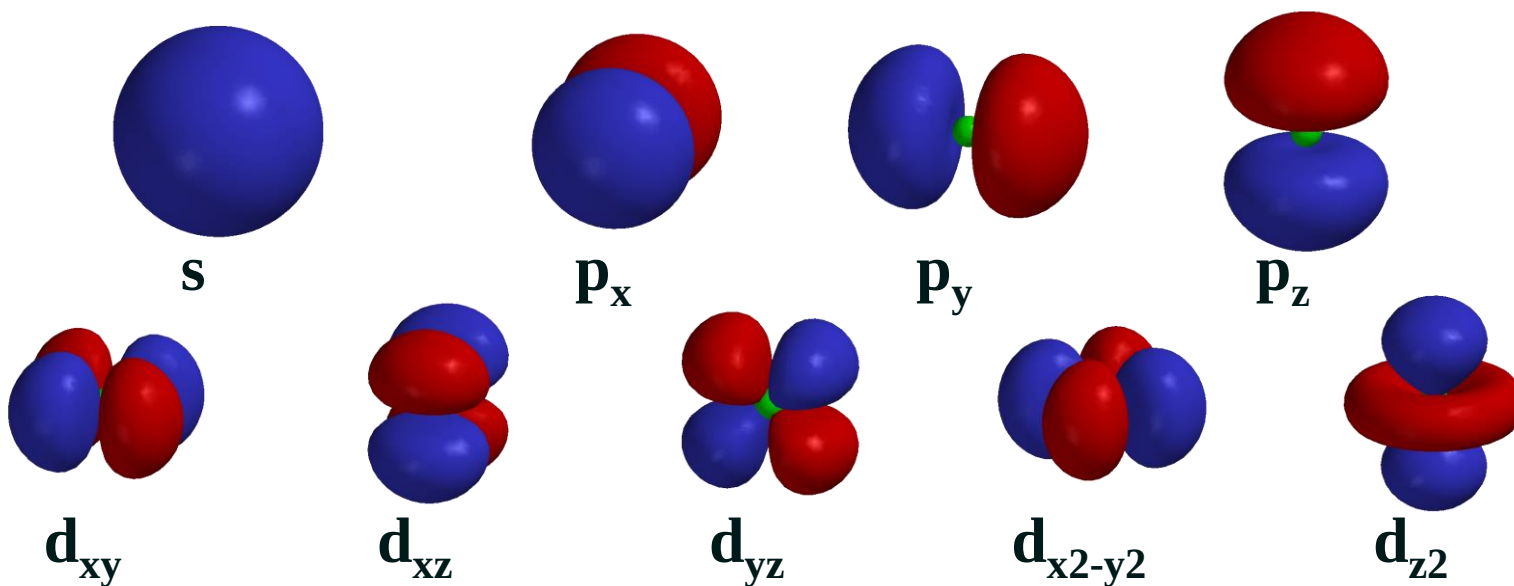
* **Biological Science. Organometallic chemistry may help us to understand some enzyme-catalyzed reactions.**

e.g. B_{12} catalyzed reactions.



18 Electron Rule (Sidgwick, 1927)

- OM chemistry gives rise to many “stable” complexes - how can we tell by a simple method
- Every element has a certain number of valence orbitals:
 - 1 { $1s$ } for H
 - 4 { $ns, 3' np$ } for main group elements
 - 9 { $ns, 3' np, 5' (n-1)d$ } for transition metals

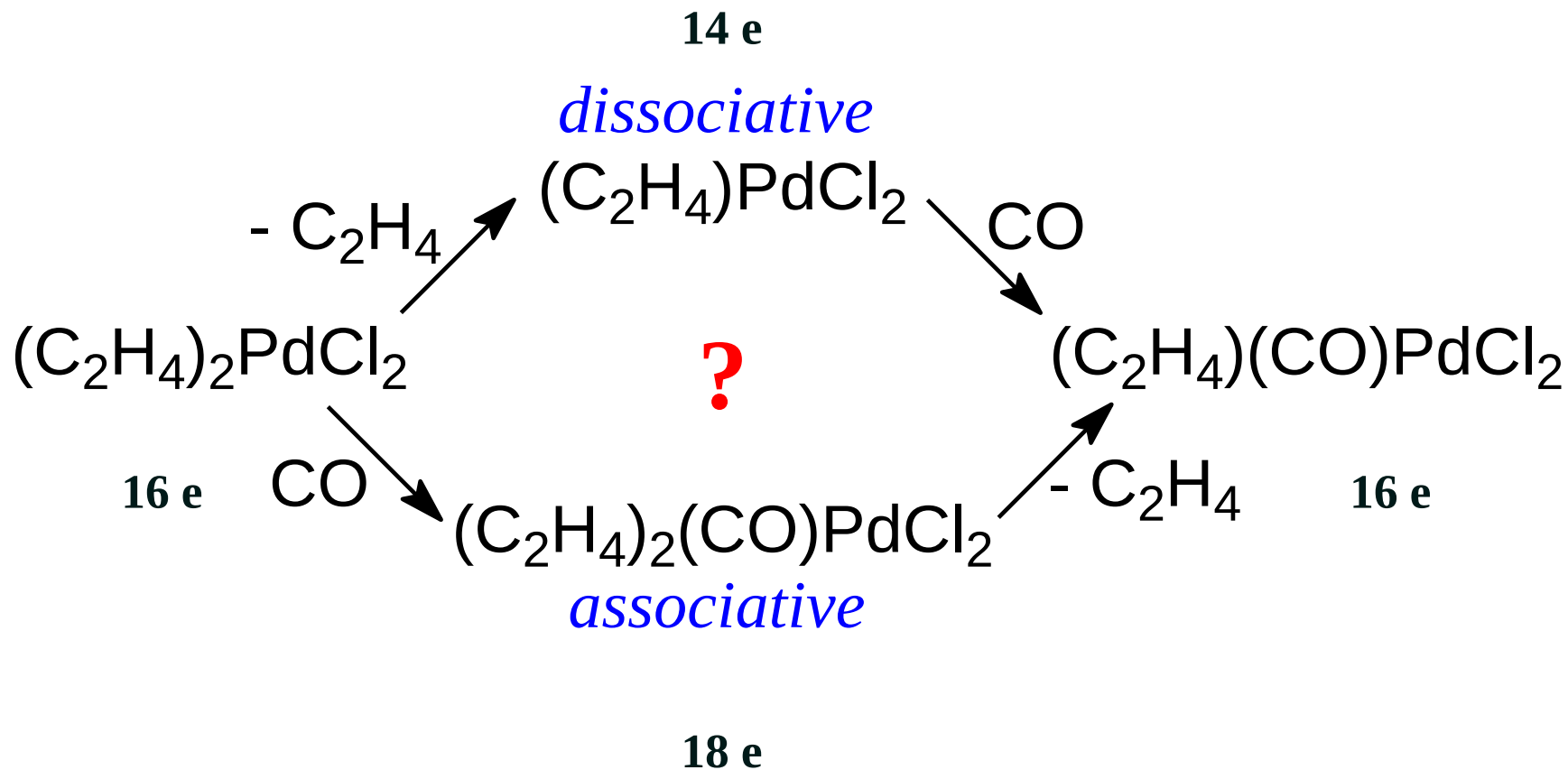


- Therefore, every element wants to be surrounded by 2/8/18 electrons
 - For main-group metals (8-*e*), this leads to the standard Lewis structure rules
 - For transition metals, we get the 18-electron rule

The strength of the preference for electron-precise structures depends on the position of the element in the periodic table

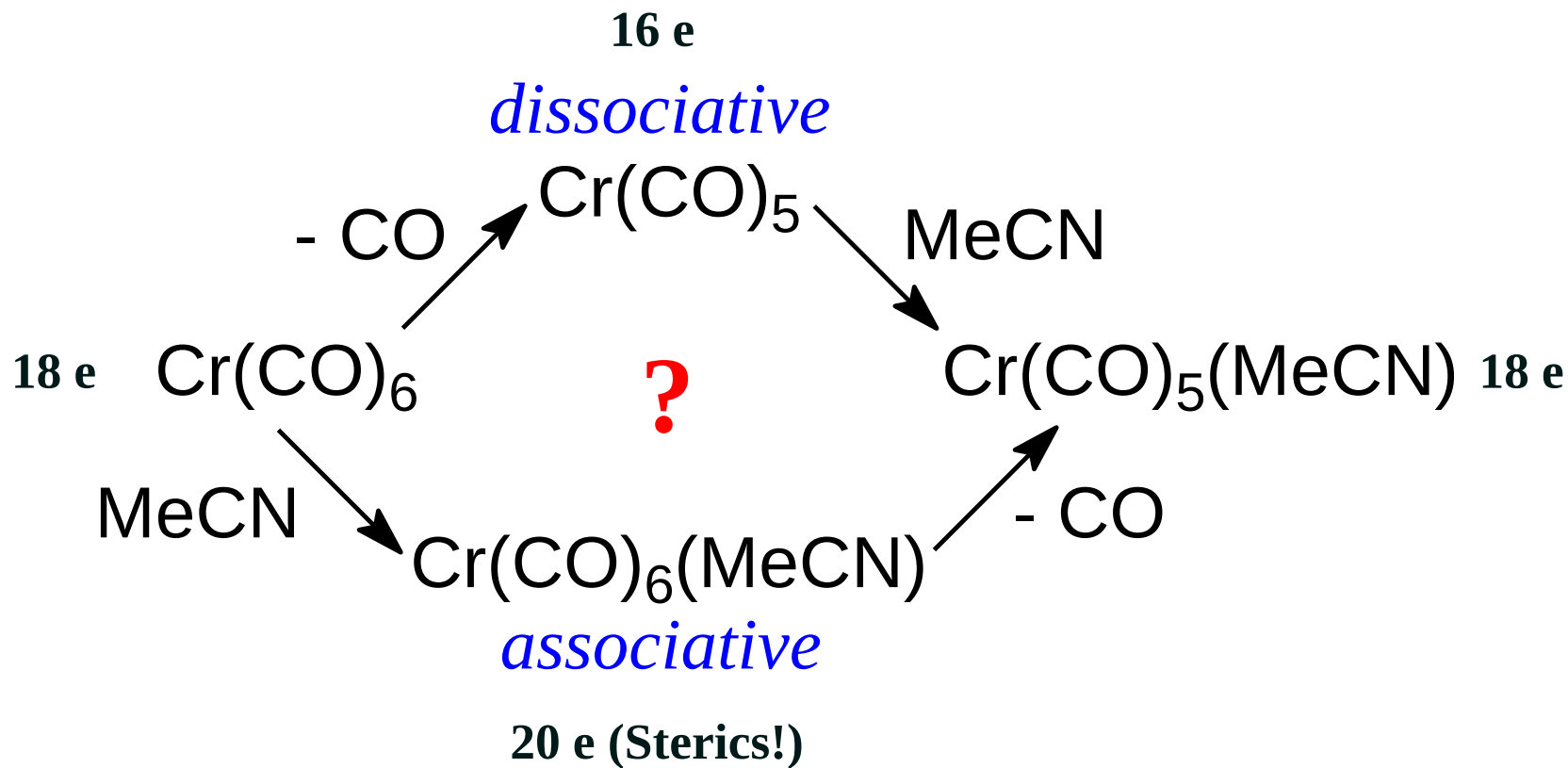
- For early transition metals, 18-*e* is often unattainable for steric reasons - the required number of ligands would not fit
- For later transition metals, 16-*e* is often quite stable (square-planar d^8 complexes)

Predicting reactivity

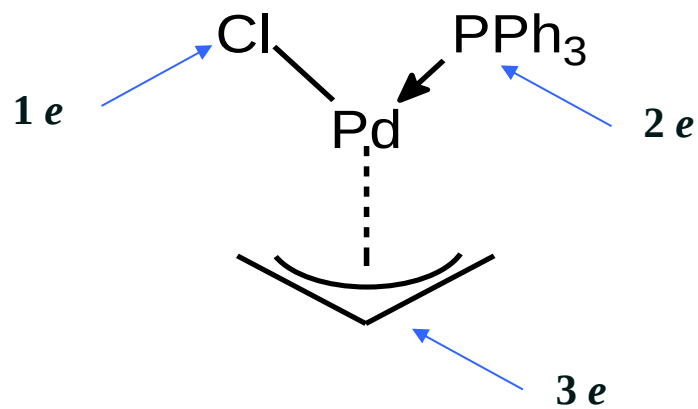
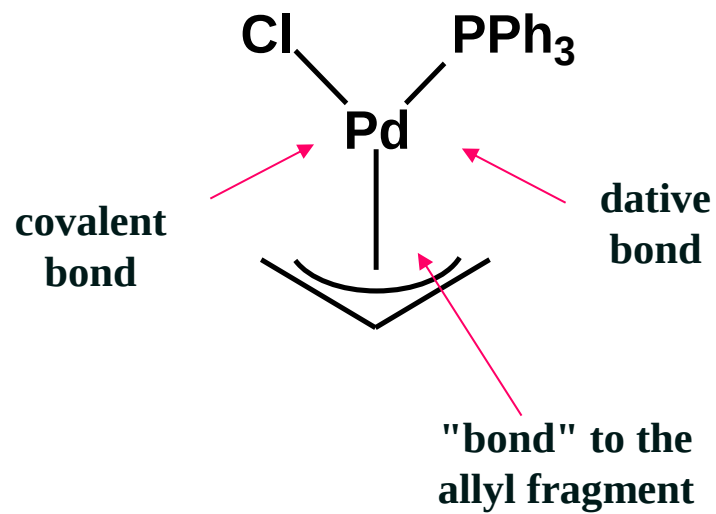


Most likely *associative*

Predicting reactivity



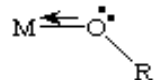
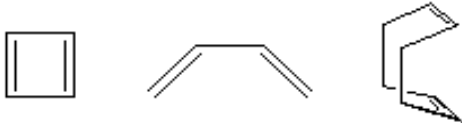
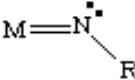
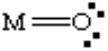
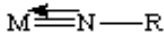
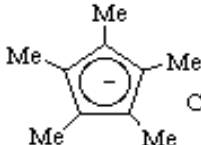
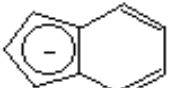
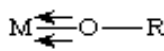
Most likely *dissociative*



Pd =	10
Cl =	1
PR₃ =	2
allyl =	3
=	=
e-count	16

"*Covalent*" count: (ionic method also useful)

1. Number of valence electrons of central atom.
 - from periodic table
2. Correct for charge, if any
3. Count 1 *e* for every *covalent* bond to another atom.
4. Count 2 *e* for every *dative* bond *from* another atom.
5. Delocalized carbon fragments: usually 1 *e* per C (hapticity)
6. Three- and four-center bonds need special treatment
7. Add everything

	Neutral/Covalent	Oxid State/Ionic
R (H, Me, Et, Pr, H) -CN, OH, Cl, OR (<i>sp</i> ³), acyl, NO (bent)	1	2
PR ₃ , amines, ethers, NH ₃ , R ₂ S, CO, RCN, RNC, olefins, acetylenes (sometimes), ketones	2	2
 (allyl)	3	4
 (<i>sp</i> ² alkoxide)		
NO (linear)	3	2 (NO ⁺)
 Acetylenes (sometimes) (1,5-COD)	4	4
 (bent imido)  (oxo)	2	4
 (linear imido)  (oxo)	4	6
 Cp  Cp* 	5	6
 (Indenyl)  (linear alkoxide)		
Arenes (benzene etc.)	6	6
C ₇ H ₇	7	6 (as a cation)
		8 (as an anion)
cyclooctatetraene (COT)	8 (max)	10 (as 2-)

Ligand

Electrons

R (H, Me, Et, Pr)

1

CN, OH, Cl, OR, NO (bent)

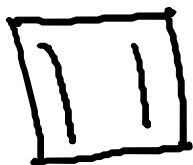
PR₃, Amines, Ethers, NH₃

2

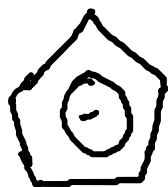
R₂S, CO, RCN, olefins,

NO (linear)

3



4



cp

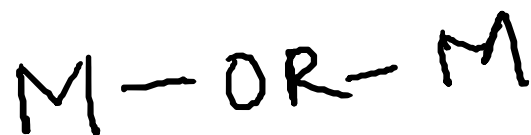
5



3

X = Halogen

M-alkoxide

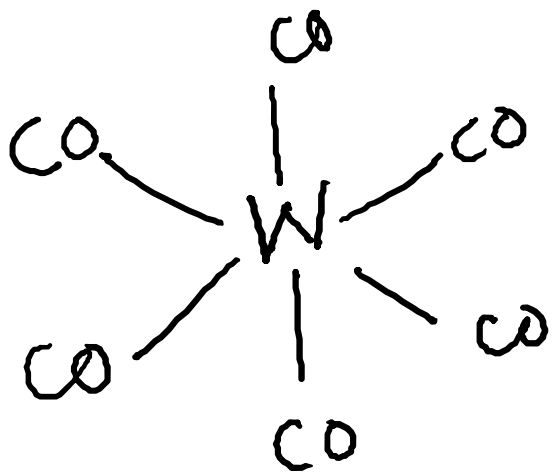


3

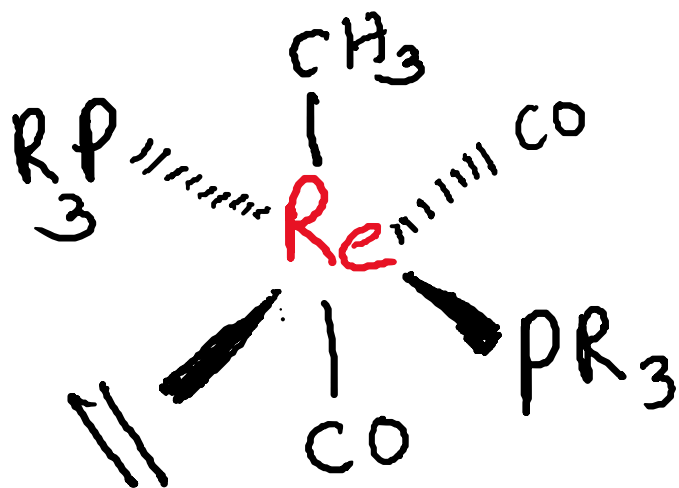
Hapto (h) Number (hapticity)

For some molecules the molecular formula provides insufficient information with which to classify the metal carbon interactions

The hapto number (h) gives the number of carbon (conjugated) atoms bound to the metal



$$\begin{aligned}
 W &= 6e \\
 6 \text{ CO} &= 12e \\
 \hline
 &18e
 \end{aligned}$$



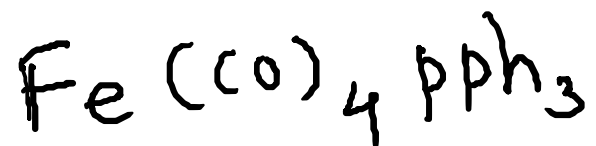
$$\text{Re}^{(+1)} = d^6$$

$$2 \text{ PR}_3 = 4e$$

$$2 \text{ CO} = 4e$$

$$\text{CH}_3^- = 2e$$

$$\begin{aligned}
 \text{CH}_2=\text{CH}_2 &= 2e \\
 &\hline
 &18e
 \end{aligned}$$

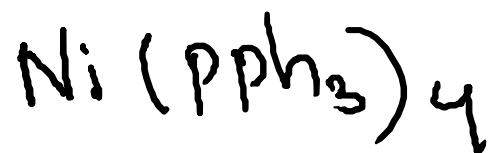


$$\text{Fe} = d^8$$

$$4\text{CO} = 8e$$

$$\text{PPh}_3 = 2e$$

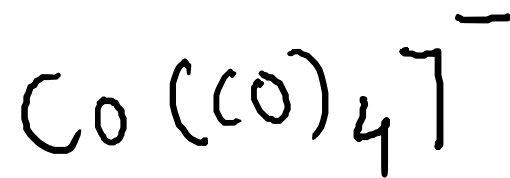
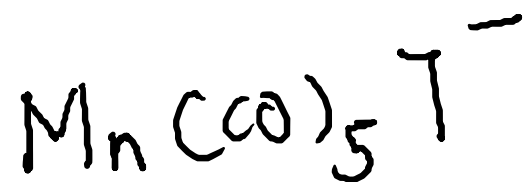
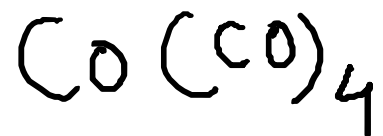
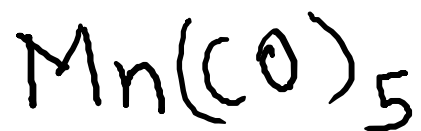
$$\hline 18e$$

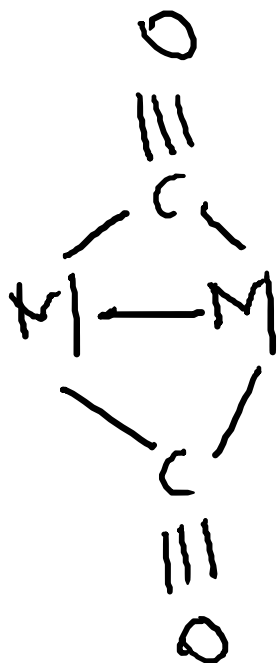
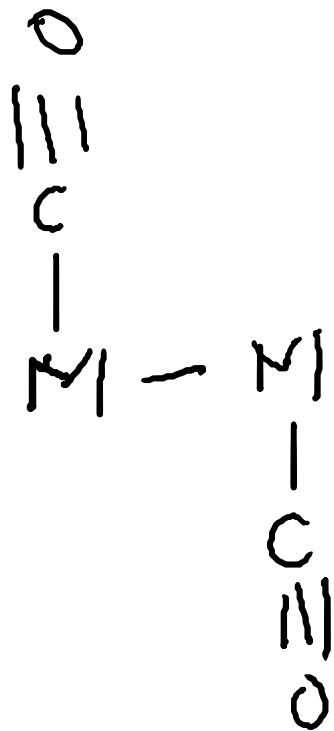
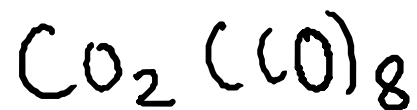


$$\text{Ni} = d^{10}$$

$$4\text{PPh}_3 = 8e$$

$$\hline 18e$$





$$2 \text{Co} = 18 e^-$$

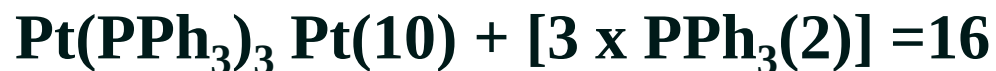
$$8 \text{CO} = 16 e^-$$

$$\text{Co} - \text{Co} = 2 e^-$$

$$36 e^-$$

$$18 e^- / \text{Co}$$

Exceptions to the 18 Electron Rule



What features do these complexes possess?

- **Early transition metals (Zr, Ta, W)**
- **Several bulky ligands (PPh₃)**
- **Square planar *d*8 e.g. Pt(II), Ir(I)**
- **σ -donor ligands (Me)**

Bonding of selected Organometallic ligands

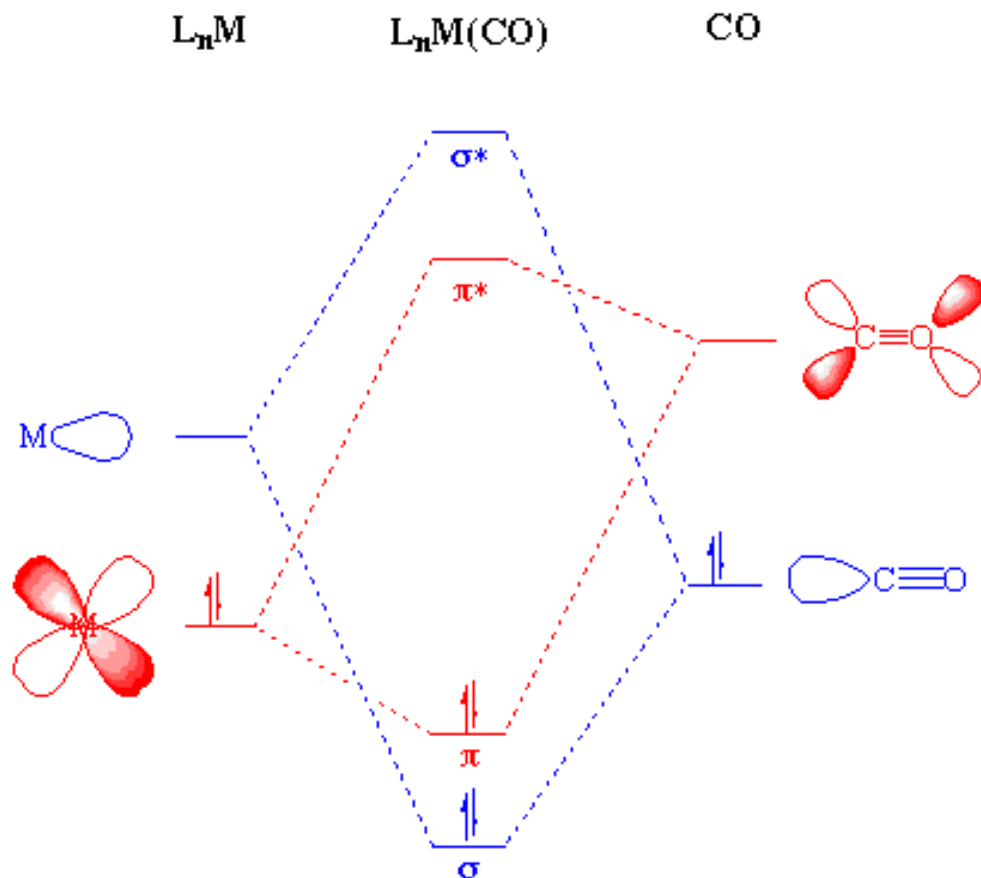
Carbonyl Ligand: CO

Carbonyl Complexes

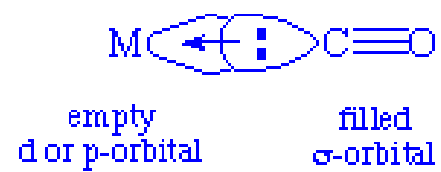
Bonding of CO

Electron donation of the lone pair on carbon. This electron donation makes the metal more electron rich - compensate for this increased electron density, a filled metal *d*-orbital may interact with the empty *p** orbital on the carbonyl ligand

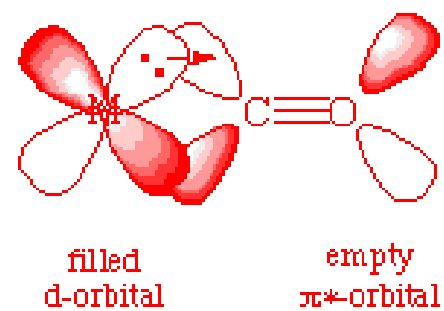
p-backbonding or p-backdonation or synergistic bonding



σ bond:



π backbond:



What stabilizes CO complexes is $M \rightarrow C$ π -bonding

The lower the formal charge on the metal ion the more willing it is to donate electrons to the π -orbitals of the CO

IR spectra and metal-carbon bonds

IR spectra and metal-carbon bonds

The ν_{CO} stretching frequency of the coordinated CO is very informative

Recall that the stronger a bond gets, the higher its stretching frequency

$\text{M}=\text{C}=\text{O}$ (C=O is a double bond) canonical structure

Lower the ν_{CO} stretching frequency as compared to the $\text{M}-\text{C}\equiv\text{O}$ structure (triple bond)

Note: ν_{CO} for free CO is 2041 cm^{-1})

IR spectra and metal-carbon bonds

ν_{CO} for free CO is 2041 cm^{-1})



ν_{CO} 1748 1858 1984 2094 2204 cm^{-1}



increasing M=C double
bonding



decreasing M=C double
bonding

Bridging versus terminal carbonyls

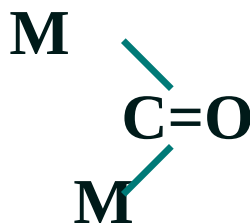
Bridging CO groups can be regarded as having a double bond C=O group, as compared to a terminal C≡O, which is more like a triple bond:

~ triple bond



terminal carbonyl
(~ 1850-2125 cm⁻¹)

~ double bond

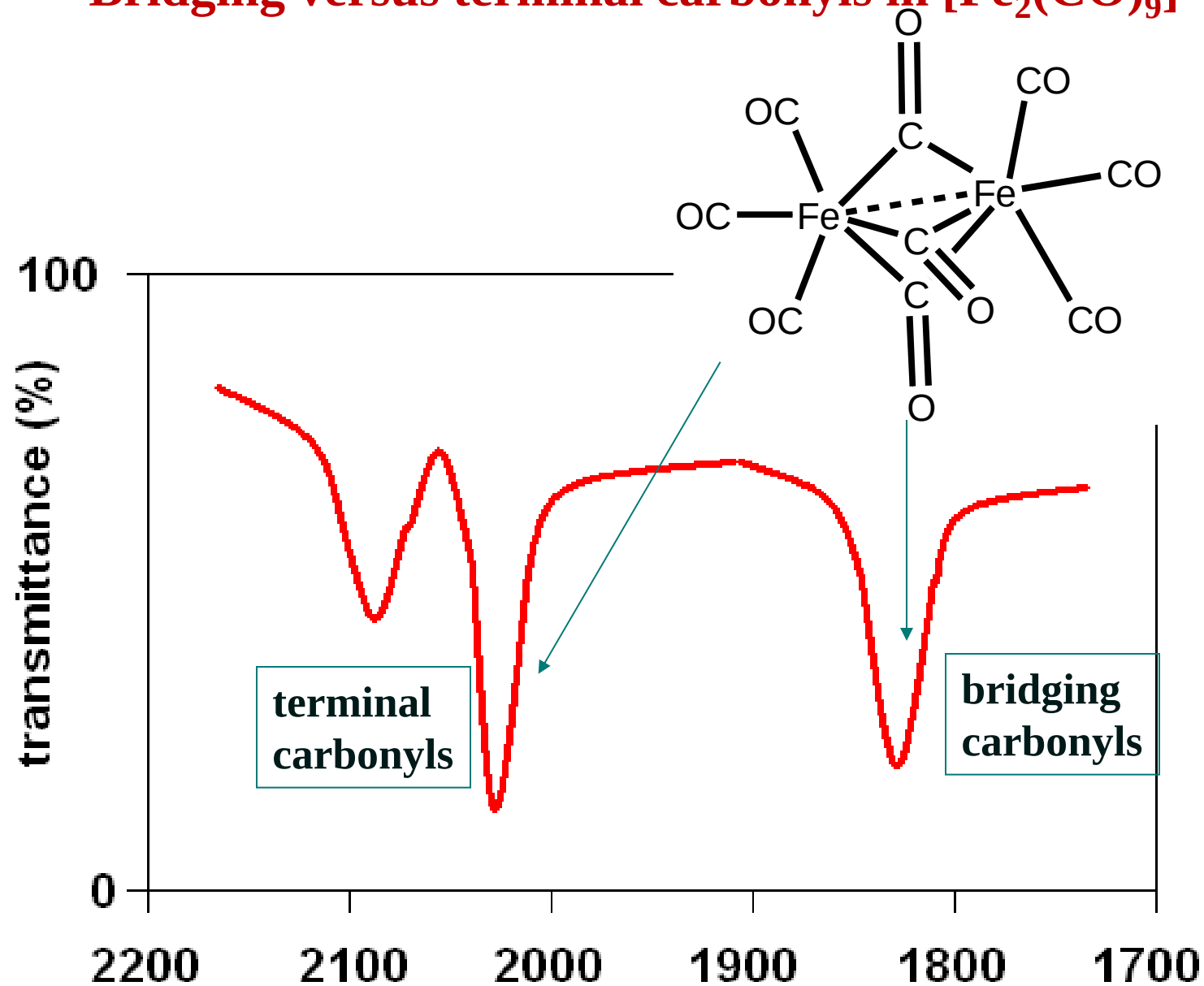


bridging carbonyl
(~1700-1860 cm⁻¹)

the C=O group
in a bridging
carbonyl is more
like the C=O in
a ketone, which
typically has
 $\nu_{\text{C=O}} = 1750 \text{ cm}^{-1}$

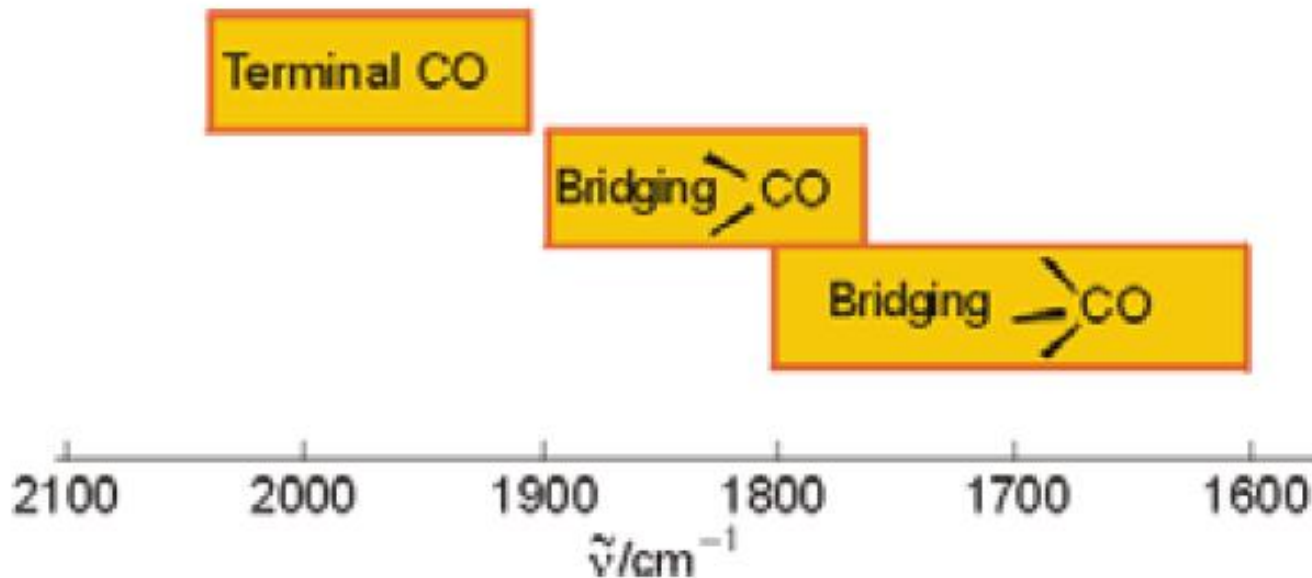
Bridging CO between 1700 and 2200 cm⁻¹

Bridging versus terminal carbonyls in $[\text{Fe}_2(\text{CO})_9]$



Summary

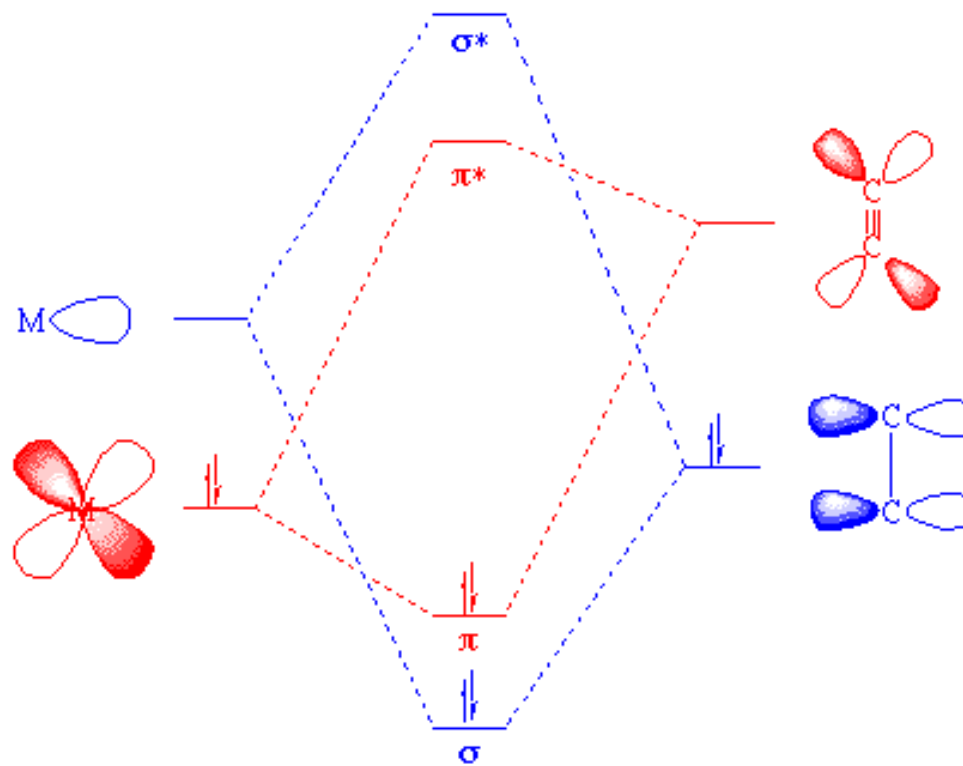
1. As the CO bridges more metal centers its stretching frequency drops – same for all pi ligands
– More back donation

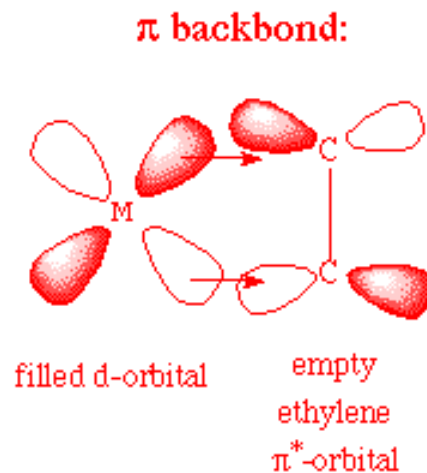
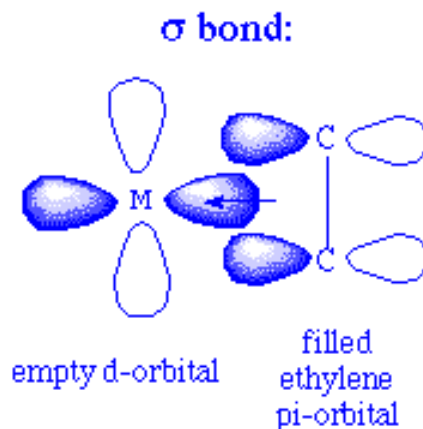


2. As the metal center becomes increasingly electron rich the stretching frequency drops

The influence of coordination and charge on CO stretching frequency	
Compound	Frequency (cm ⁻¹)
CO _(g)	2143
[Mn(CO) ₆] ⁺	2090
[Cr(CO) ₆]	2000
[V(CO) ₆] ⁻	1860
[Ti(CO) ₆] ²⁻	1750

Alkene ligands

L_nM $L_nM(C_2H_4)$ C_2H_4 

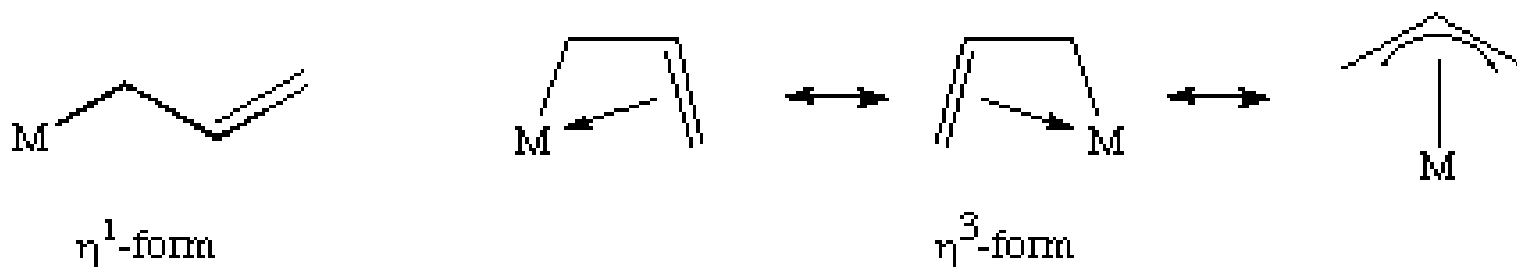


The greater the electron density back-donated into the p^* orbital on the alkene, the greater the reduction in the C=C bond order

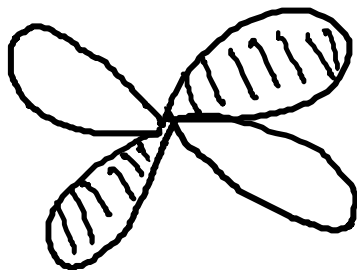
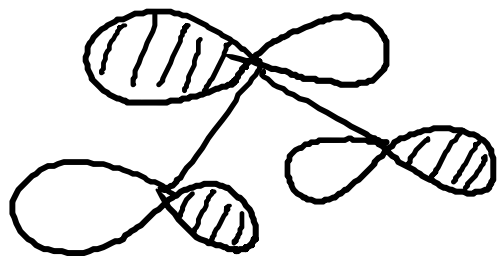
Stability of alkene complexes also depends on steric factors as well

Allyl ligands:

Allyl ligands are ambidentate ligands that can bind in both a monohapto and trihapto form. The trihapto form can be expressed as a number of different resonance forms as shown here for an unsubstituted allyl ligand: Important applications

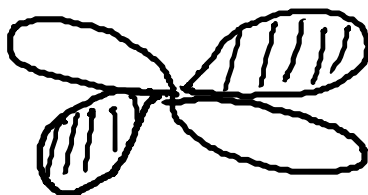
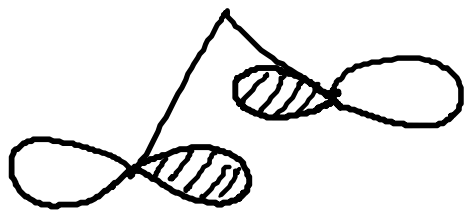


A.



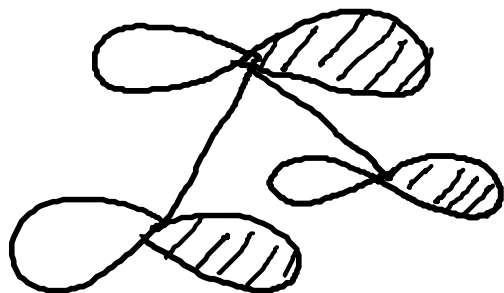
dxz

N.



dyz

B.



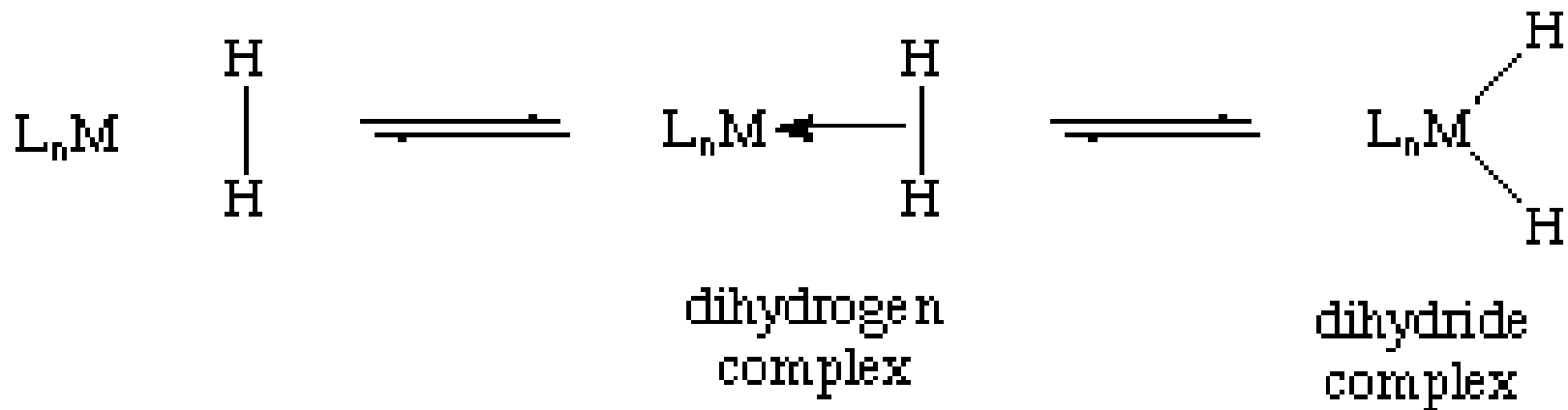
dz^2

Dihydrogen Ligand: H₂

Dihydrogen Ligands:

Metal is more electropositive than hydrogen

Hydrogen acts as a two electron sigma donor to the metal center.

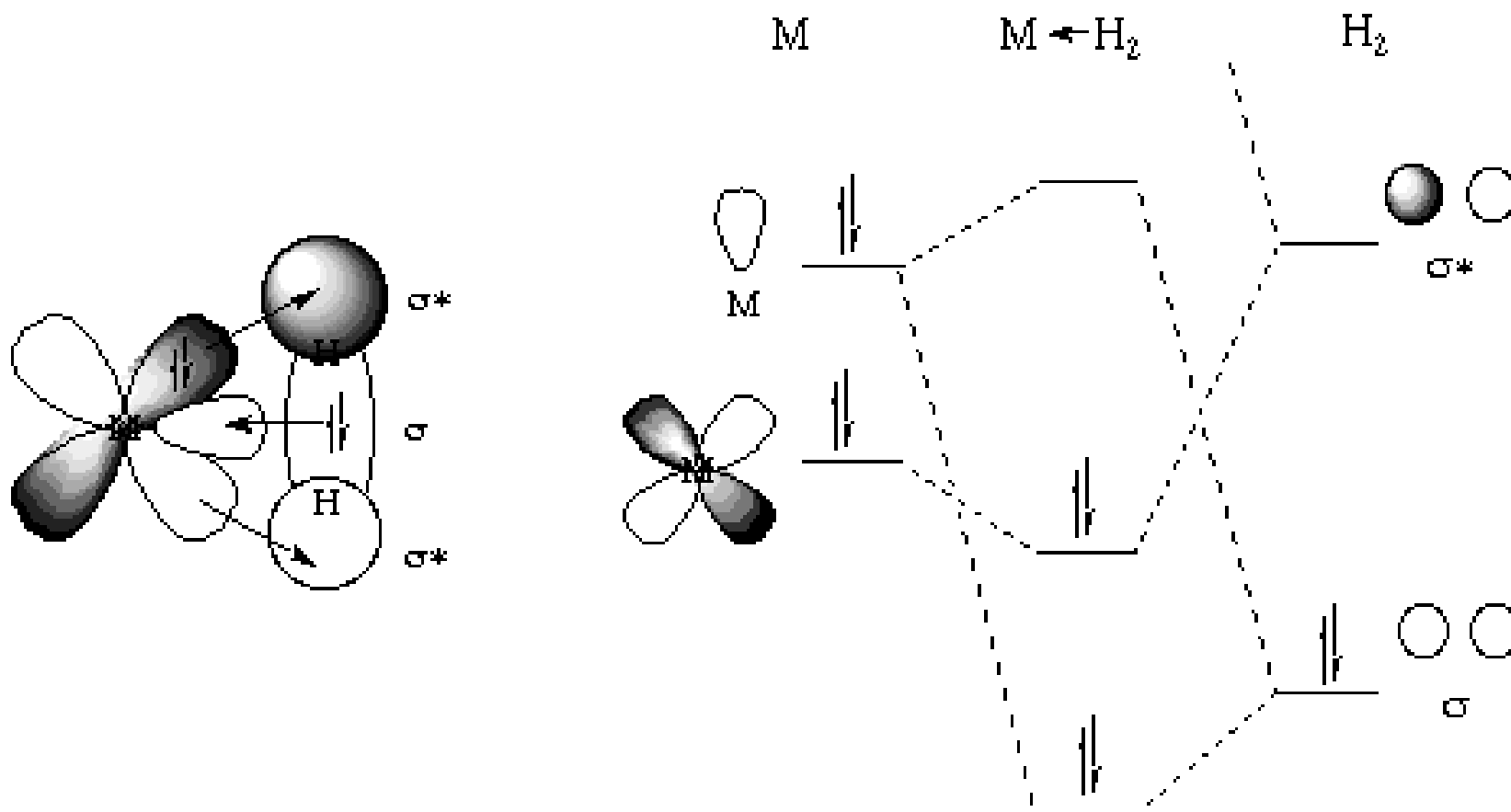


How does this affect the oxidation state of the metal?

Dihydrogen complexes Bonding

H_2 - neutral two electron sigma donor

One could also describe a back-donation of electrons from a filled metal orbital to the sigma-* orbital on the dihydrogen



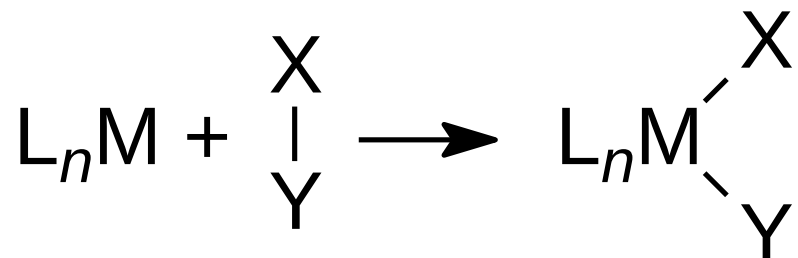
Reaction chemistry of Organometallic complexes

Reaction chemistry of complexes

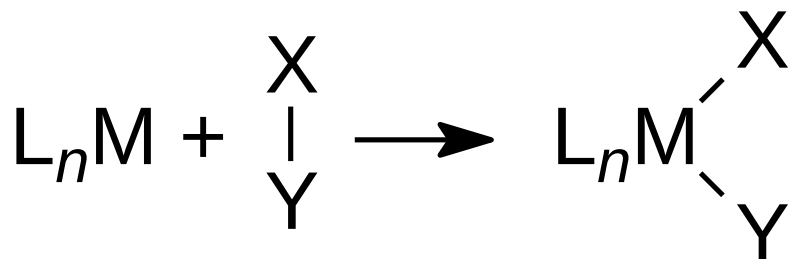
Three general forms:

- 1. Reactions involving the gain and loss of ligands**
 - a. Oxidative Addition**
 - b. Reductive Elimination**
- 2. Reactions involving modifications of the ligand**
 - a. Carbonyl insertion (alkyl migration)**
- 3. Catalytic processes by the complexes**
 - Wilkinson**
 - Monsanto**
 - Ziegler Natta catalyst**

a) Oxidative Addition



a) Oxidative Addition



- **Oxidation state changes by + 2**
- **The new M-X and M-Y bonds are formed using:**
 - **the electron pair of the X-Y bond**
 - **one metal-centered lone pair**

b) Reductive elimination

This is the reverse of oxidative addition - Expect *cis* elimination

Rate depends strongly on types of groups to be eliminated.

Usually easy for:

- H + alkyl / aryl / acyl**

Modifications of the ligand

a) Insertion reactions *Migratory* insertion!

The ligands involved must be *cis*

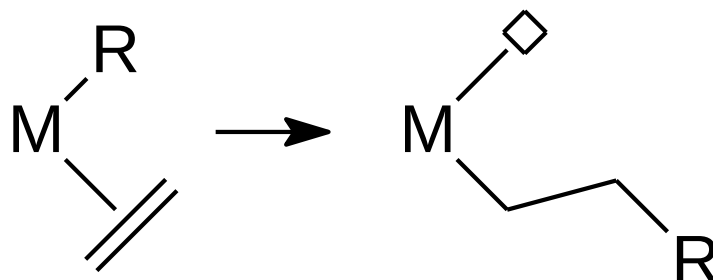
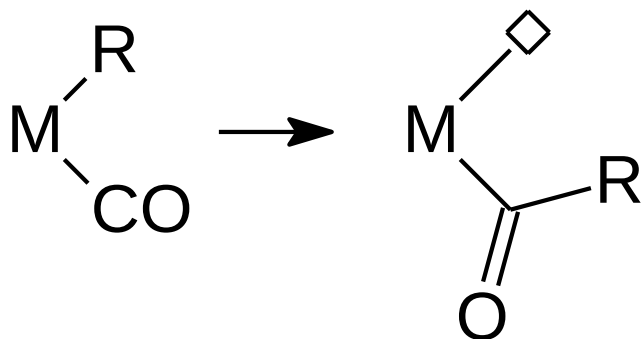
If at a metal center you have a sigma-bound group (hydride, alkyl, aryl) and a ligand containing a pi-system (olefin, alkyne, CO) the sigma-bound group can *migrate* to the pi-system

Modifications of the ligand

a) Insertion reactions *Migratory insertion!*

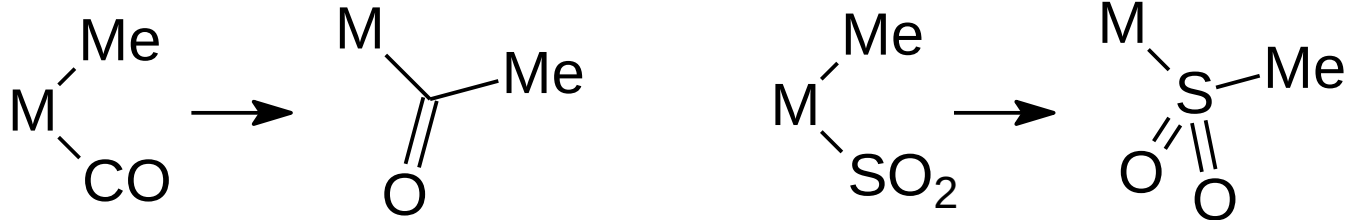
1. CO : 1,1-insertion

2. Olefins: 1,2-insertion,



In a 1,1-insertion, metal and X group "move" to the same atom of the inserting substrate.

The metal-bound substrate atom increases its valence



CO, isonitriles (RNC) and SO₂ often undergo 1,1-insertion

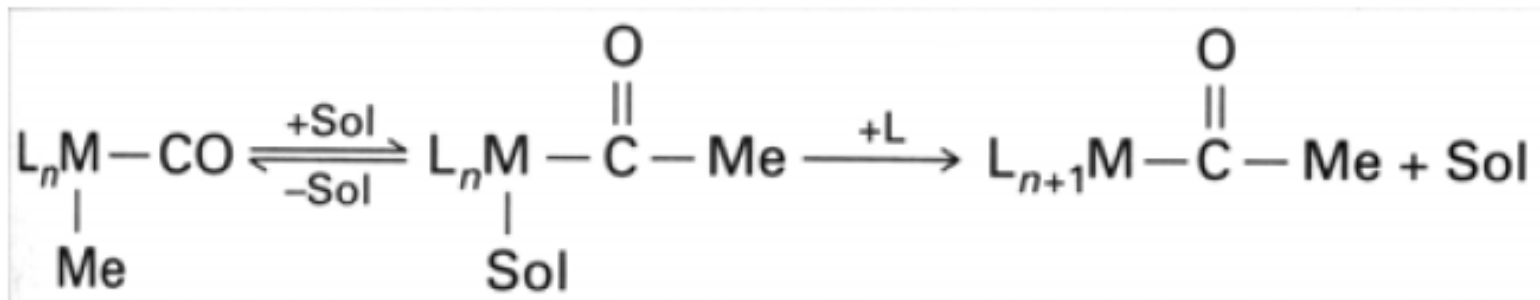
1,2 insertion (olefins)

Insertion of an olefin in a metal-alkyl bond produces a new alkyl

Thus, the reaction leads to *oligomers* or *polymers* of the olefin

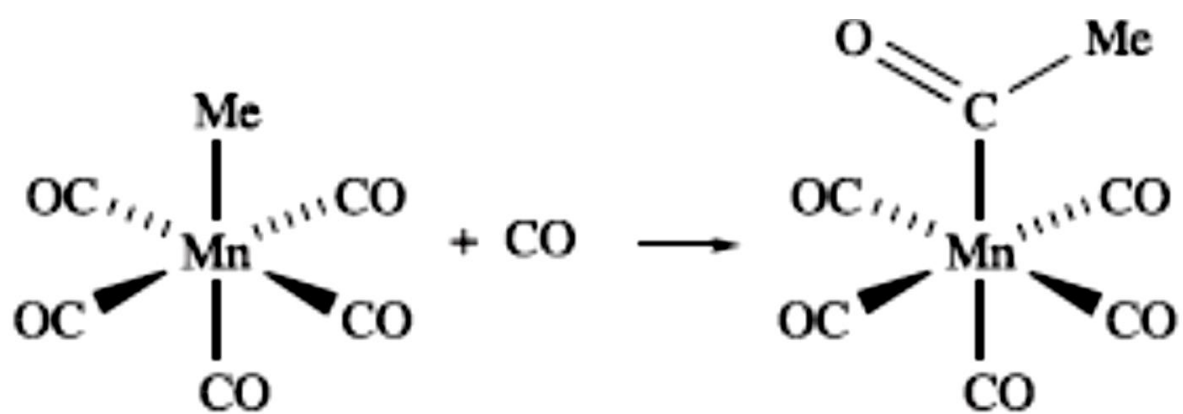
- polyethene (polythene)

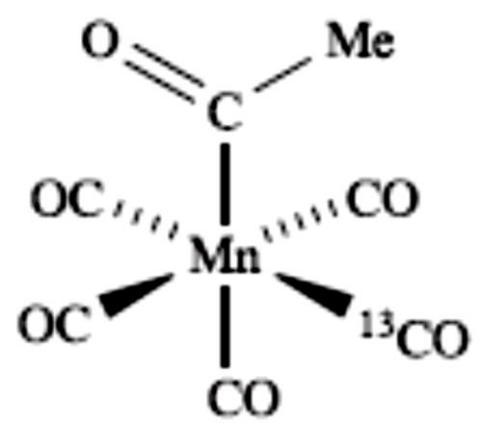
1,1 Insertion

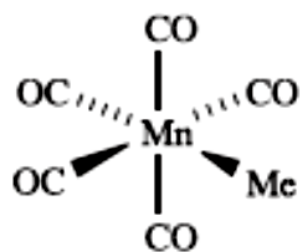
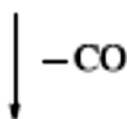
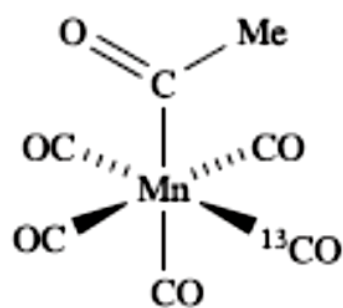


if you only see the result, it *looks like* CO has *inserted* into the M-X bond, hence the name *insertion*

To emphasize that it is actually (mostly) the X group that moves, we use the term *migratory insertion*.

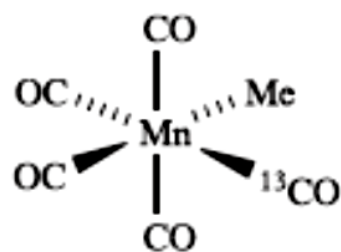






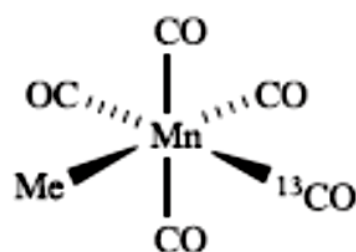
25%

+



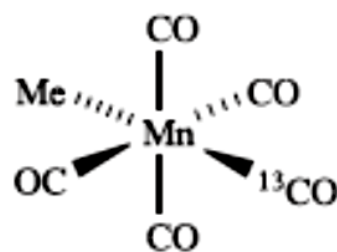
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+



25%

Equivalent products

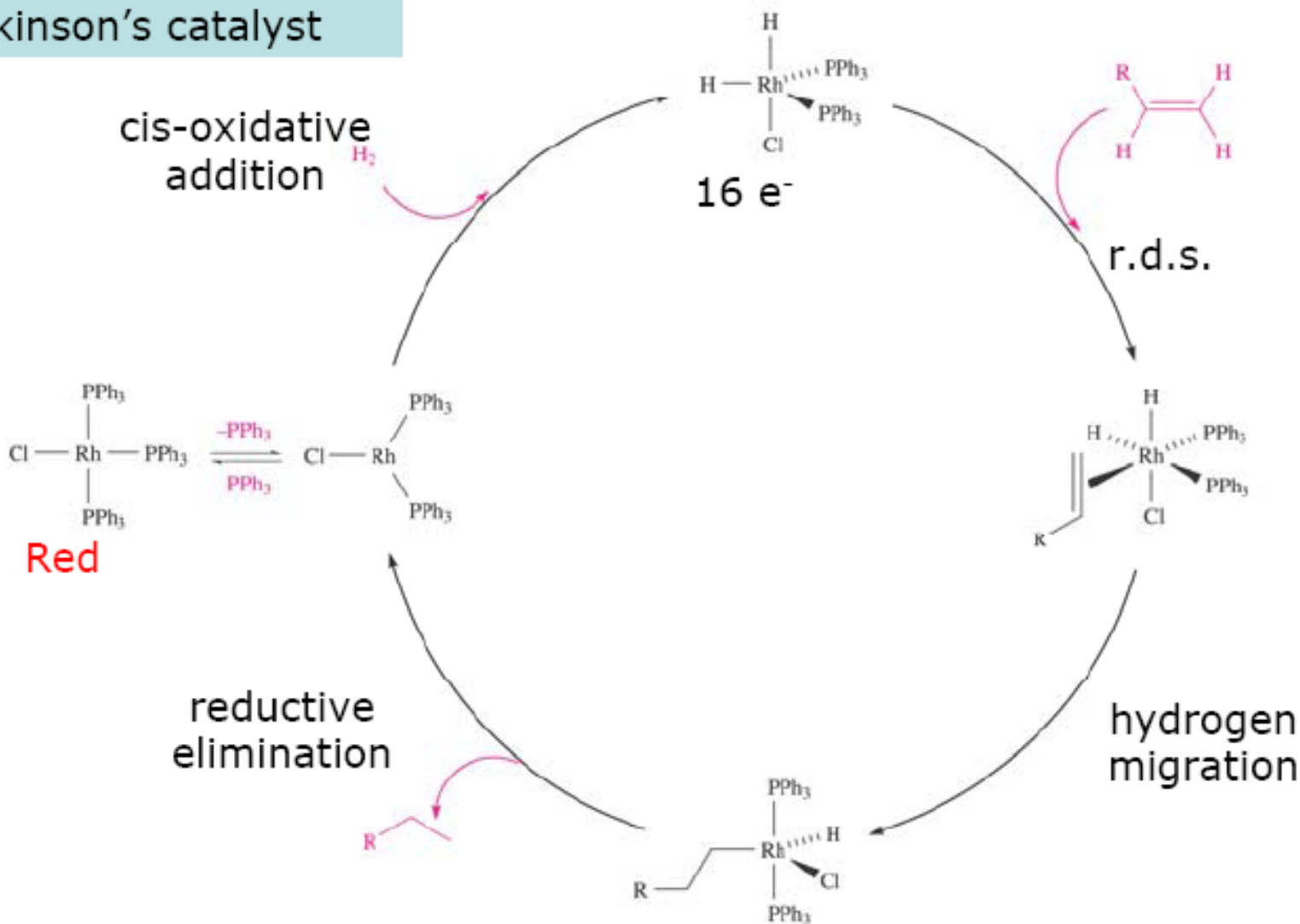
Catalysis (homogeneous)

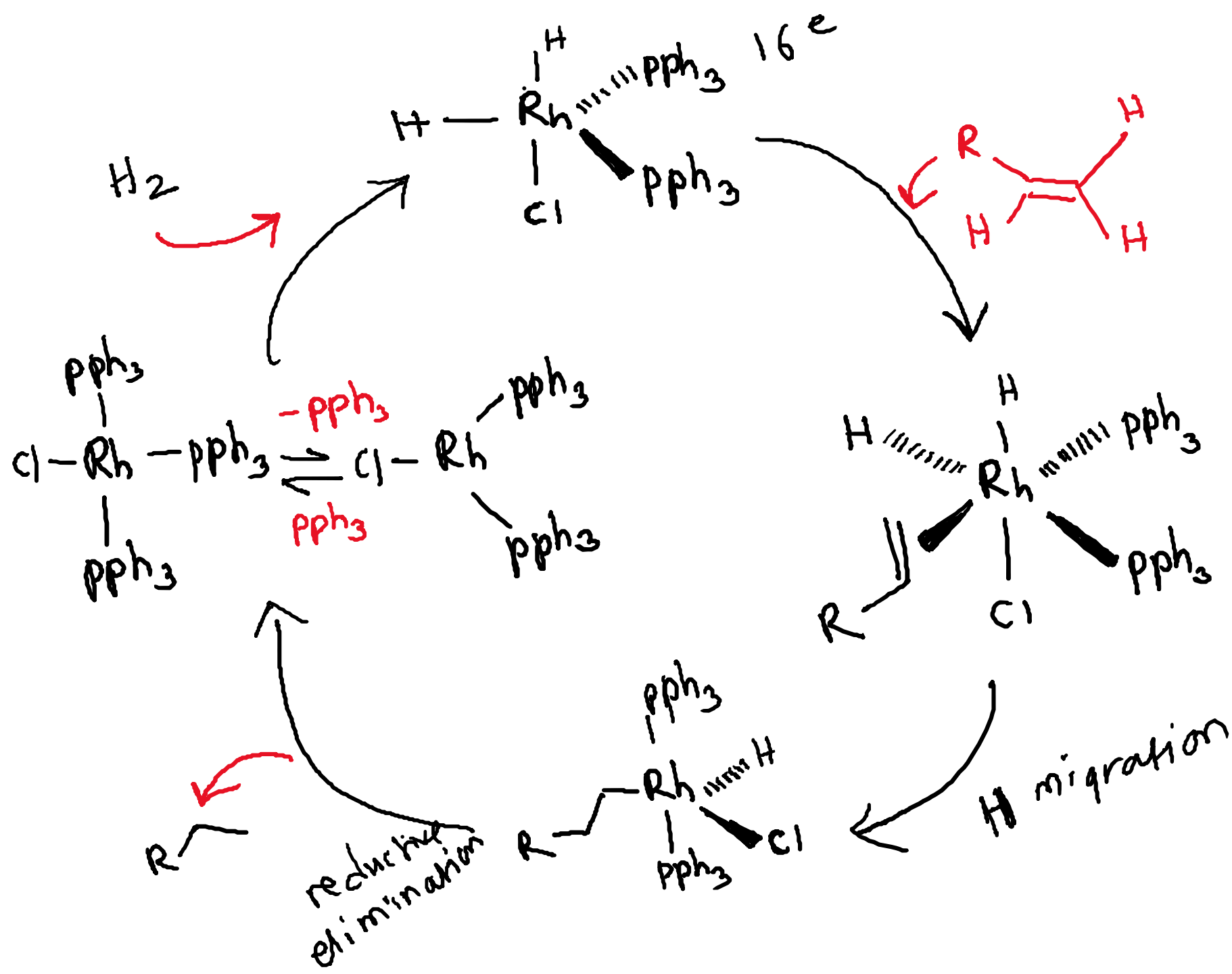
Alkene Hydrogenation

Catalysis (homogeneous)

Alkene Hydrogenation

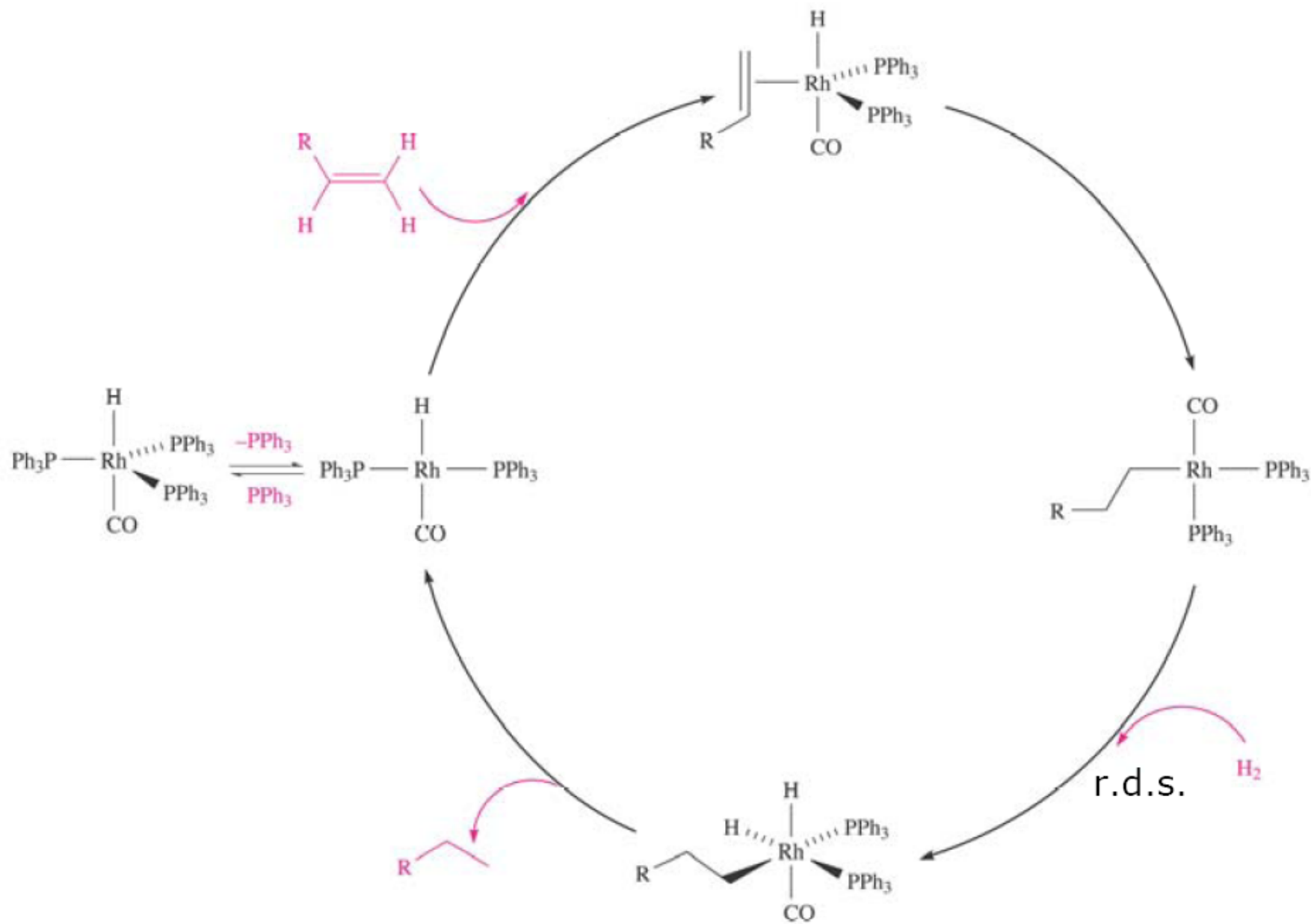
Wilkinson's catalyst





The size of the substrate has an effect on the rate of reaction

Same reaction different catalyst



g. 27.8 Catalytic cycle for the hydrogenation of $RCH=CH_2$ using $HRh(CO)(PPh_3)_3$ as catalyst.

The Monsanto acetic acid process

Monsanto synthesis

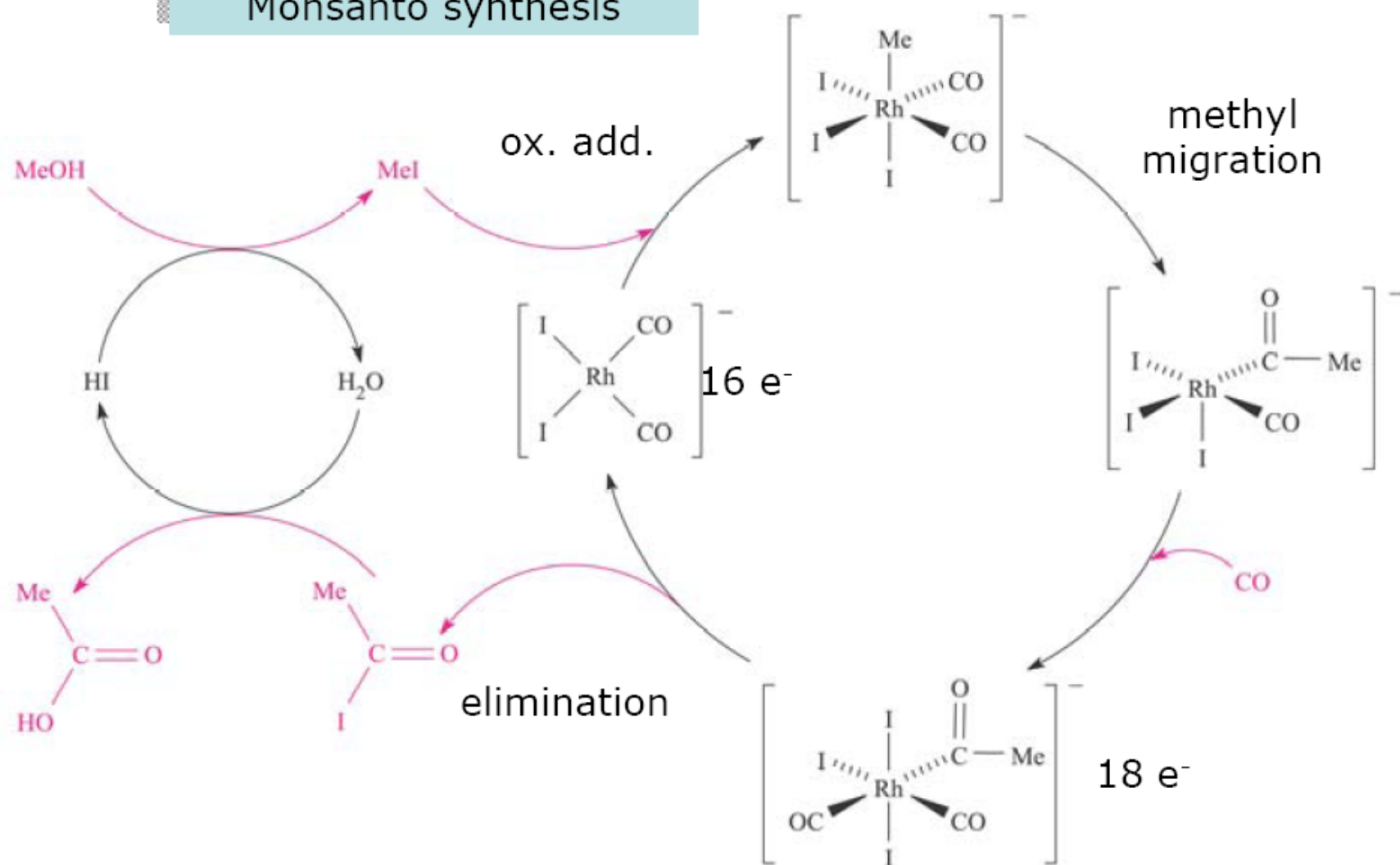
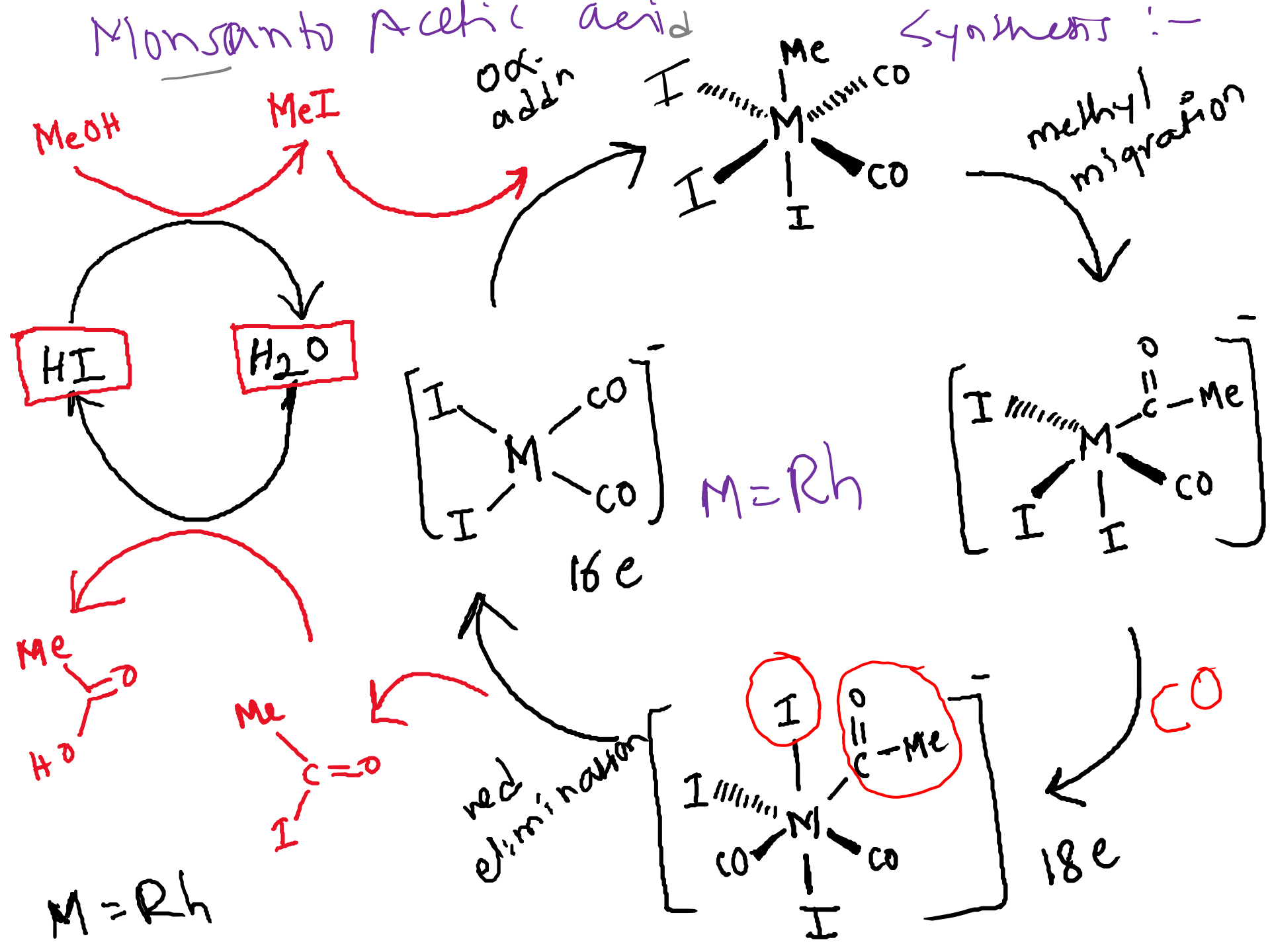


Fig. 27.9 The Monsanto acetic acid process involves two interrelated catalytic cycles.

Monsanto Acetic acid

Synthesis :-



The Monsanto acetic acid process

Methanol - reacted with carbon monoxide in the presence of a catalyst to afford acetic acid

Insertion of carbon monoxide into the C-O bond of methanol

The catalyst system - iodide and rhodium

Iodide promotes the conversion of methanol to methyl iodide,

Methyl iodide - the catalytic cycle begins:

- 1. Oxidative addition of methyl iodide to $[\text{Rh}(\text{CO})_2\text{I}_2]^-$**
- 2. Coordination and insertion of CO - intermediate 18-electron acyl complex**
- 3. Can then undergo reductive elimination to yield acetyl iodide and regenerate our catalyst**

Ziegler-Natta catalysis:

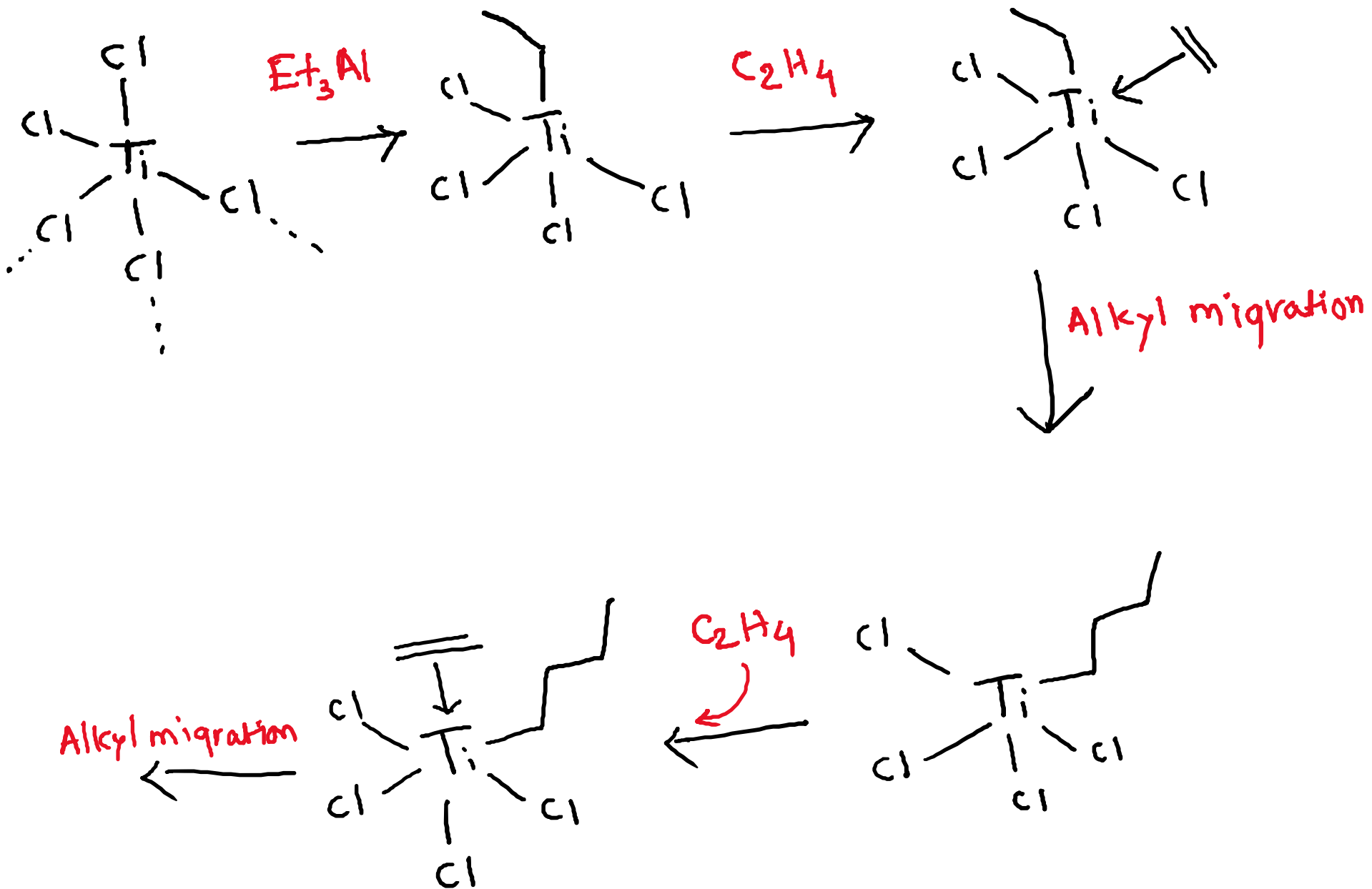
Alkene polymerization

Ziegler-Natta catalysis:

Alkene polymerization

The 1963 the Nobel Prize in Chemistry was awarded to Karl Ziegler and Giulio Natta 'for their discoveries in the field of the chemistry and technology of high polymers

First generation Ziegler–Natta catalysts were made by reacting TiCl_4 with Et_3Al



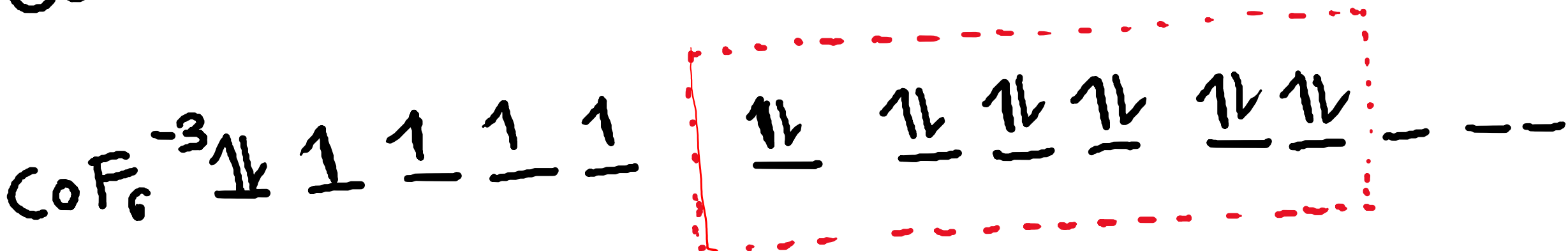
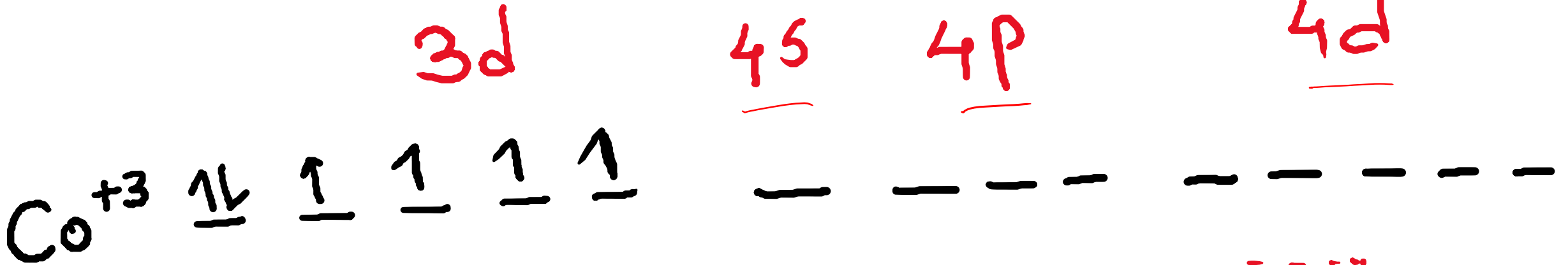
Bonding in Coordination Compounds

Valence Bond Theory

This theory was developed by Pauling. The model utilizes hybridization of metal valence orbitals to account for the observed structures and magnetic properties of complexes.

Pauling suggested that $(n-1)d$, ns and np undergo hybridization to give hybridized orbitals.

An empty hybrid orbital on the metal center can accept a pair of electrons from a ligand to form a σ -bond.





5d

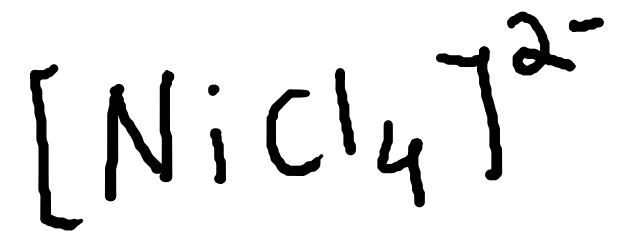


6s



6p

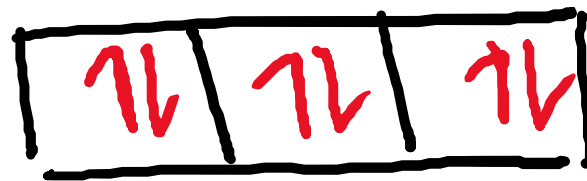
dsp^2



3d



4s



4p

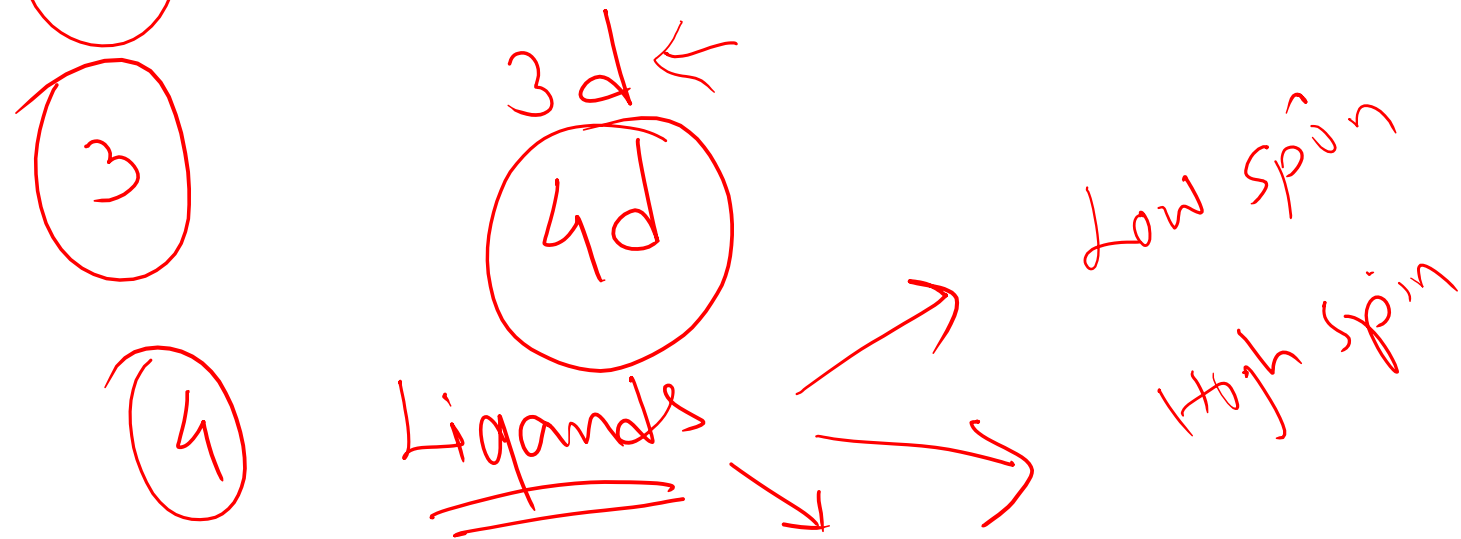


C.N.	Geometry	Description	Examples
2	Linear	sp	$[\text{Ag}(\text{NH}_3)_2]^+$
3	Trigonal planar	sp^2	$[\text{HgI}_3]^-$
4	Tetrahedral	sp^3	$[\text{FeBr}_4]^{2-}$
5	Square planar	sp^2d	$[\text{Ni}(\text{CN})_4]^{2-}$
4	Trigonal bipyramidal	sp^3d	$[\text{CuCl}_5]^{3-}$
5	Square-based pyramidal	sp^3d	$[\text{Ni}(\text{CN})_5]^{3-}$
6	Octahedral	sp^3d^2	$[\text{Co}(\text{NH}_3)_6]^{3+}$

Limitations

-1

- ① Electronic Spectroscopy
- ② "High spin - Low Spin" complexes



Crystal Field Theory

In view of the weaknesses of VBT, an alternative bonding model was applied to transition metal complexes. This is known as crystal field theory (CFT).

It has been originally proposed by Hans Bethe and Van Vleck in 1929.

The assumptions of the crystal field theory are as follows:

The interaction between the metal ion and the ligand is assumed to be purely electrostatic (ionic) in nature.

The ligands are treated as point charges.

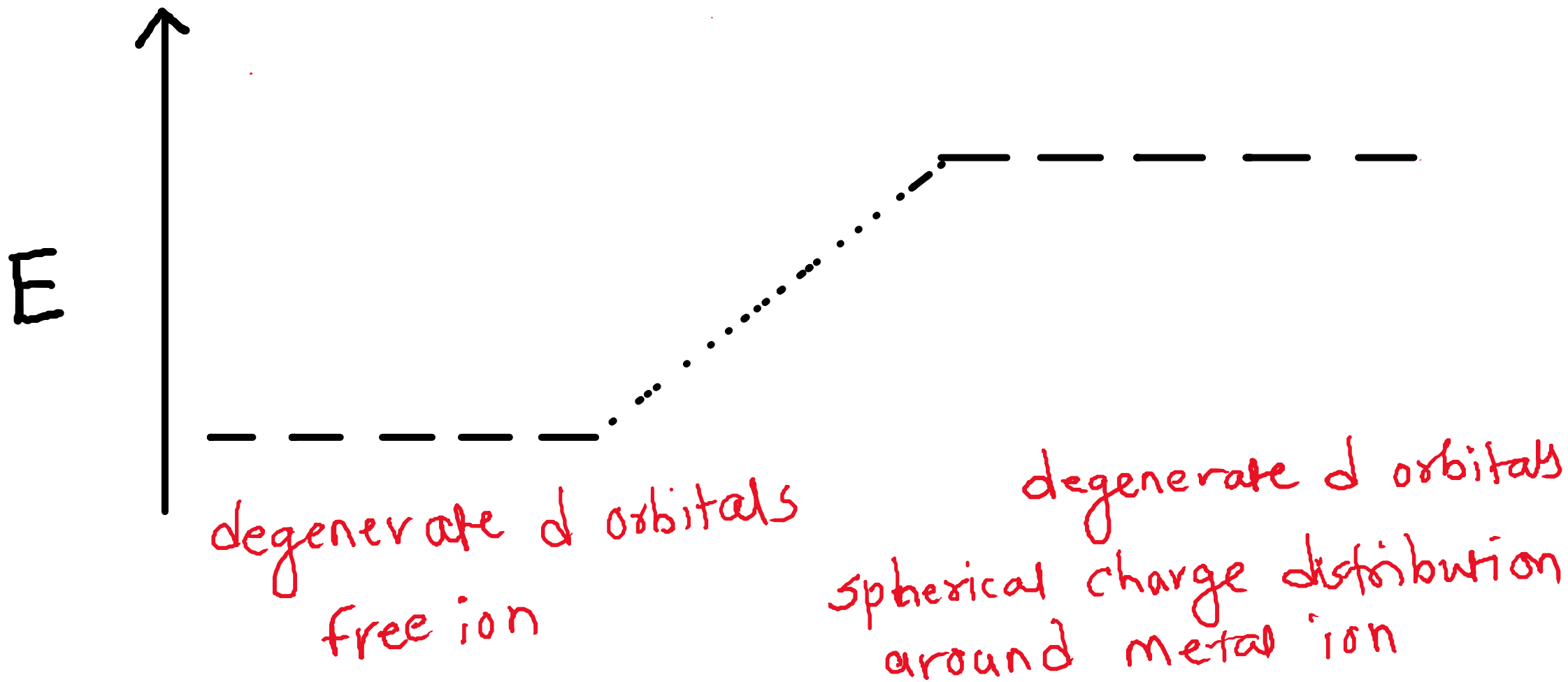
Consequent to this assumption, the CFT does not allow for any overlap between the metal orbitals and the ligand orbitals.

Consequences of the metal ion interaction with ligands

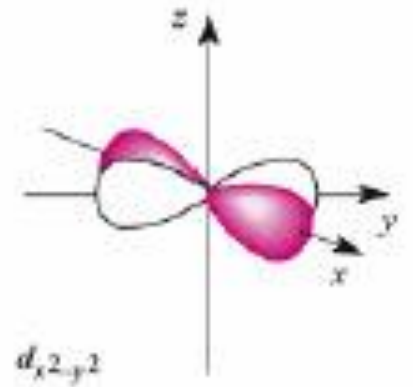
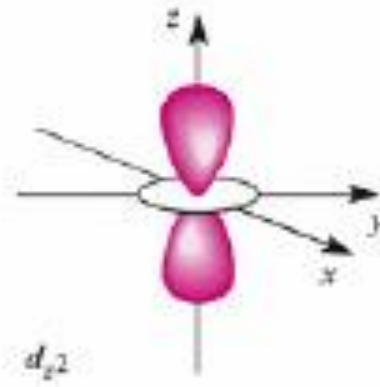
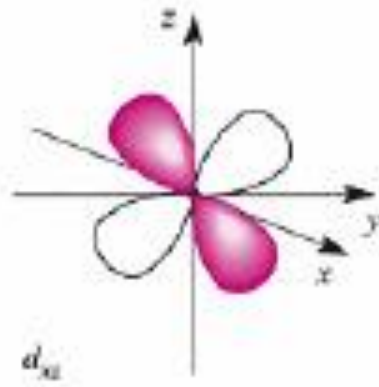
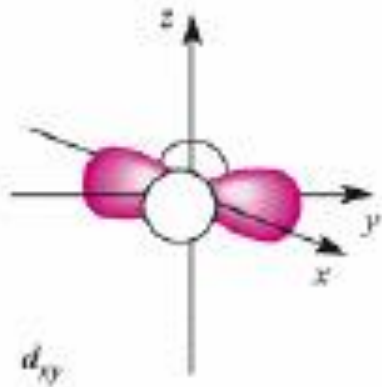
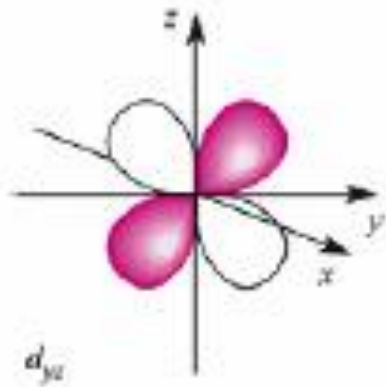
The five d orbitals in an isolated, gaseous metal ion are degenerate (are of the same energy and symmetry).

If a spherically symmetric field of negative charges is placed around the metal ion the orbitals will remain degenerate, but all of them will be raised in energy as a result of repulsion between the negative field and the metal electrons in the orbitals.

If the field results produced by the ligands is not spherically symmetric but of a particular symmetry (octahedral, tetrahedral etc), the d-orbitals are not affected equally and the degeneracy of the d orbitals is removed.



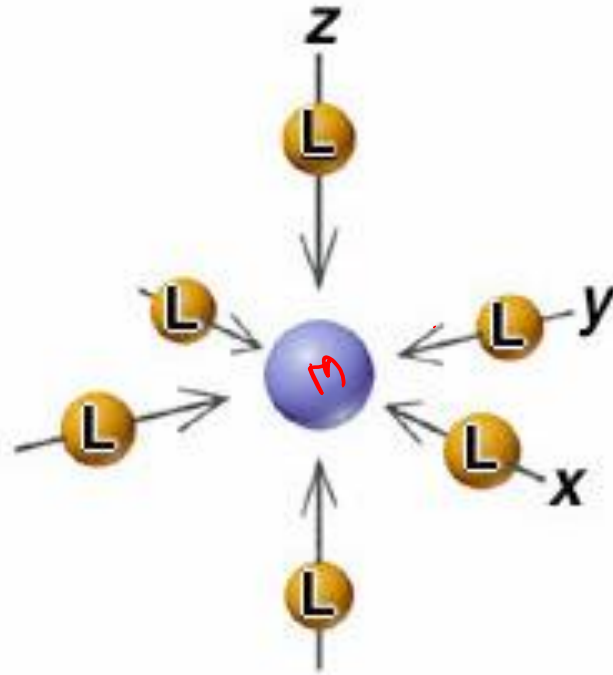
Shape of d orbitals



Octahedral Field

An octahedral field is described by six ligands at the six corners of an octahedron and the metal is at the center.

Hence, the orbitals lying along the axes are strongly repelled than those whose orbitals are lying in between the axes (d_{xz} , d_{yz} , d_{xy}).

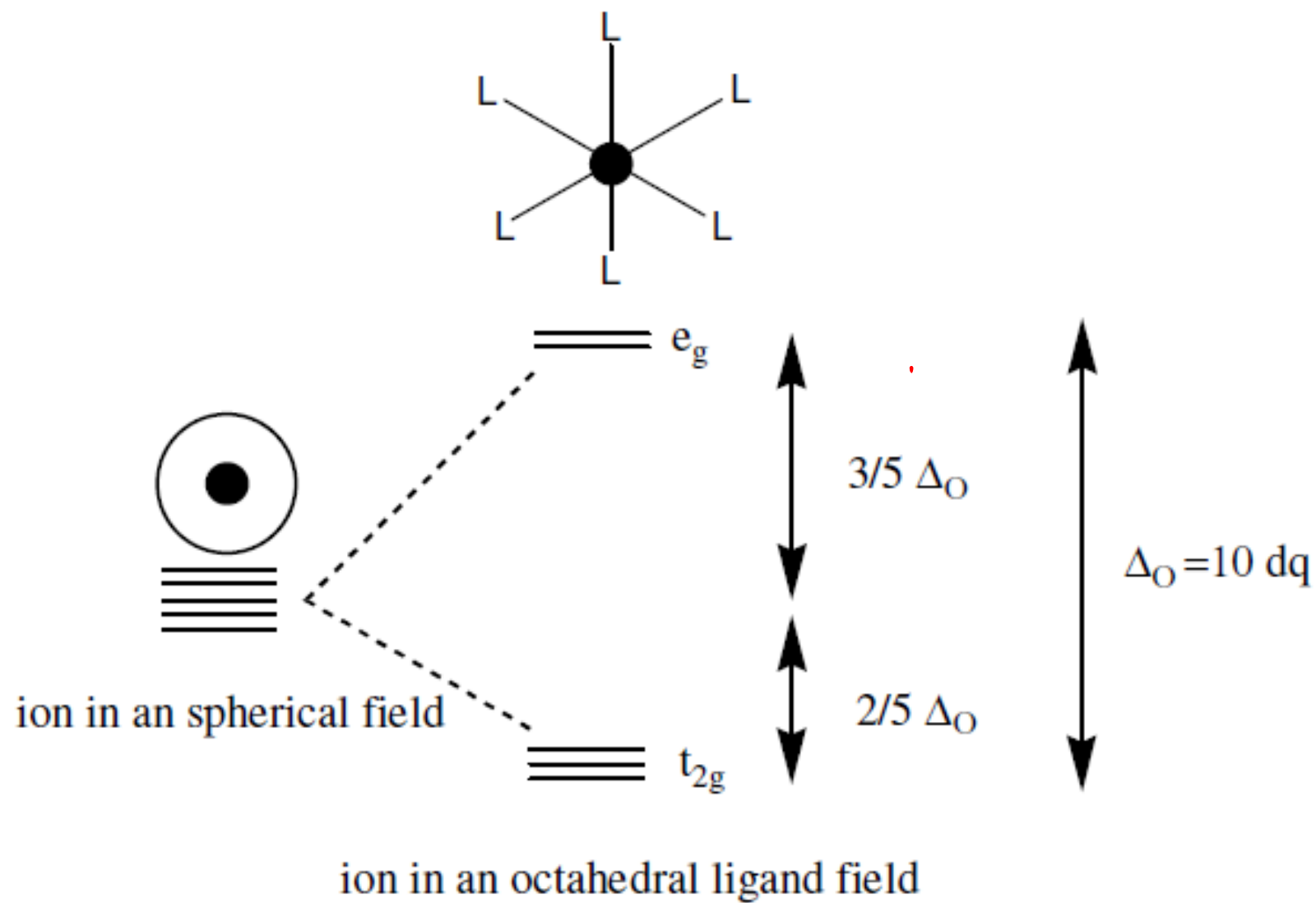


The higher energy set of orbitals (d_{z^2} and $d_{x^2-y^2}$) are labeled as e_g

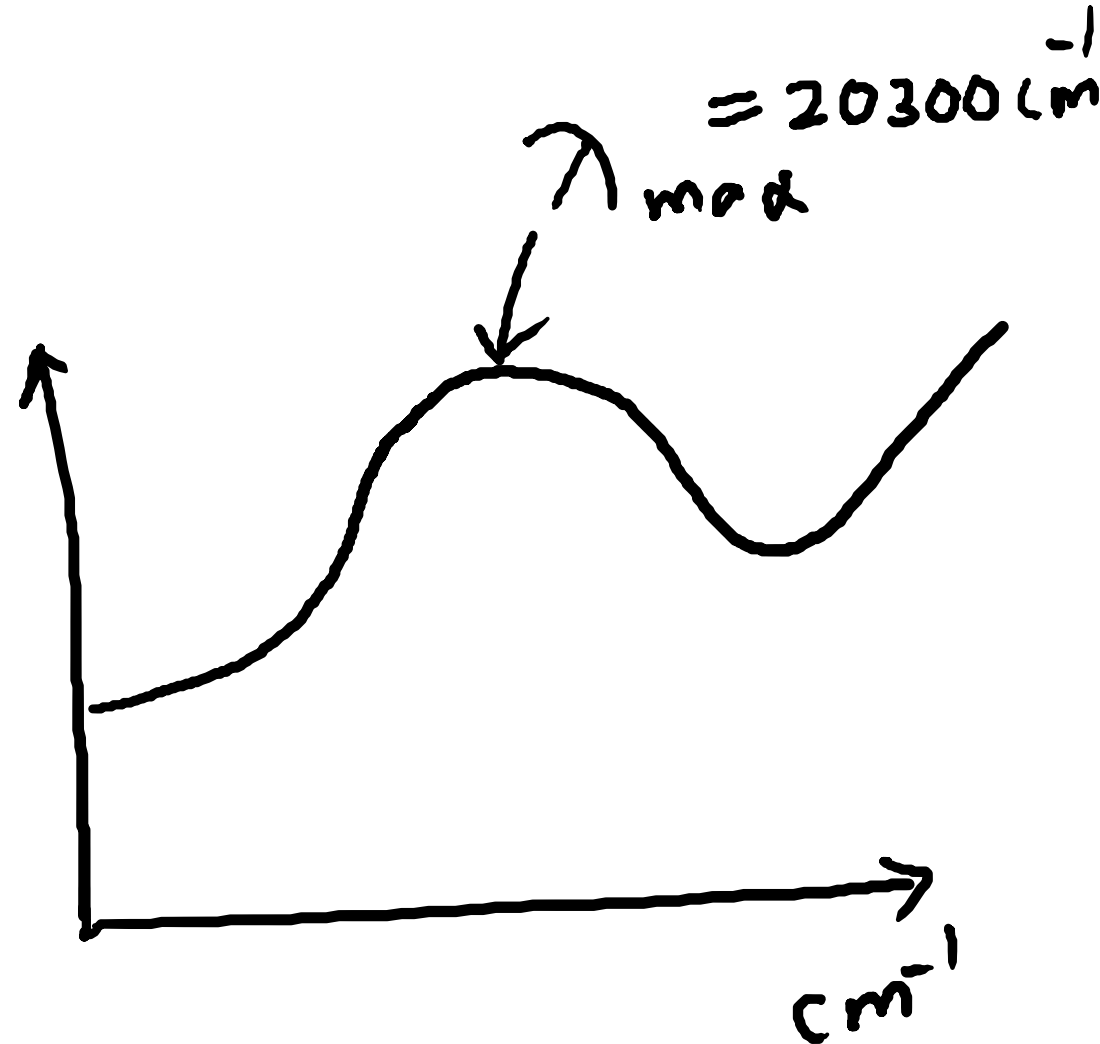
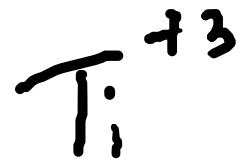
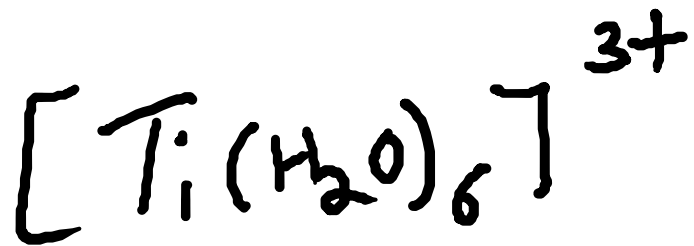
and the lower energy set is labeled as t_{2g}).

The energy separation between the two levels is denoted by Δ_o or $10 Dq$.

Splitting of d-orbitals in an Octahedral field



Splitting Energy Δ_o



Doct → Experimental quantity

Complex	Δ / cm^{-1}	Complex	Δ / cm^{-1}
$[\text{TiF}_6]^{3-}$	17 000	$[\text{Fe}(\text{ox})_3]^{3-}$	14 100
$[\text{Ti}(\text{OH}_2)_6]^{3+}$	20 300	$[\text{Fe}(\text{CN})_6]^{3-}$	35 000
$[\text{V}(\text{OH}_2)_6]^{3+}$	17 850	$[\text{Fe}(\text{CN})_6]^{4-}$	33 800
$[\text{V}(\text{OH}_2)_6]^{2+}$	12 400	$[\text{CoF}_6]^{3-}$	13 100
$[\text{CrF}_6]^{3-}$	15 000	$[\text{Co}(\text{NH}_3)_6]^{3+}$	22 900
$[\text{Cr}(\text{OH}_2)_6]^{3+}$	17 400	$[\text{Co}(\text{NH}_3)_6]^{2+}$	10 200
$[\text{Cr}(\text{OH}_2)_6]^{2+}$	14 100	$[\text{Co}(\text{en})_3]^{3+}$	24 000
$[\text{Cr}(\text{NH}_3)_6]^{3+}$	21 600	$[\text{Co}(\text{OH}_2)_6]^{3+}$	18 200
$[\text{Cr}(\text{CN})_6]^{3-}$	26 600	$[\text{Co}(\text{OH}_2)_6]^{2+}$	9 300
$[\text{MnF}_6]^{2-}$	21 800	$[\text{Ni}(\text{OH}_2)_6]^{2+}$	8 500
$[\text{Fe}(\text{OH}_2)_6]^{3+}$	13 700	$[\text{Ni}(\text{NH}_3)_6]^{2+}$	10 800
$[\text{Fe}(\text{OH}_2)_6]^{2+}$	9 400	$[\text{Ni}(\text{en})_3]^{2+}$	11 500

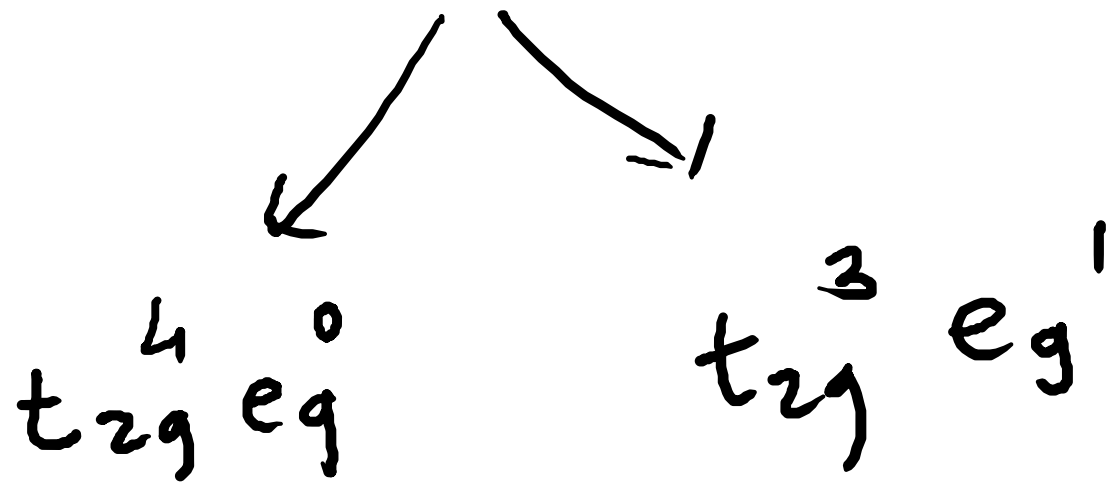
CFSE

$$d^1 \longrightarrow 0.4 \Delta_0$$

$$d^2 \longrightarrow 0.8 \Delta_0$$

$$d^3 \longrightarrow 1.2 \Delta_0$$

for d^4 two possibilities



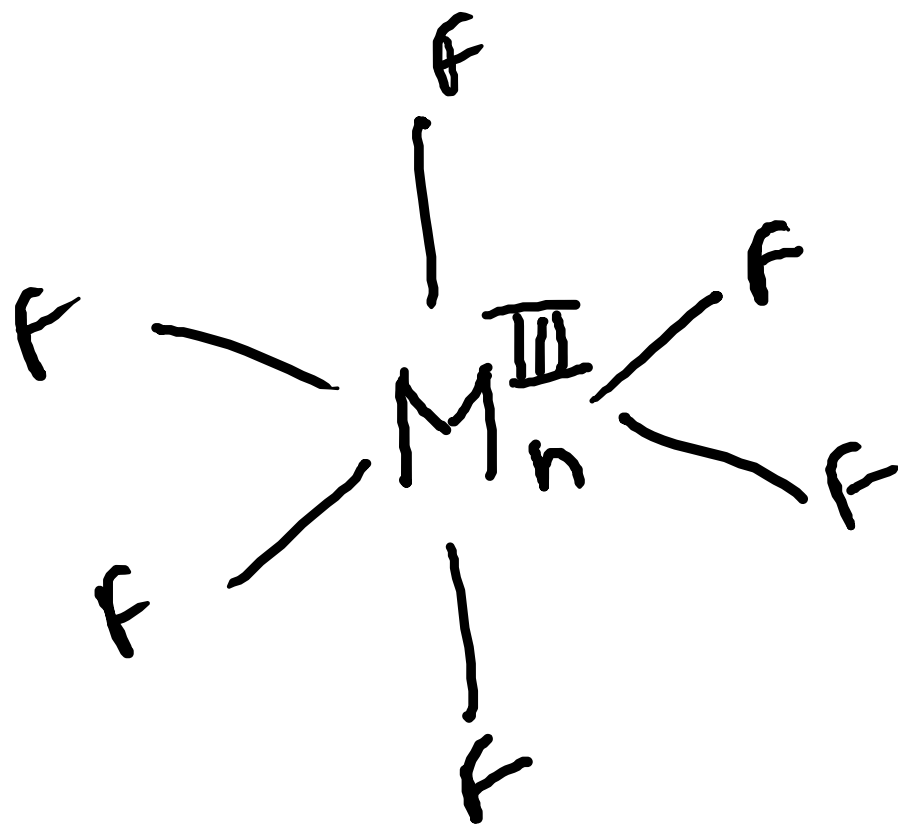
pairing energy $\rightarrow$ Coulombic repulsion
Exchange energy

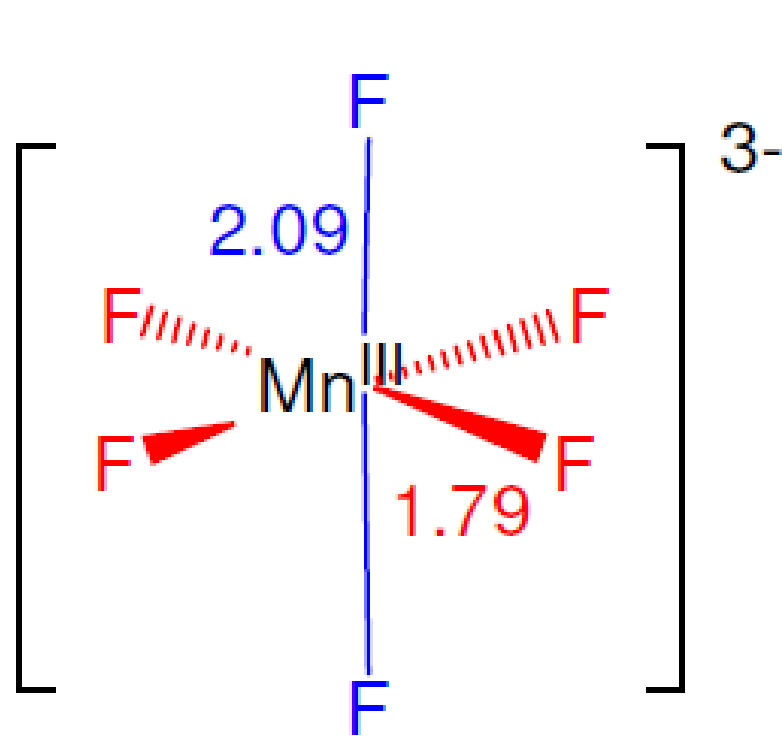
if $P > \Delta_0 \longrightarrow t_{2g}^3 e_g^1$ High Spin

if $P < \Delta_0 \longrightarrow t_{2g}^4 e_g^0$ low spin

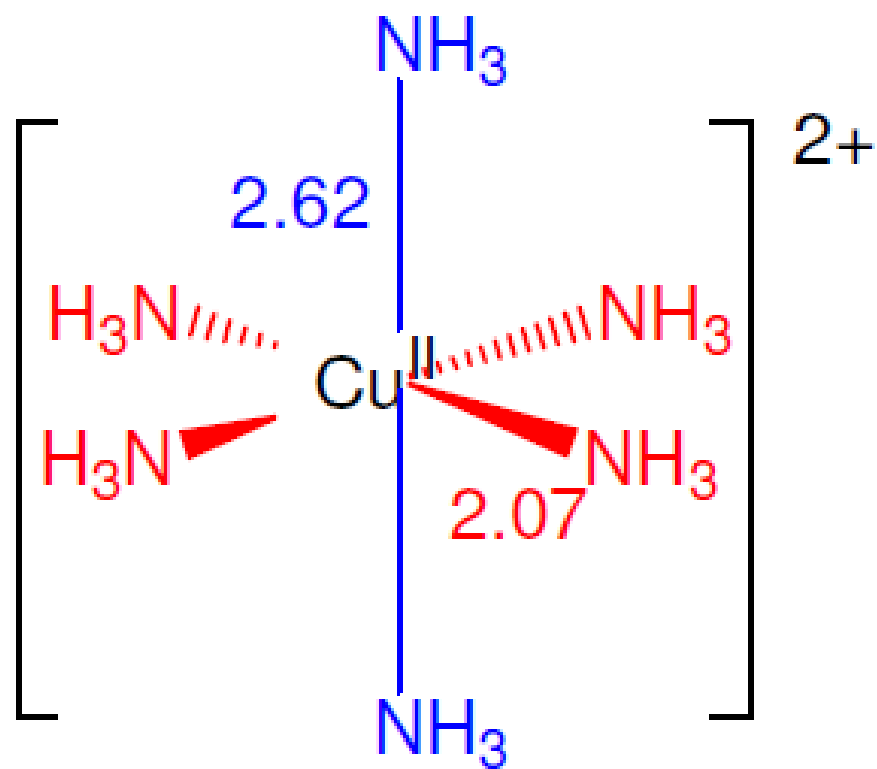
d^n	Weak field			Strong field		
	Configuration	Unpaired electrons	CFSE	Configuration	Unpaired electrons	CFSE
d^1	t_{2g}^1	1	$0.4\Delta_o$	t_{2g}^1	1	$0.4\Delta_o$
d^2	t_{2g}^2	2	$0.8\Delta_o$	t_{2g}^2	2	$0.8\Delta_o$
d^3	t_{2g}^3	3	$1.2\Delta_o$	t_{2g}^3	3	$1.2\Delta_o$
d^4	$t_{2g}^3 e_g^1$	4	$0.6\Delta_o$	t_{2g}^4	2	$1.6\Delta_o$
d^5	$t_{2g}^3 e_g^2$	5	$0.0\Delta_o$	t_{2g}^5	1	$2.0\Delta_o$
d^6	$t_{2g}^4 e_g^2$	4	$0.4\Delta_o$	t_{2g}^6	0	$2.4\Delta_o$
d^7	$t_{2g}^5 e_g^2$	3	$0.8\Delta_o$	$t_{2g}^6 e_g^1$	1	$1.8\Delta_o$
d^8	$t_{2g}^6 e_g^2$	2	$1.2\Delta_o$	$t_{2g}^6 e_g^2$	2	$1.2\Delta_o$
d^9	$t_{2g}^6 e_g^3$	1	$0.6\Delta_o$	$t_{2g}^6 e_g^3$	1	$0.6\Delta_o$
d^{10}	$t_{2g}^6 e_g^4$	0	$0.0\Delta_o$	$t_{2g}^6 e_g^4$	0	$0.0\Delta_o$

^a This table is somewhat simplified because pairing energies and electron–electron effects have been neglected.





d^4

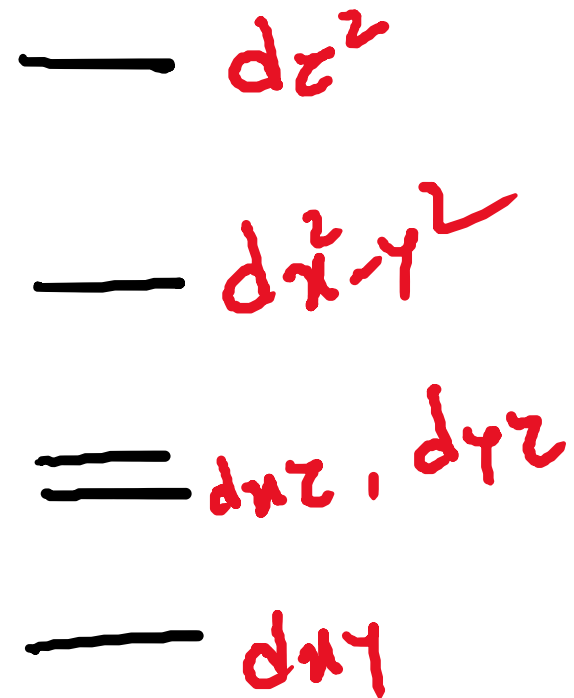
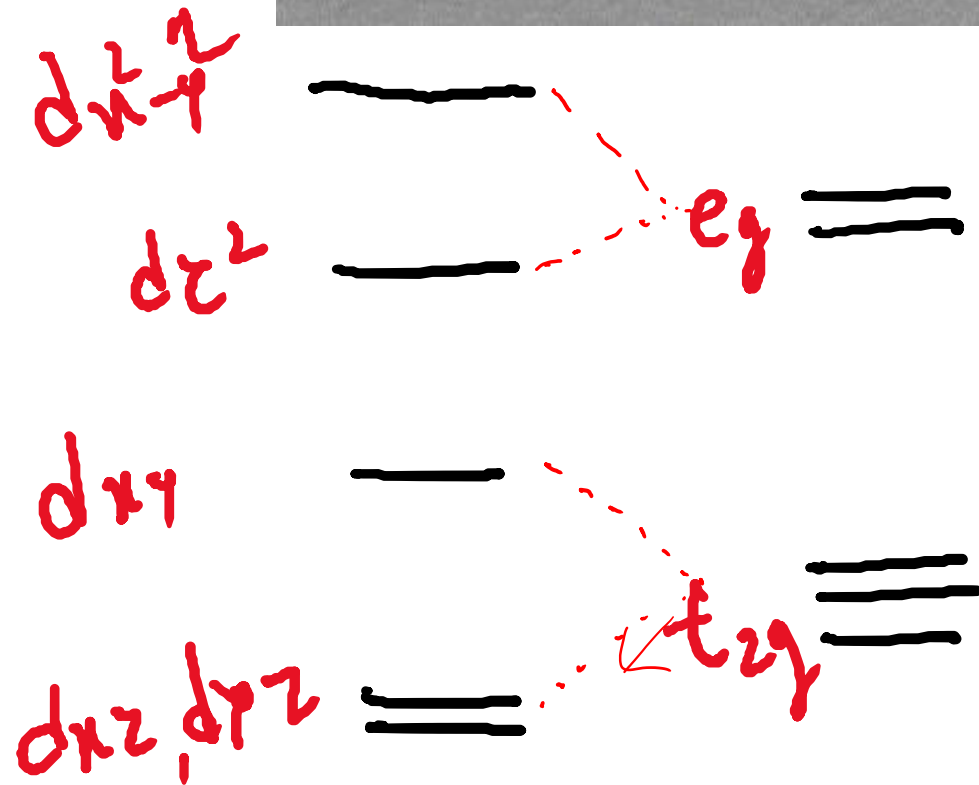
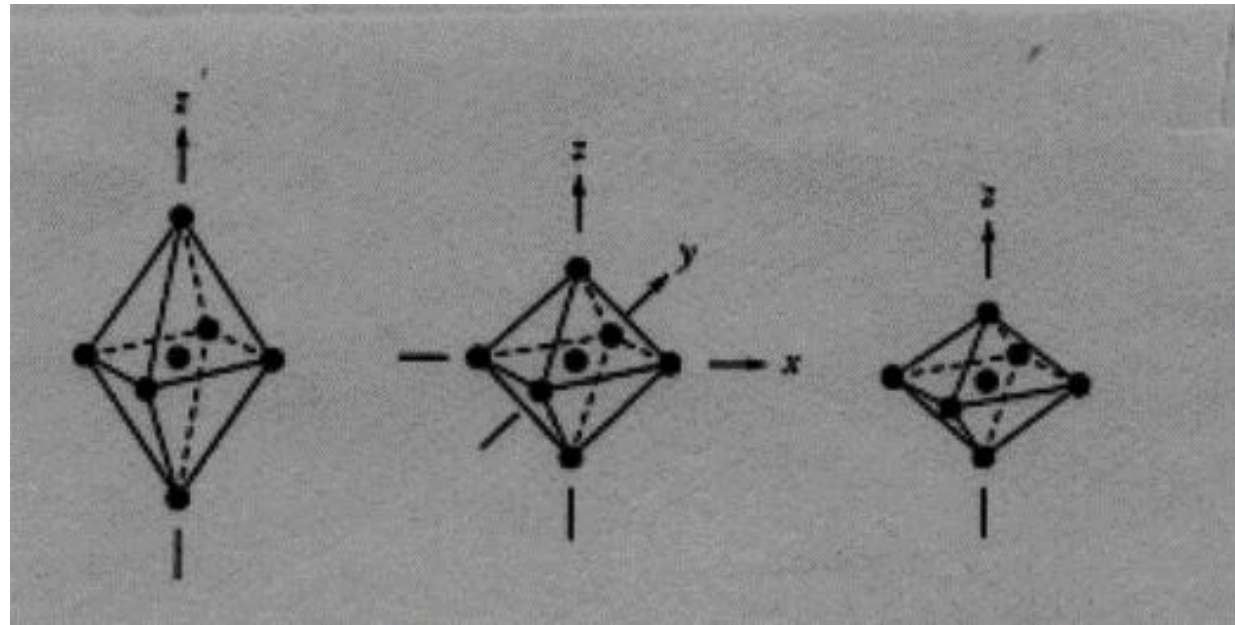


d^9

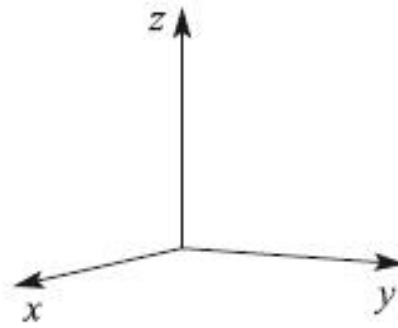
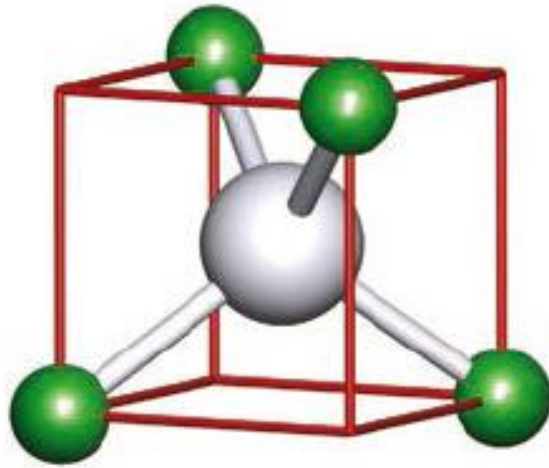
The Jahn-Teller Theorem:

For a non-linear molecule in an electronically degenerate state, distortion must occur to lower the symmetry, remove the degeneracy and lower the energy.

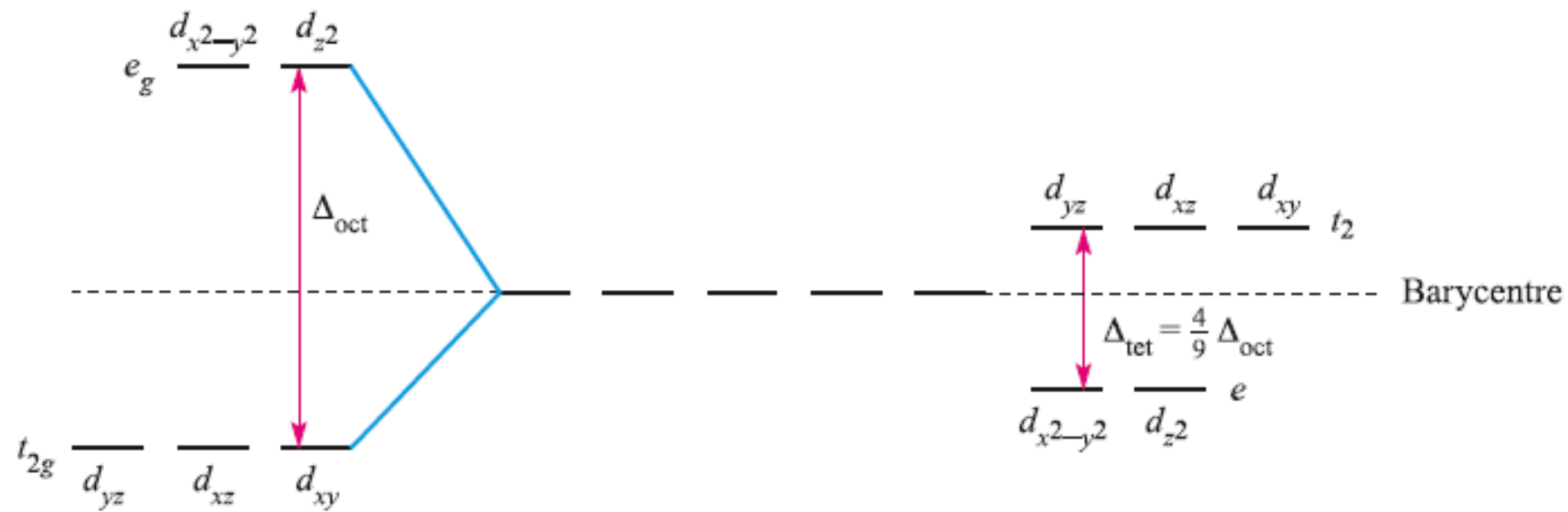
The Jahn-Teller effect is more pronounced when the e_g level is not completely filled



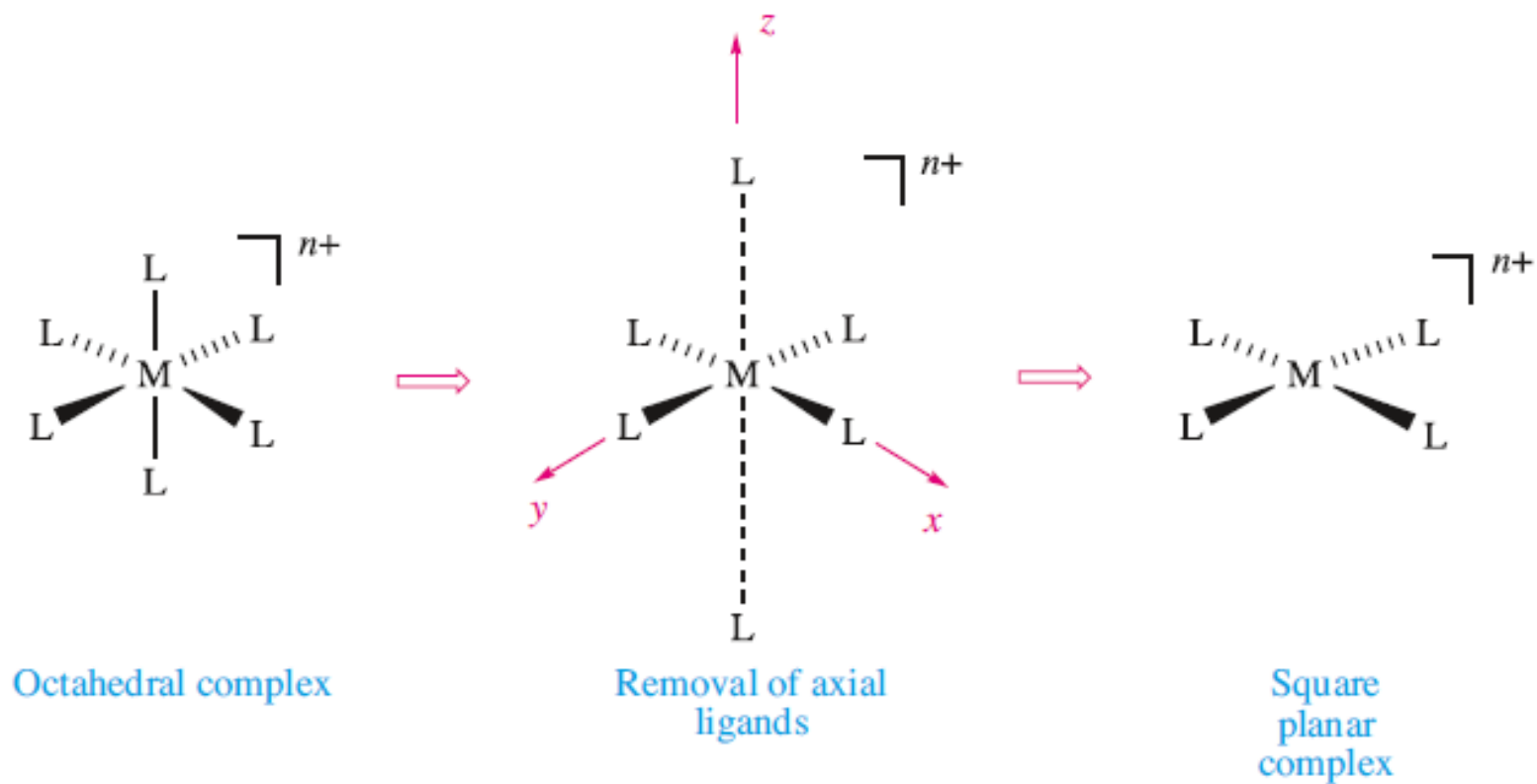
The tetrahedral crystal field

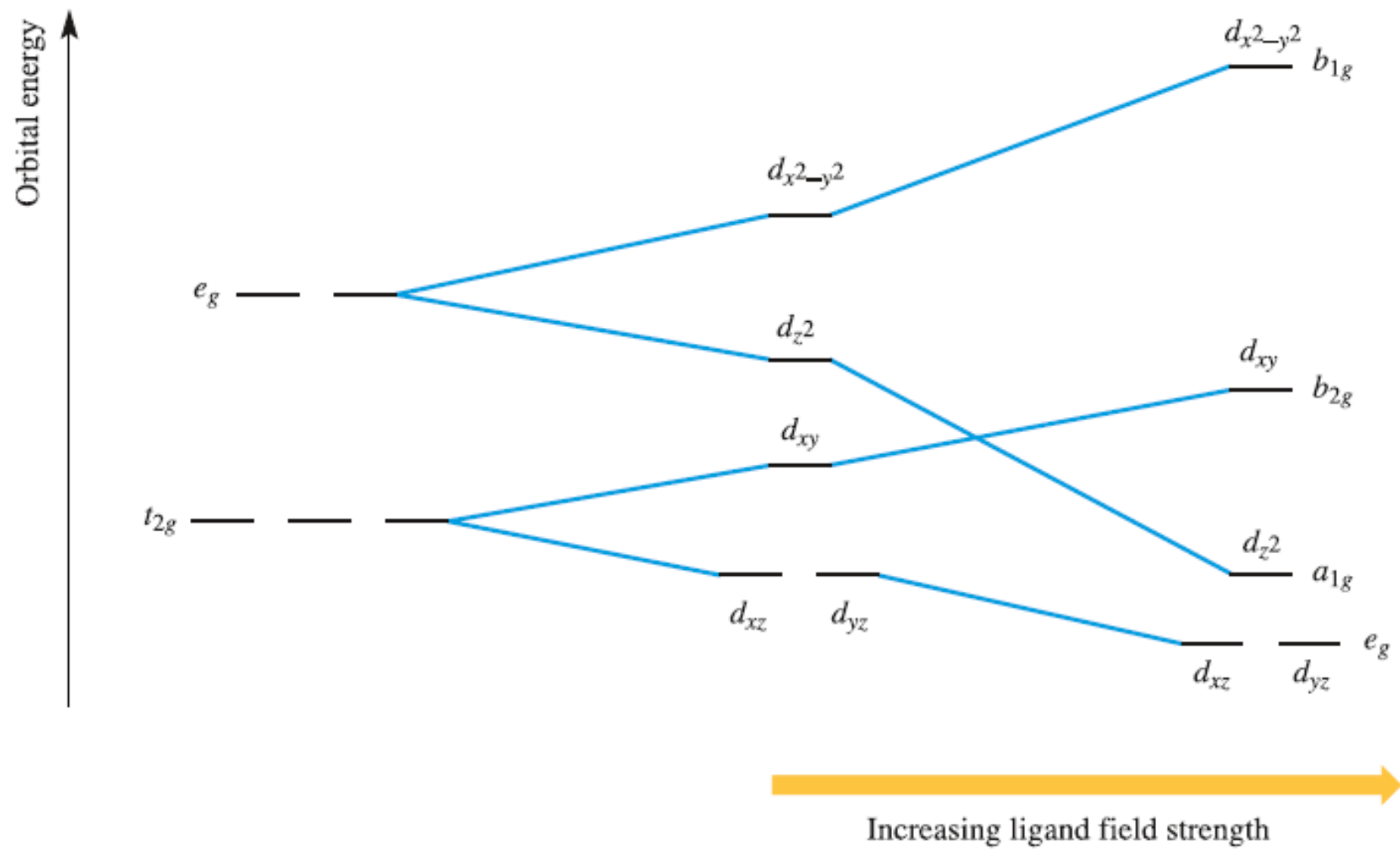


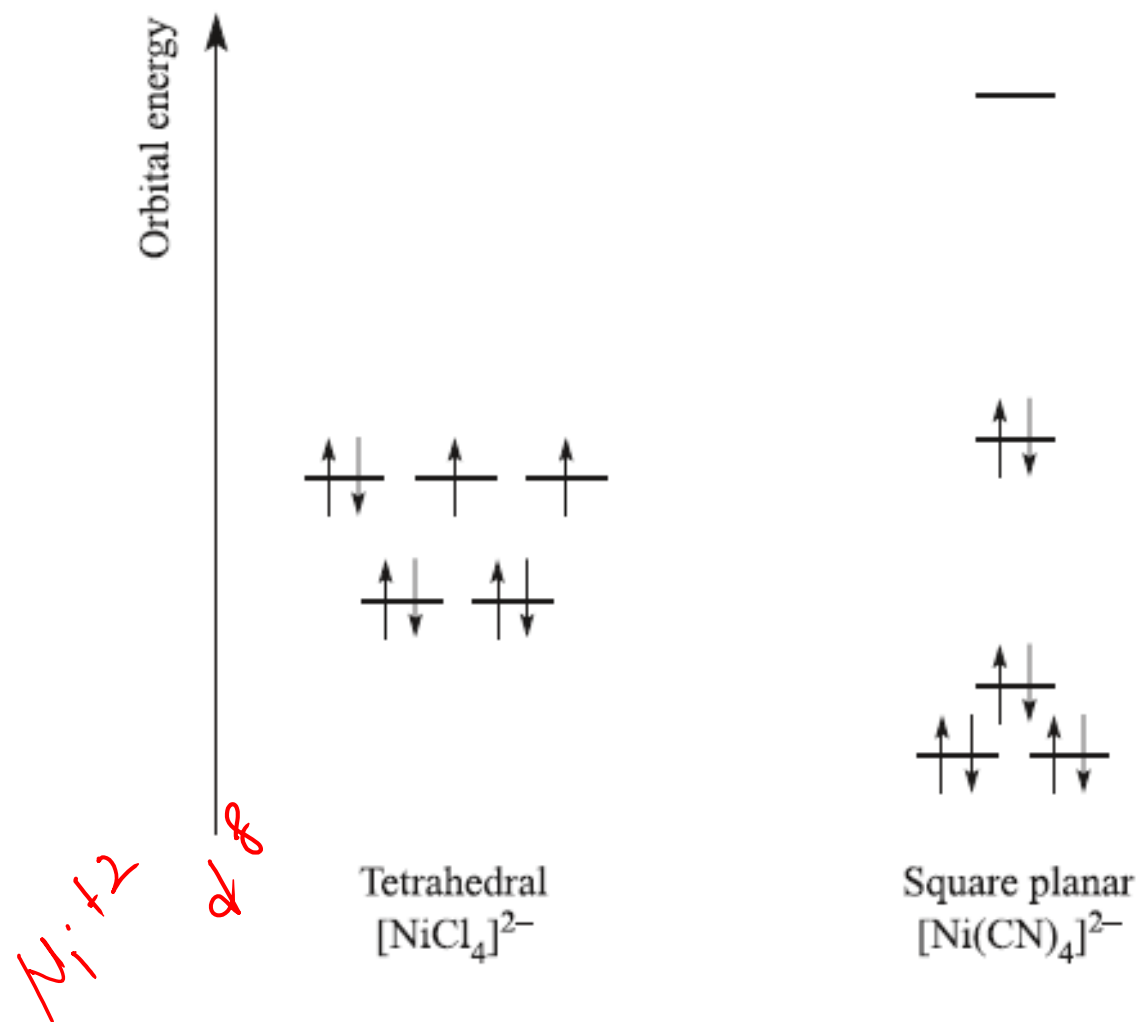
Orbital
energy ↑

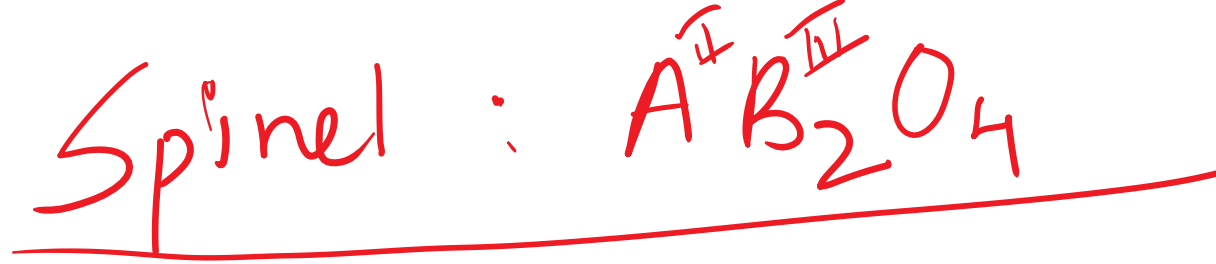


The square planar crystal field







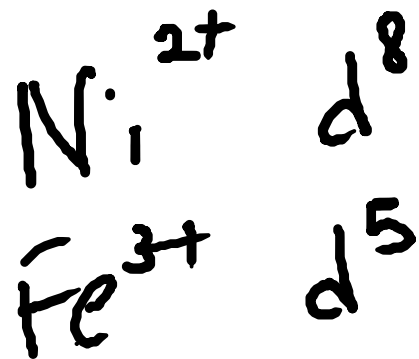


The distribution of metal ions between tetrahedral and octahedral sites in a spinel can be rationalized in terms of CFSEs.

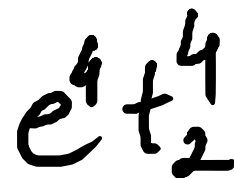
In a normal spinel the tetrahedral sites are occupied by the A^{2+} ions and the octahedral sites by B^{3+} ions: $(A^{\text{II}})^{\text{tet}} (B^{\text{III}})^{\text{oct}} O_4$

In an inverse spinel, the distribution is





Ni^{2+} In tetrahedral



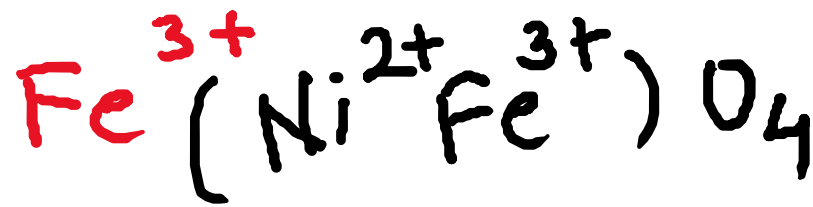
CFSE $0.8 \Delta_t$

Ni^{2+} In octahedral



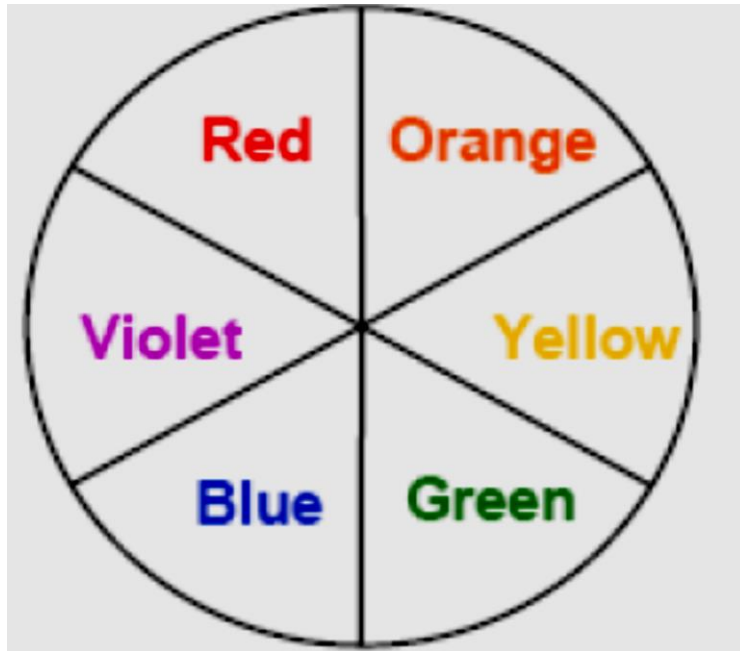
$\sim 0.4 \Delta_o$

$t_{2g}^6 e_g^2$ CFSE $1.2 \Delta_o$



Color



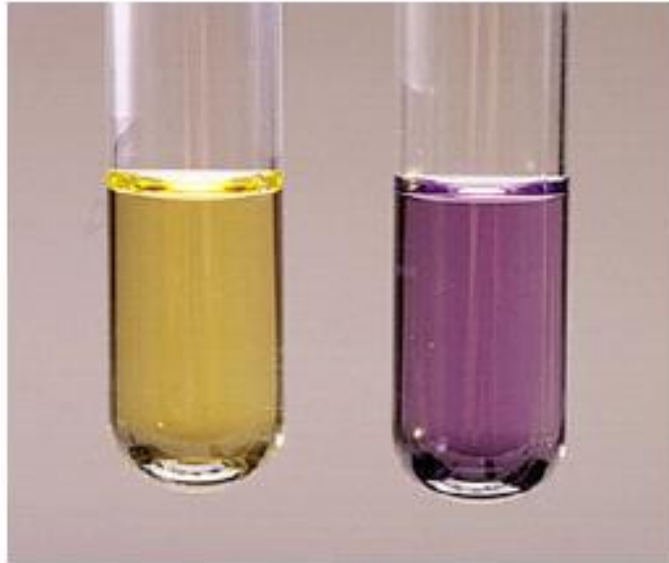
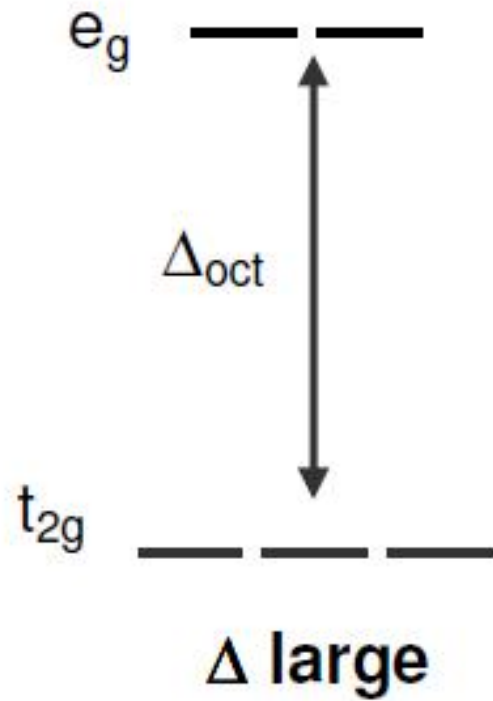


Colour of light absorbed	Approx. λ ranges / nm	Colour of light transmitted
Red	700-620	Green
Orange	620-580	Blue
Yellow	580-560	Violet
Green	560-490	Red
Blue	490-430	Orange
Violet	430-380	Yellow

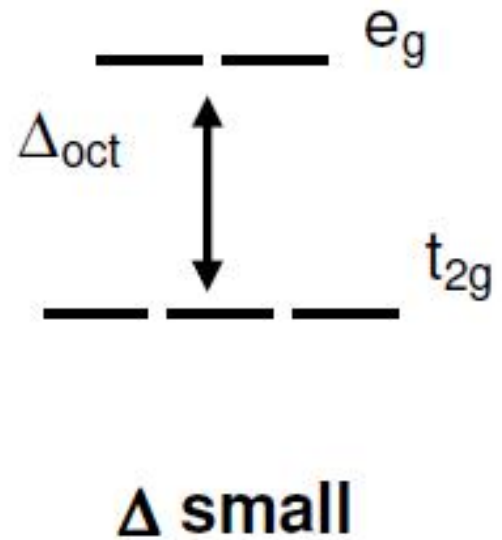
Color of Coordination Complexes

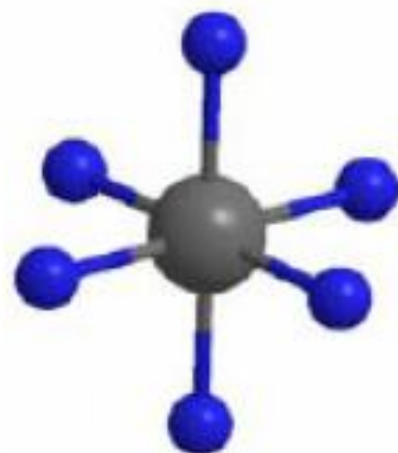
The color of coordination complexes arises from electronic transitions between levels whose spacing corresponds to the wavelengths available in the visible light. In complexes, these transitions are frequently referred to as d-d transitions because they involve the orbitals that are mainly d in character (for examples: t_{2g} and e_g for the octahedral complexes and e and t_2 for the tetrahedral complexes).

violet light absorbed
complex appears yellow



yellow light absorbed
complex appears violet



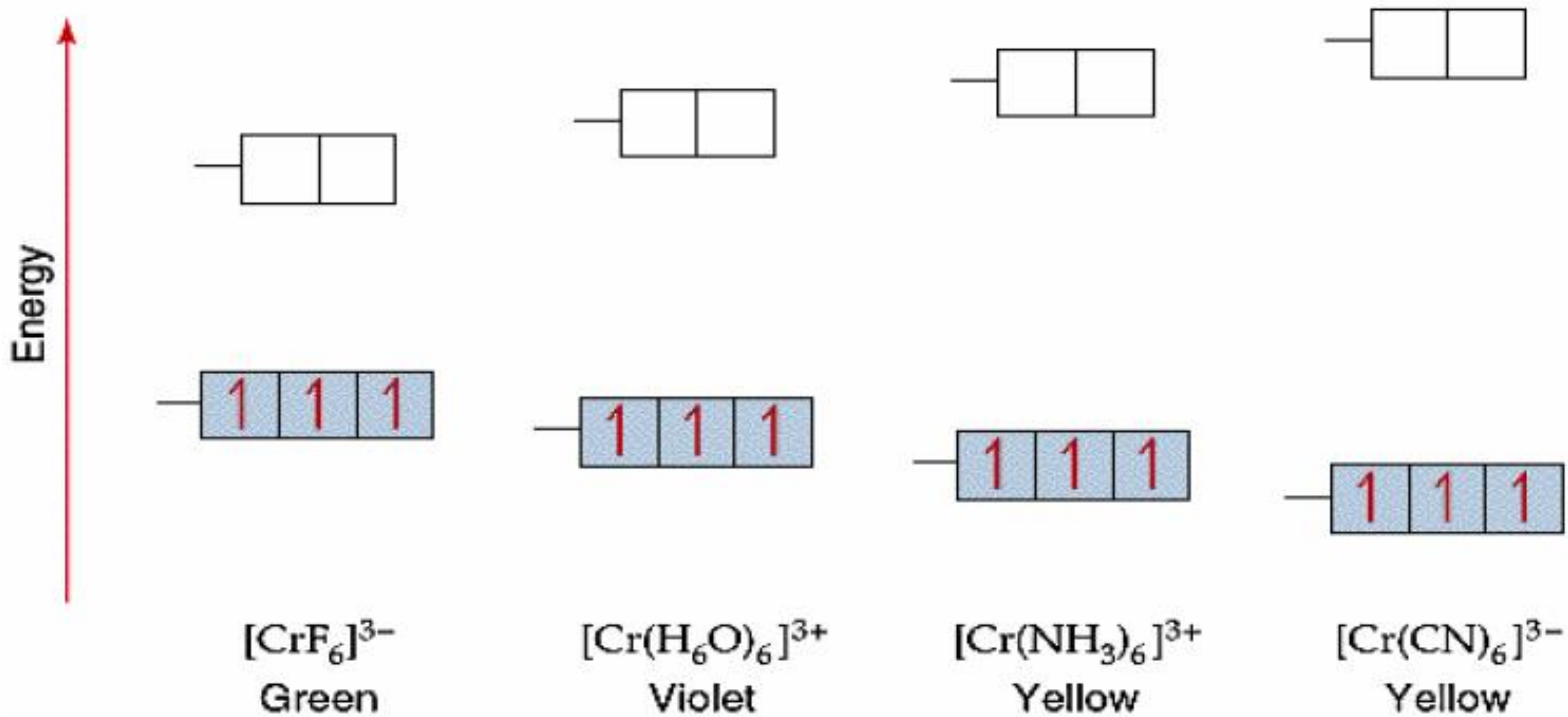


Strong Δ_{oct}
High energy,
violet light absorbed
Complex: yellow

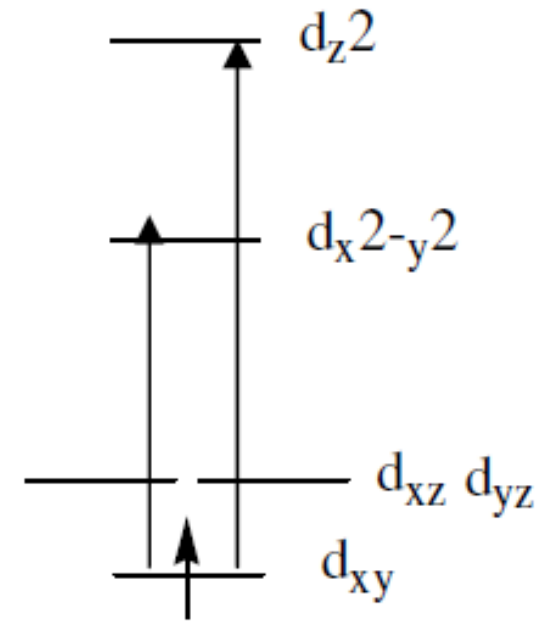
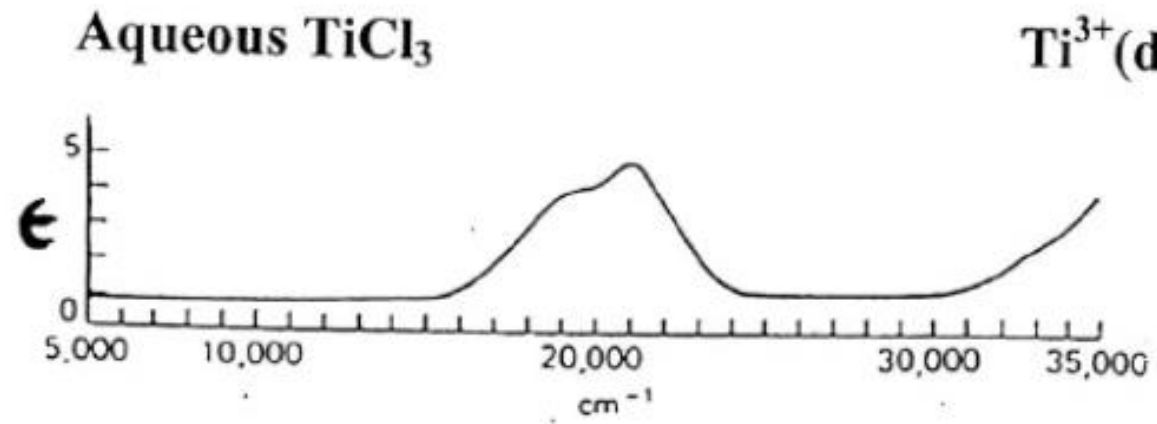


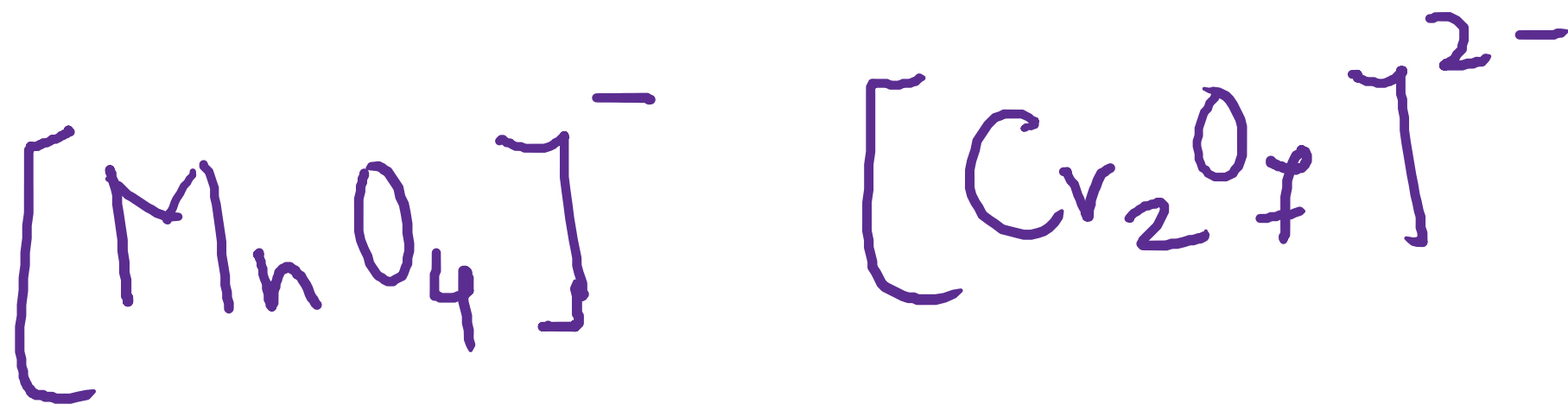
Weak Δ_{oct}
Low energy,
yellow light absorbed
Complex: magenta





appearance of a shoulder in the case of $[\text{Ti}(\text{H}_2\text{O})_6]^{3+}$





Selection rules for electronic transitions

The Beer-Lambert Law

$$A = \log_{10}(I_0/I) = \epsilon c l$$

where ϵ is the molar extinction coefficient (in L cm⁻¹ mole⁻¹),

c is concentration in mole L⁻¹

l is the path length in cm.

A is known as 'Absorbance'

The Laporte Rule.

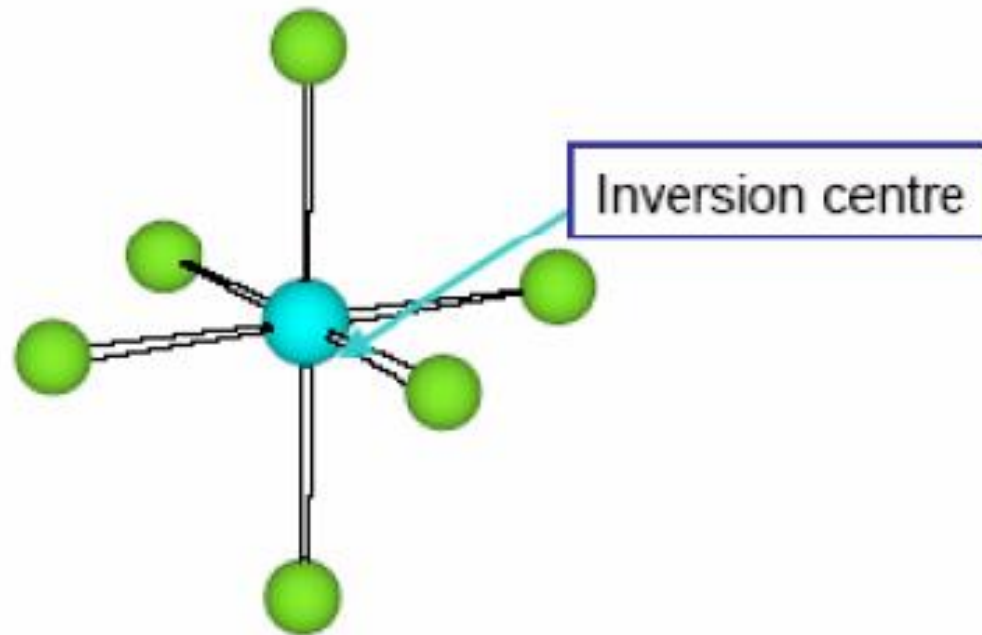
In a molecule or ion possessing a centre of symmetry, transitions are not allowed between orbitals of the same parity, for example d to d. In other words, there must be change in parity ($\Delta l = \pm 1$), i.e. the orbital quantum number should differ by 1.

The rule is not applicable to tetrahedral complexes as it does not contain a center of symmetry.

Octahedral complexes

(Centrosymmetric)

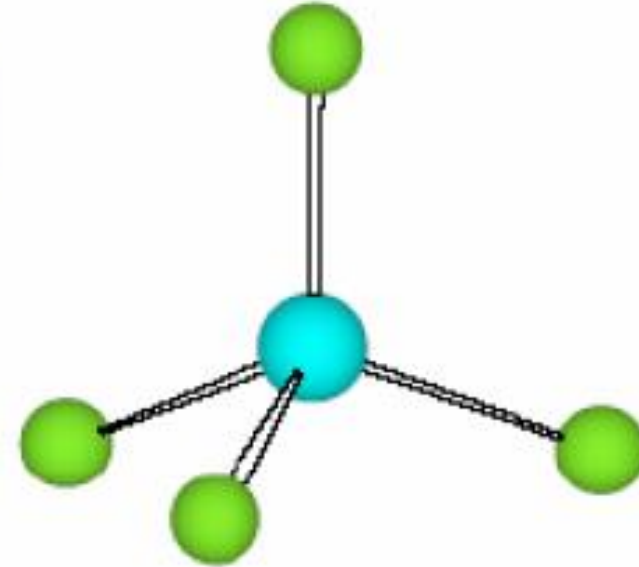
Laporte rule applies



Tetrahedral complexes

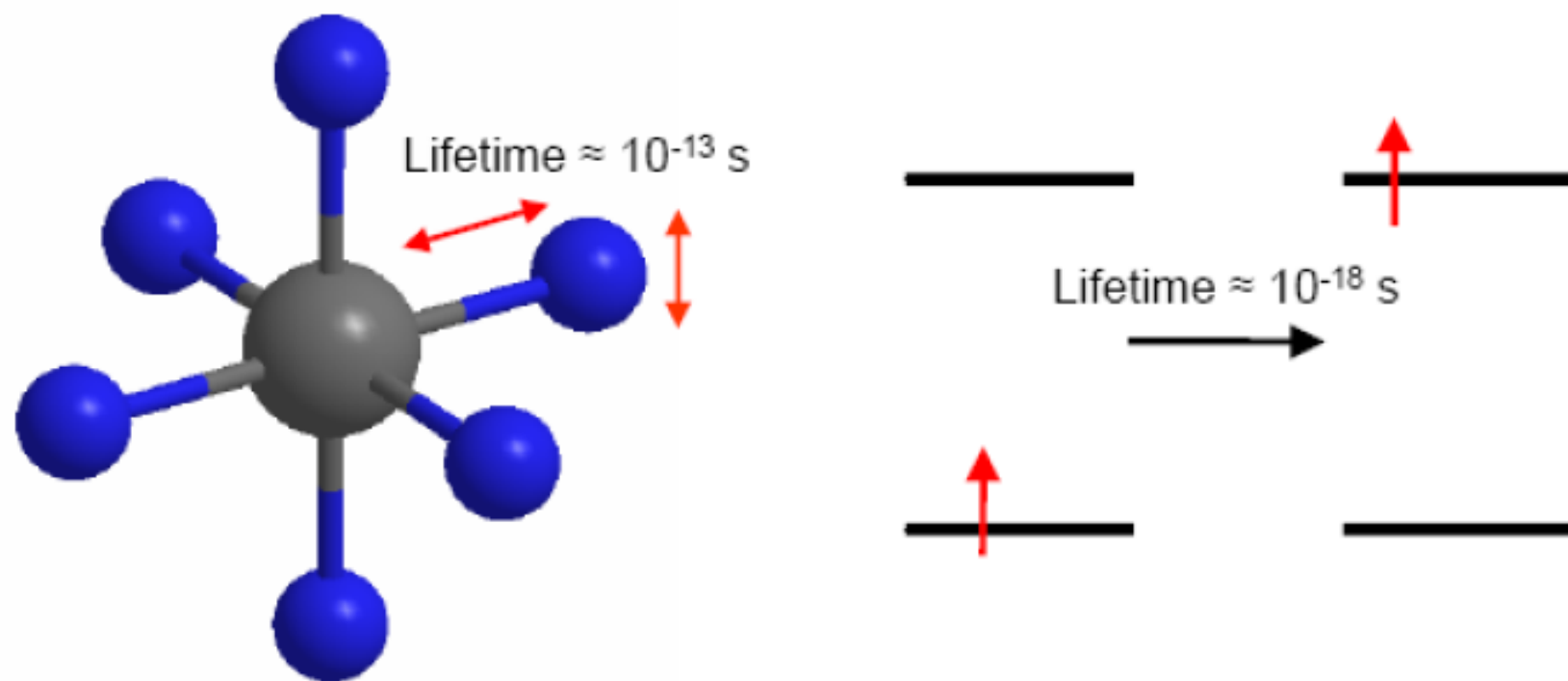
(Non-centrosymmetric)

Laporte rule relaxed



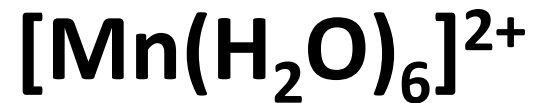
$\epsilon (T_d)$ is a 100 times greater than $\epsilon (O_h)$

Molecular vibrations: Vibronic coupling

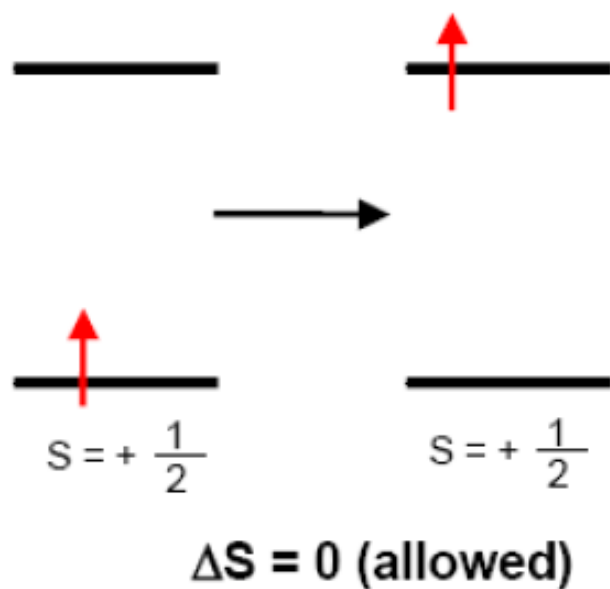


Spin Allowed - Spin Forbidden

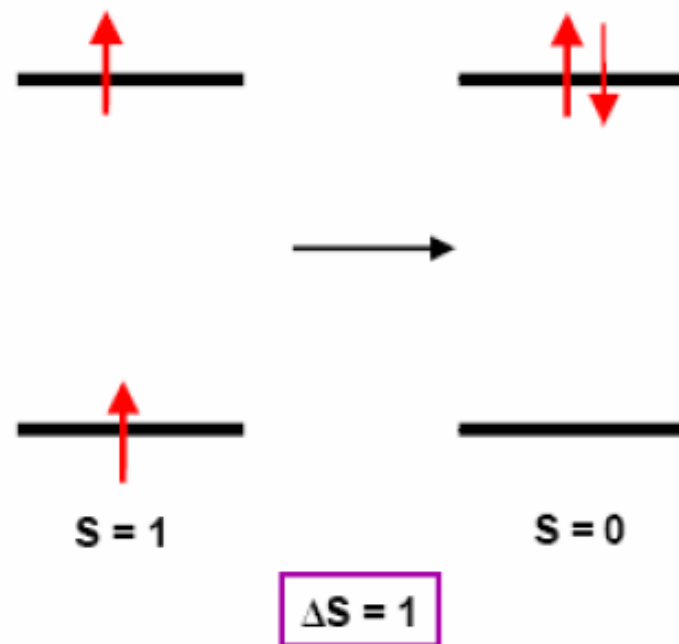
Any transition for which $\Delta S \neq 0$ is *strongly* forbidden; that is, in order to be allowed, a transition must involve no change in spin state.



Promotion of an electron can only proceed if the spin orientation is conserved, i.e. $\Delta S = 0$



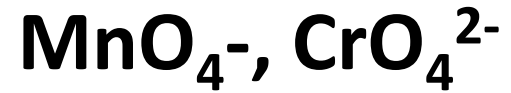
'forbidden' transitions



Charge-Transfer (CT) Bands.

Similar to d-d transitions, charge-transfer (CT) transitions also involve the metal d-orbitals. CT bands are observed if the energies of empty and filled ligand- and metal-centered orbitals are similar. The direction of the electron transfer is determined by the relative energy levels of these orbitals:

Ligand to-Metal charge transfer (LMCT)



For MnO_4^- , the d-electron count on Mn(VII) is d^0

The origin of the color in this species is not due to d-d transition, rather, charge transfer from O_2^- to Mn(VII), described as LMCT band.

Metal-to Ligand charge transfer (MLCT)



In this complex the charge transfer occurs from Fe(II) to the empty π^* orbitals of bpy ligand

LMCT $\Rightarrow$ favourable when metal centre is in high oxidation state

MLCT $\Rightarrow$ favoured when electron rich metal centre is present.

Expected intensities of electronic transitions

<u>Type of transition</u>	<u>ϵ value</u> <u>$\text{dm}^3 \text{mol}^{-1} \text{cm}^{-1}$</u>	<u>Example</u>
spin forbidden d-d	< 1	$[\text{Mn}(\text{H}_2\text{O})_6]^{2+}$ High spin d^5

Laporte forbidden
Spin Allowed
d-d

1 - 10 centrosymmetric
 $[\text{Ti}(\text{H}_2\text{O})_6]^{+3}$

10 - 1000 Non centrosymmetric

charge transfer
fully Allowed

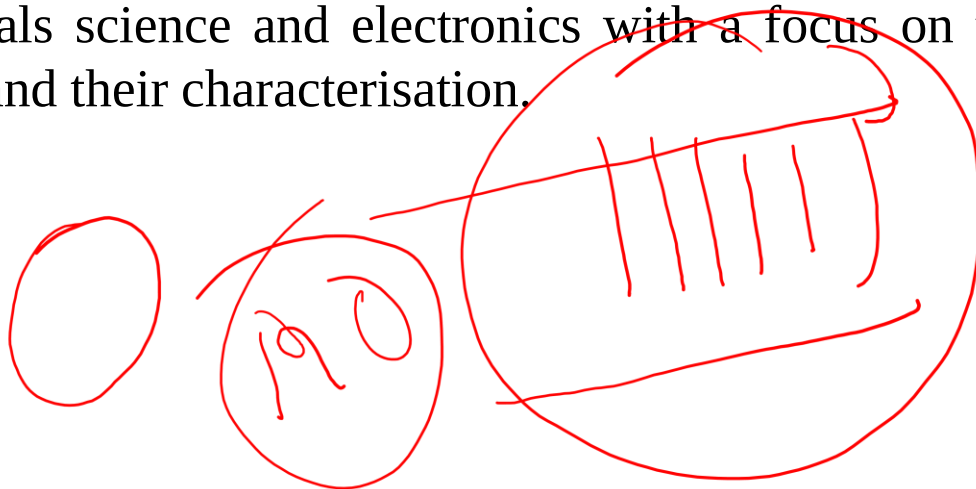
1000 - 50000 MnO_4^-

Solid State Chemistry

Solid state

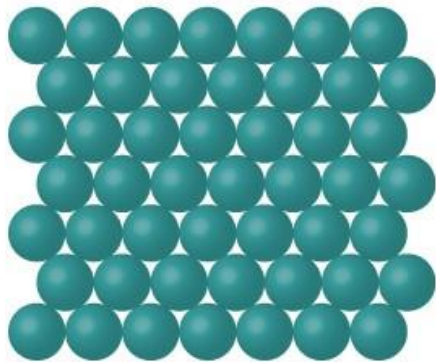
- **Definite volume and shape**
- **Strong intermolecular forces**
- **Low kinetic energy**
- **Solid-state chemistry** is the study of the synthesis, structure, and properties of solid phase materials, particularly, but not necessarily exclusively of, non-molecular solids.
- It has a strong overlap with solid state physics, mineralogy, crystallography, ceramics, metallurgy, materials science and electronics with a focus on the synthesis of novel materials and their characterisation.
- **Materials chemistry**

Solid state compound.

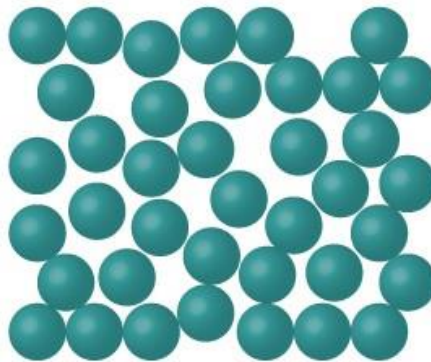


Classification of Solids

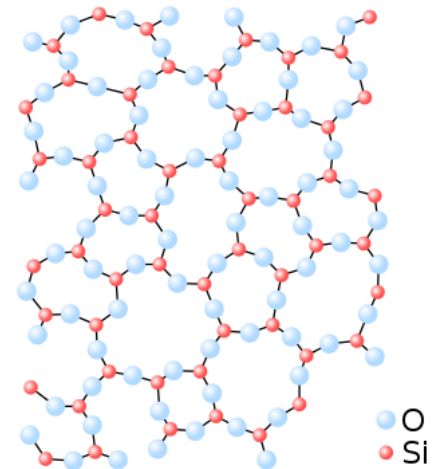
- Solids can be classified as crystalline or amorphous.
(on the basis of arrangement of constituent atoms/molecules/ions)
- A **crystalline solid** is composed of one or more crystals; each crystal has a well-defined, ordered structure in three dimensions. They have sharp melting point
Examples include sodium chloride and sucrose.
- An **amorphous solid** has a disordered structure. It lacks the well-defined arrangement of basic units found in a crystal. They melt over a wide range of temperature.
- Glass is an example of amorphous solid.



Crystalline



Amorphous



Classification of Solids

*Solids can be classified into following types
(based on nature of bonding)*

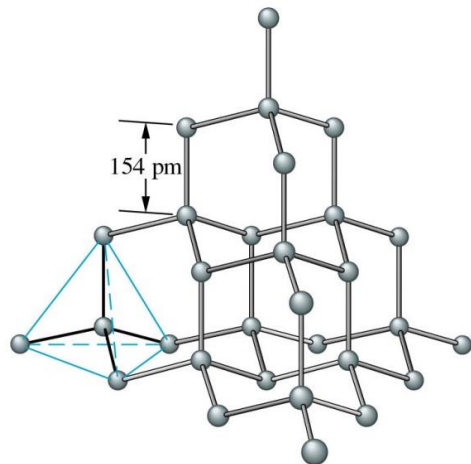
- **Molecular solids (Van der Waals forces)**
- **Covalent (Covalent bond)**
- **Metallic (Metallic bond)**
- **Ionic (Ionic bond)**

Molecular solids

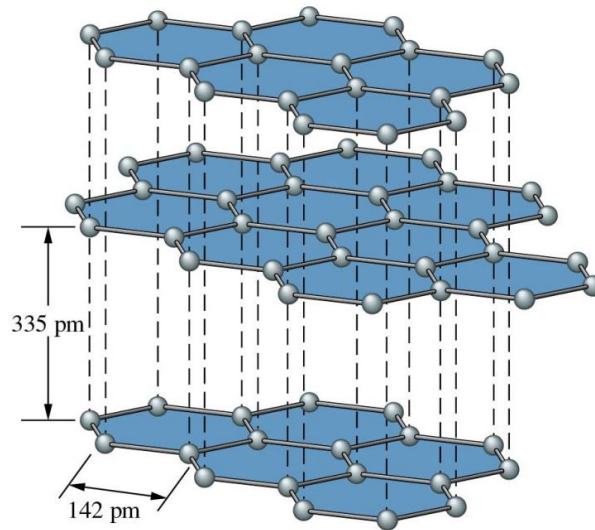
- A **molecular solid** is a solid that consists of atoms or molecules held together by weak intermolecular forces.
- Soft, low melting point, volatile, electrical insulators.
- Solid neon (melting point of -248°C), Solid CO_2 (dry ice), Solid H_2O (ice), Iodine, Sugar
- In non-polar molecules, the inter- molecular forces are **only weak van der Waals forces**.
- **Polar molecules** have dipoles and so have **slightly stronger attractions** between them meaning that the **melting point of a polar molecular solid is a little higher** because more energy is needed to overcome the slightly stronger forces.

Covalent Solids

- A **covalent network solid** is a solid that consists of atoms held together in large networks or chains by covalent bonds.
- Very hard, high melting point, poor conductors of heat and electricity.
- Examples include carbon, in its forms as diamond or graphite, asbestos, and silicon carbide.



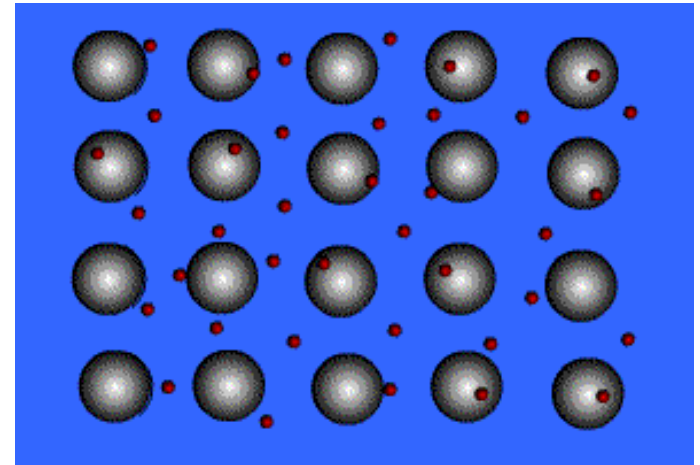
Diamond



Graphite

Metallic Solids

- A **metallic solid** is a solid that consists of positive metal ions held together by a surrounding “sea of electrons” (metallic bonding).



- In this kind of bonding, positively charged metal ions form the lattice which is submerged in the “sea of electrons”
Or, the metal ions occupy fixed position in the lattice structure and they are surrounded by delocalized electrons.
- Very soft to very hard, low to high melting point, good conductors of electricity and heat.
- **Malleable and ductile** due to **non-directional metallic bonds**
- **Good conductors of heat and electricity** due to high density and mobile electrons
- **High melting and boiling points** due to strong metallic bonds

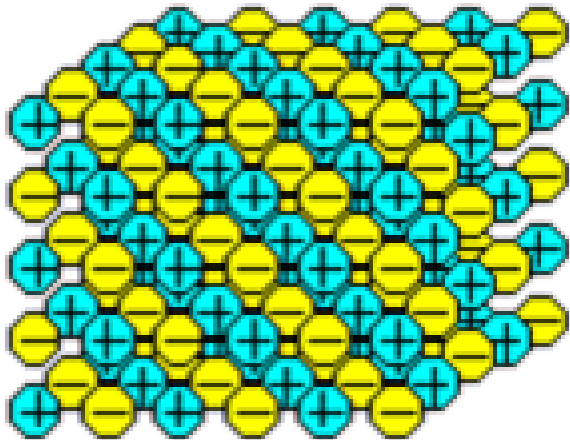
iron, copper, and silver

Ionic Solids

An **ionic solid** is a solid that consists of cations and anions held together by electrostatic attraction of opposite charges (ionic bond).

- Brittle, high melting point, good conductors in the aqueous solution or fused state, high heats of fusion.

Salts like NaCl, KNO_3 , LiF, BaSO_4 etc.



Ionic lattice structure

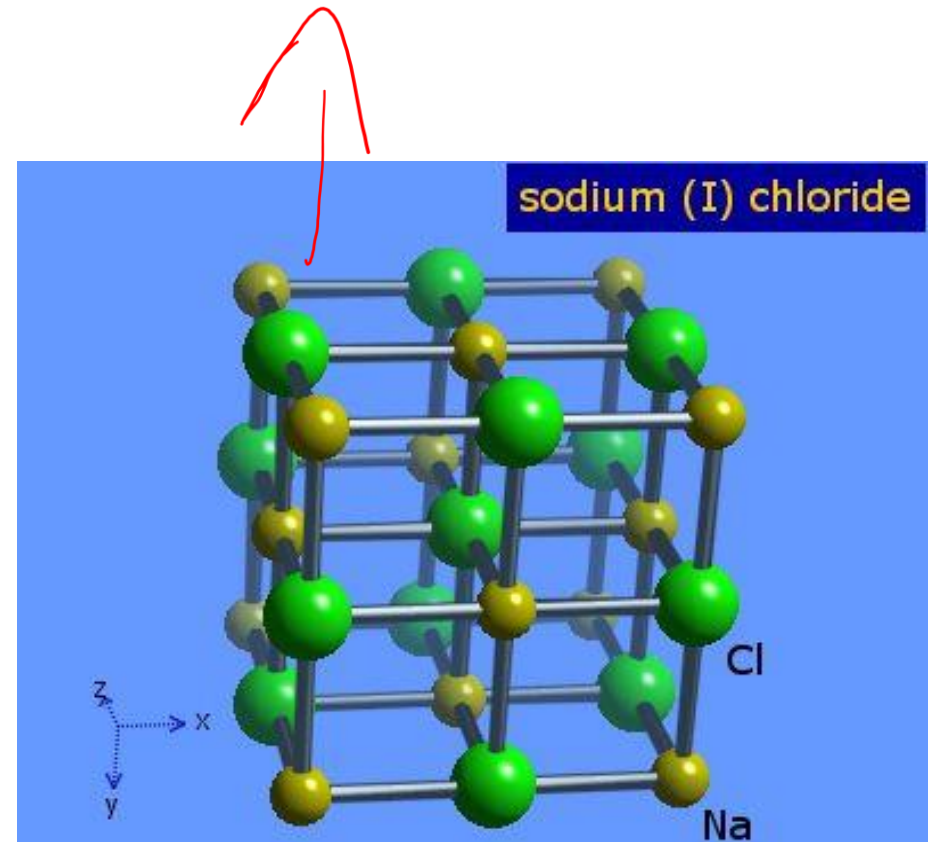
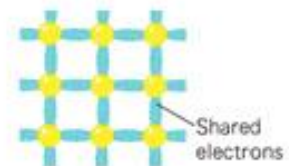
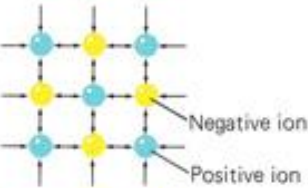
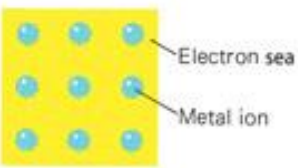
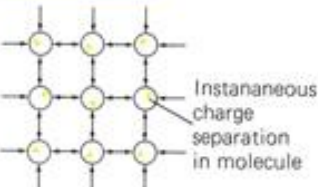
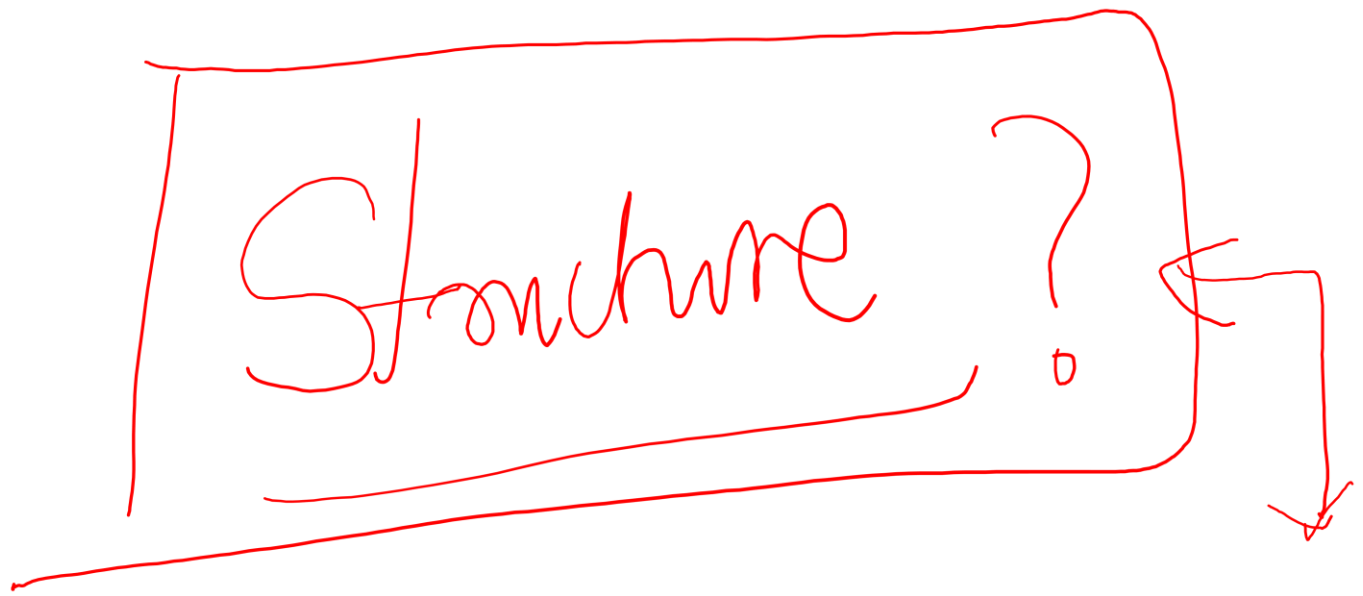


Table 10-1

Crystal Types (Bonds Are Strongest in Covalent Crystals, Weakest in Molecular Crystals)

Type		Bond	Example	Properties
Covalent		Shared electrons	Diamond C	Very hard; high melting point
Ionic		Electrical attraction	Sodium chloride NaCl	Hard; high melting point
Metallic		Electron sea	Copper Cu	Can be deformed; metallic luster; high electrical and thermal conductivity
Molecular		Van der Waals forces	Ice H ₂ O	Soft; low melting point

Structure?

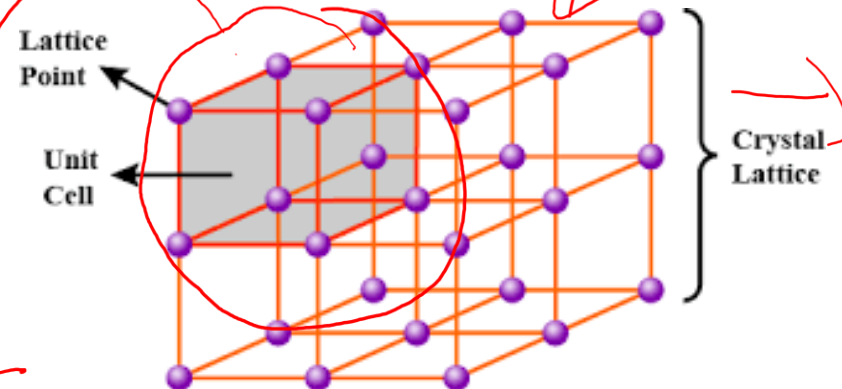
A hand-drawn red box containing the word "Structure?" in a cursive script. A red arrow points downwards from the right side of the box.

Crystal Lattices and Unit cell

- The idea of lattice was developed from internal regularity of a crystal structure.
- A highly ordered three dimensional structure formed by atoms ions or molecules are called crystal lattice. If the constituents are denoted by points, then a lattice may be regarded as an infinite set of points repeated at a regularly through space.
- 1-D Lattice: along lines
- 2D Lattice: along a plane
- 3D Lattice: along three coordinate axis
- Unit cell is the smallest building unit of a crystal, which when stacked together with pure translational repetition reproduces the whole crystal. It is an essential feature of crystal structure.
- Any point placed in unit cell must occupy The same relative position in every unit cell

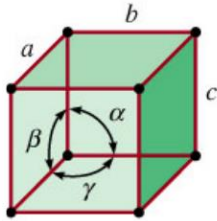
crystal structure

unit cell



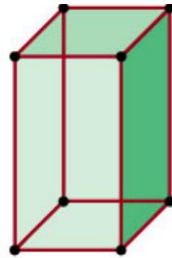
The 7 Crystal Systems

All 3D crystals belong to one of 7 *crystal systems*.



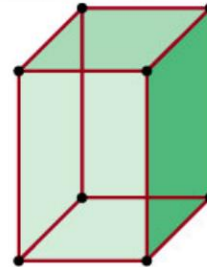
Cubic

$$a = b = c$$
$$\alpha = \beta = \gamma = 90^\circ$$



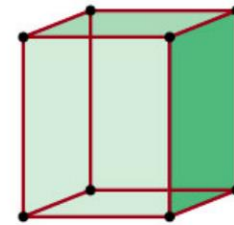
Tetragonal

$$a = b \neq c$$
$$\alpha = \beta = \gamma = 90^\circ$$



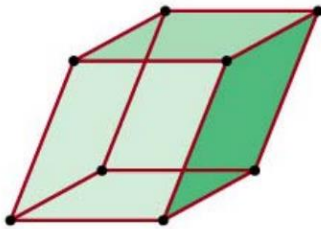
Orthorhombic

$$a \neq b \neq c$$
$$\alpha = \beta = \gamma = 90^\circ$$



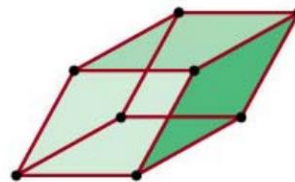
Monoclinic

$$a \neq b \neq c$$
$$\gamma \neq \alpha = \beta = 90^\circ$$



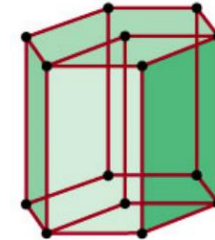
Triclinic

$$a \neq b \neq c$$
$$\alpha \neq \beta \neq \gamma \neq 90^\circ$$



Rhombohedral

$$a = b = c$$
$$\alpha = \beta = \gamma \neq 90^\circ$$



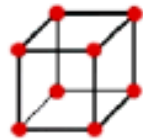
Hexagonal

$$a = b \neq c$$
$$\alpha = \beta = 90^\circ, \gamma = 120^\circ$$

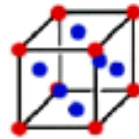
Bravais Lattices

All 3D crystals belong to one of 14 Bravais lattices.

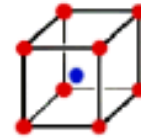
Bravais lattice: An infinite array of points with an arrangement and orientation that looks exactly the same from any lattice point.



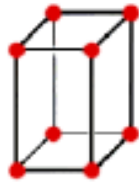
Simple
cubic



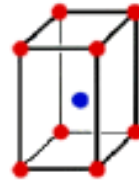
Face-centered
cubic



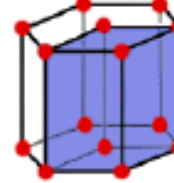
Body-centered
cubic



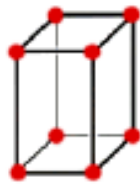
Simple
tetragonal



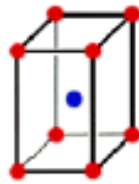
Body-centered
tetragonal



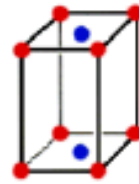
Hexagonal



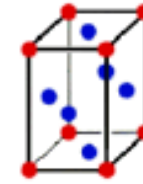
Simple
orthorhombic



Body-centered
orthorhombic



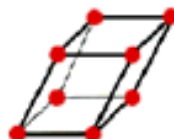
Base-centered
orthorhombic



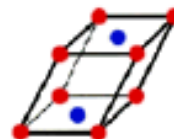
Face-centered
orthorhombic



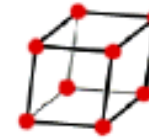
Rhombohedral



Simple
Monoclinic



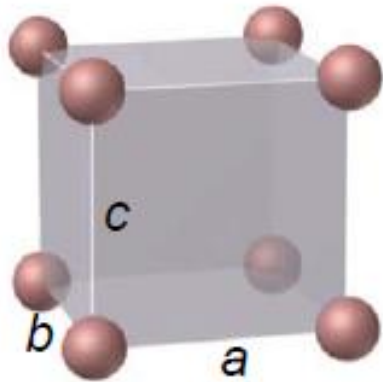
Base-centered
monoclinic



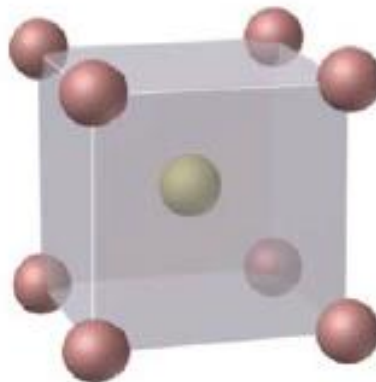
Triclinic

Simple (P)
Bodycentered (I)
Face centered (F)

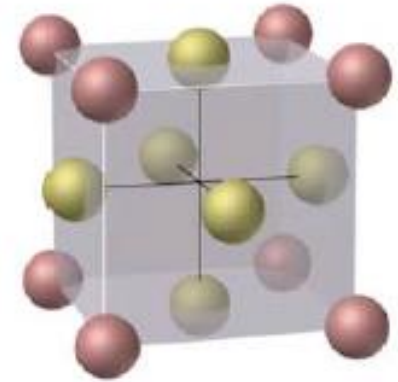
Cubic Lattices



Simple Cubic



Body-Centered Cubic



Face-Centered Cubic

- The lengths of the unit cell edges (a, b, c) are called *lattice constants*.
- For cubic crystals, $a = b = c$, so there is only one lattice constant (a).

Contents of Unit cell in orthorhombic

An important feature of the unit cell is the number of lattice points it contains. Atoms/ions/molecules are often located at lattice points.

Atoms

corner

face center

body center

edge center

Shared Between:

8 cells

2 cells

1 cell

4 cells

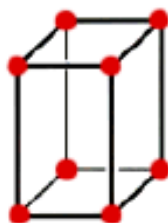
Each atom counts:

$1/8$

$1/2$

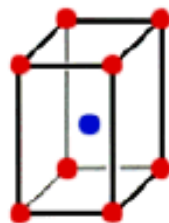
1

$1/4$



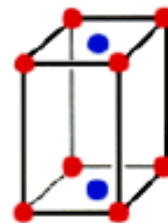
Simple
orthorhombic

$$8 \times 1/8 = 1$$



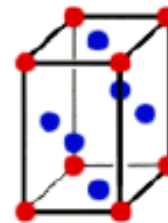
Body-centered
orthorhombic

$$8 \times 1/8 + 1 \times 1 = 2$$



Base-centered
orthorhombic

$$8 \times 1/8 + 2 \times 1/2 = 2$$

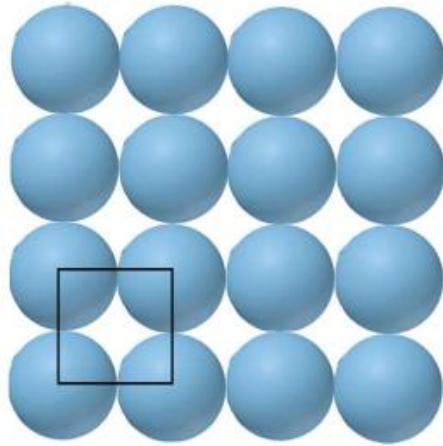


Face-centered
orthorhombic

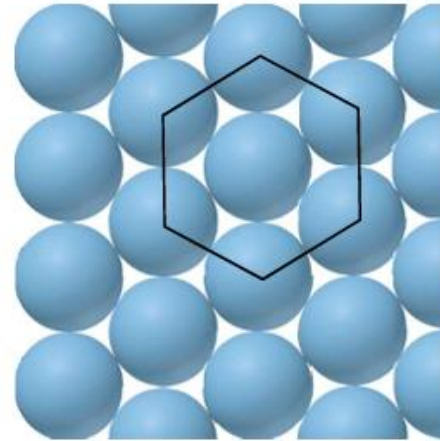
$$8 \times 1/8 + 4 \times 1/2 = 4$$

Packing of Spheres

Close-Packing

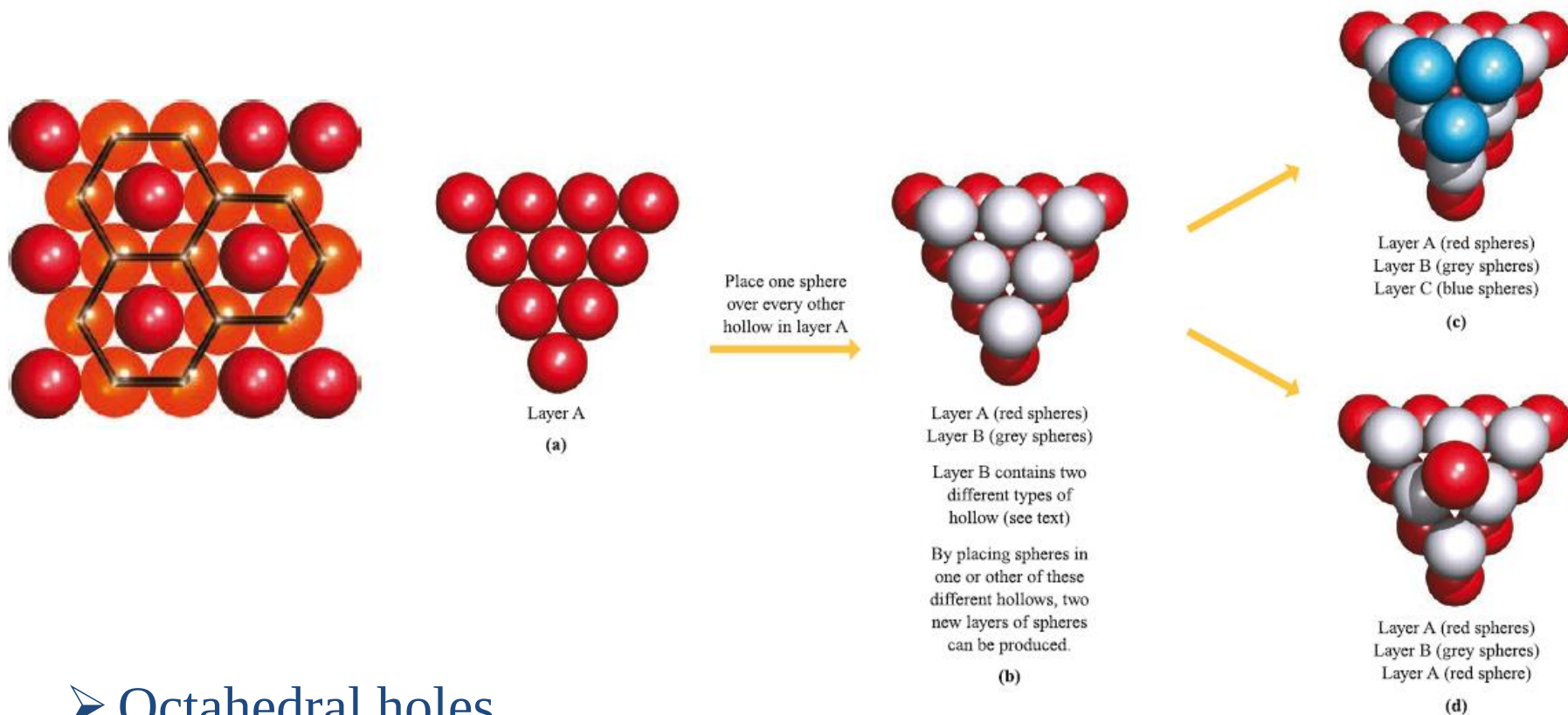


Square array of circles



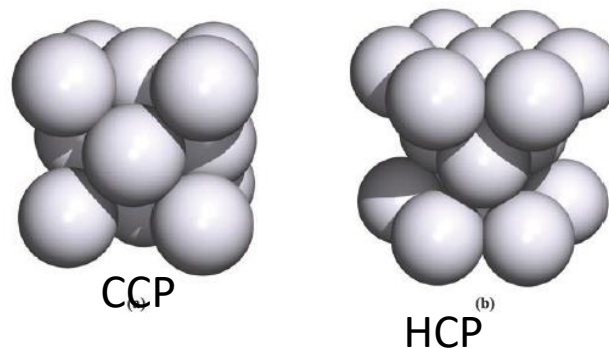
Close-packed array of circles

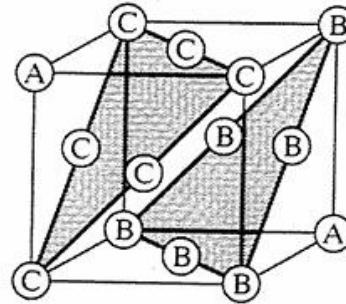
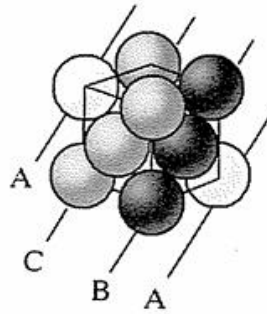
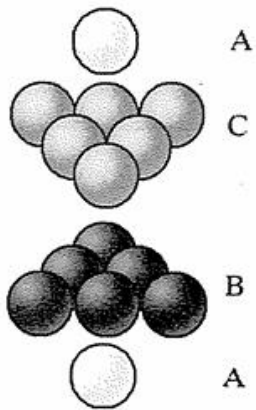
Packing of Spheres



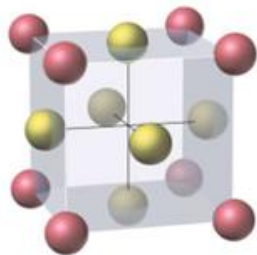
➤ Octahedral holes

➤ Tetrahedral holes

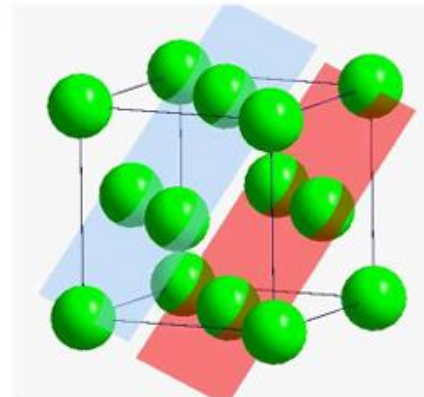




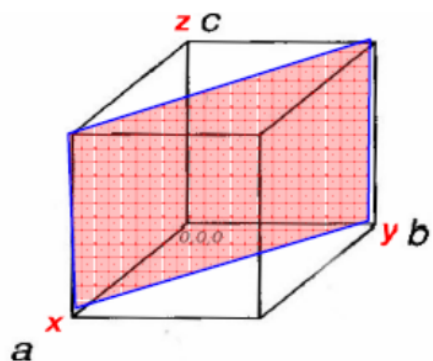
Face centered cubic (fcc)
has cubic symmetry.

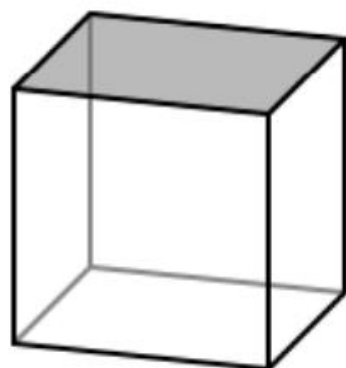


Face-centered cubic

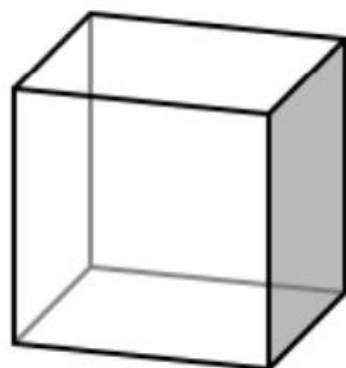


Miller Indices

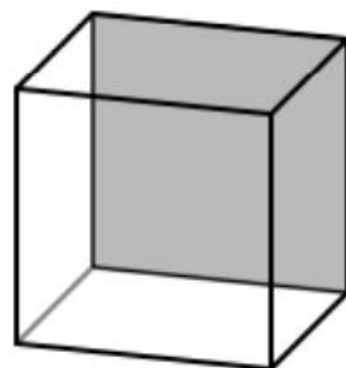
		x (a)	y (b)	z (c)
	Intercept on axes	1	1	∞
	Reciprocal	1/1	1/1	$1/\infty = 0$
	Integer Clear	1	1	0
	Miller Indices	(110)		



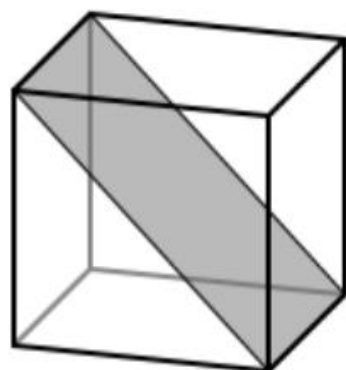
(001)



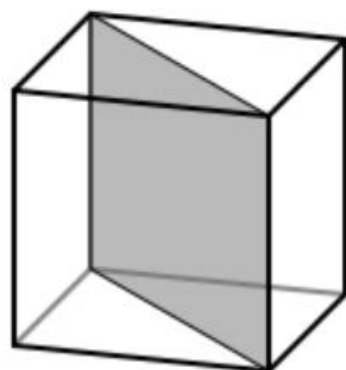
(100)



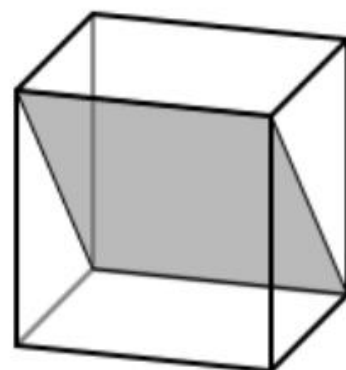
(010)



(101)



(110)



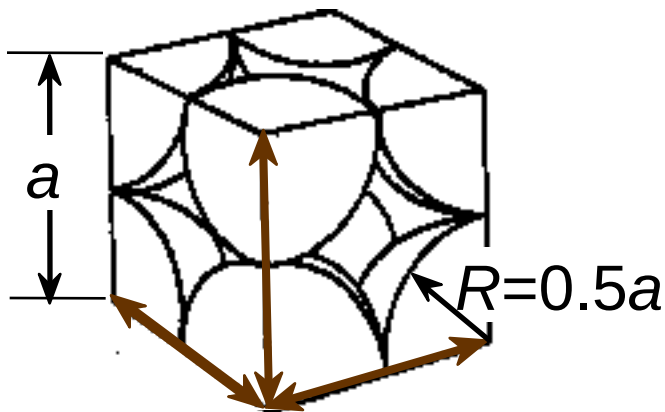
(011)

Atomic Packing Factor (APF)

$$APF = \frac{\text{Volume of atoms in unit cell}^*}{\text{Volume of unit cell}}$$

*assume hard spheres

- APF for a simple cubic structure = 0.52



close-packed directions

contains $8 \times 1/8 =$

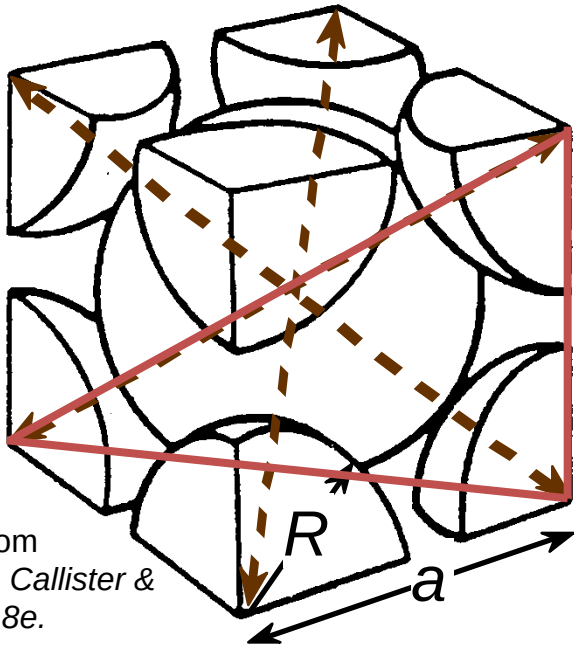
1 atom/unit cell

Adapted from Fig. 3.24,
Callister & Rethwisch 8e.

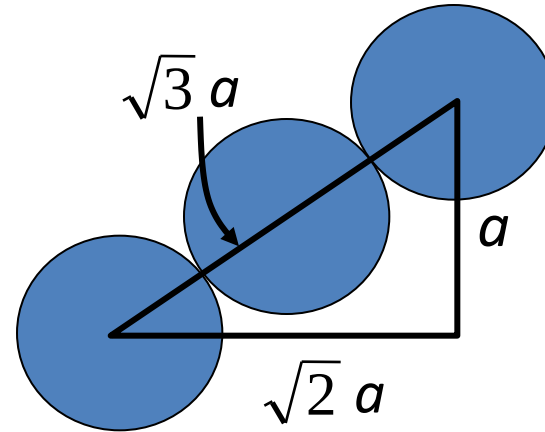
$$APF = \frac{\overbrace{1}^{\text{atoms}} \overbrace{\frac{4}{3} \pi (0.5a)^3}^{\text{volume atom}}}{\underbrace{a^3}_{\text{volume unit cell}}}$$

Atomic Packing Factor: BCC

- APF for a body-centered cubic structure = 0.68



Adapted from
Fig. 3.2(a), Callister &
Rethwisch 8e.



Close-packed directions:
length = $4R = \sqrt{3}a$

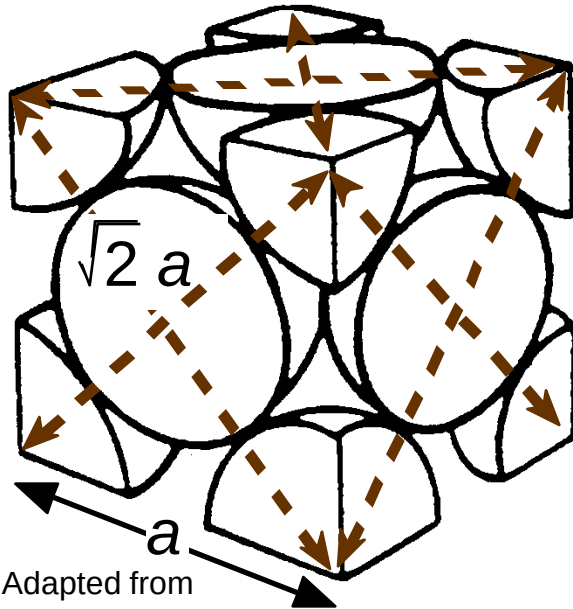
$$\text{APF} = \frac{\text{atoms unit cell} \times \text{volume atom}}{\text{volume unit cell}}$$

$$\text{APF} = \frac{2 \times \frac{4}{3} \pi (\sqrt{3}a/4)^3}{a^3}$$

Diagram illustrating the calculation of the Atomic Packing Factor (APF) for a BCC structure. The formula is shown as a fraction: the numerator is the product of the number of atoms per unit cell (2) and the volume of one atom ($\frac{4}{3} \pi (\sqrt{3}a/4)^3$), and the denominator is the volume of the unit cell (a^3). Arrows point from the labels to the corresponding parts of the formula: 'atoms unit cell' points to 2, 'volume atom' points to $\frac{4}{3} \pi (\sqrt{3}a/4)^3$, and 'volume unit cell' points to a^3 .

Atomic Packing Factor: FCC

- APF for a face-centered cubic structure = 0.74
maximum achievable APF



Adapted from
Fig. 3.1(a),
Callister &
Rethwisch 8e.

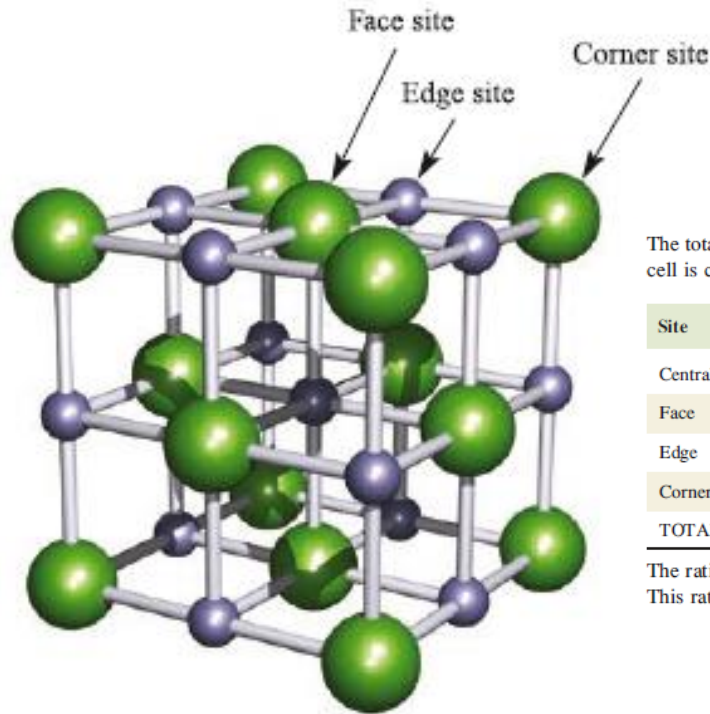
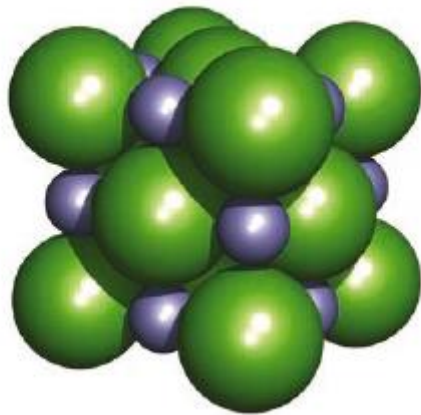
Close-packed directions:
length = $4R = \sqrt{2} a$

Unit cell contains:
 $6 \times 1/2 + 8 \times 1/8$
= 4 atoms/unit cell

$$\text{APF} = \frac{\begin{array}{c} \text{atoms} \\ \text{unit cell} \end{array} \quad 4 \quad \frac{4}{3} \pi (\sqrt{2}a/4)^3 \quad \begin{array}{c} \text{volume} \\ \text{atom} \end{array}}{\begin{array}{c} a^3 \\ \text{volume} \\ \text{unit cell} \end{array}}$$

Simple Ionic Compounds

Rocksalt (NaCl) structure type



The total number of Na^+ and Cl^- ions belonging to the unit cell is calculated as follows:

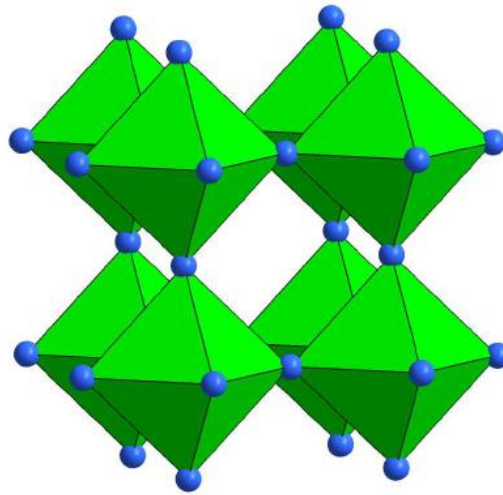
Site	Number of Na^+	Number of Cl^-
Central	1	0
Face	0	$(6 \times \frac{1}{2}) = 3$
Edge	$(12 \times \frac{1}{4}) = 3$	0
Corner	0	$(8 \times \frac{1}{8}) = 1$
TOTAL	4	4

The ratio of $\text{Na}^+ : \text{Cl}^-$ ions is $4:4 = 1:1$
This ratio is consistent with the formula NaCl.

Cl- Fcc arrangements , Na^+ occupies octahedral holes

Among the many compounds that crystallize with the NaCl structure type are NaF, NaBr, NaI, NaH, halides of Li, K and Rb, CsF, AgF, AgCl, AgBr, MgO, CaO, SrO, BaO, MnO, CoO, NiO, MgS, CaS, SrS and BaS.

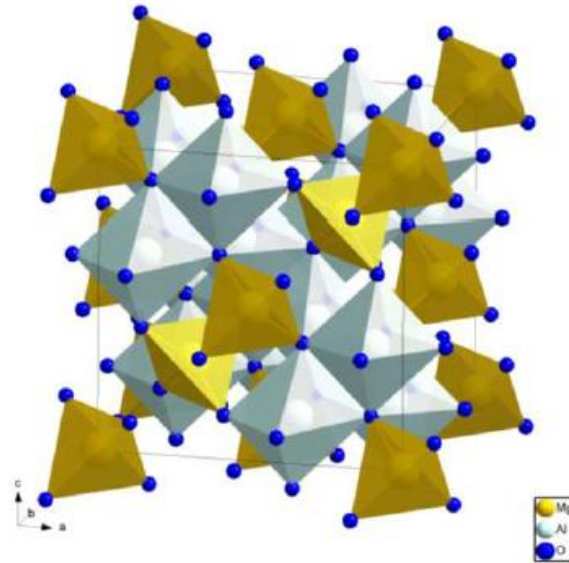
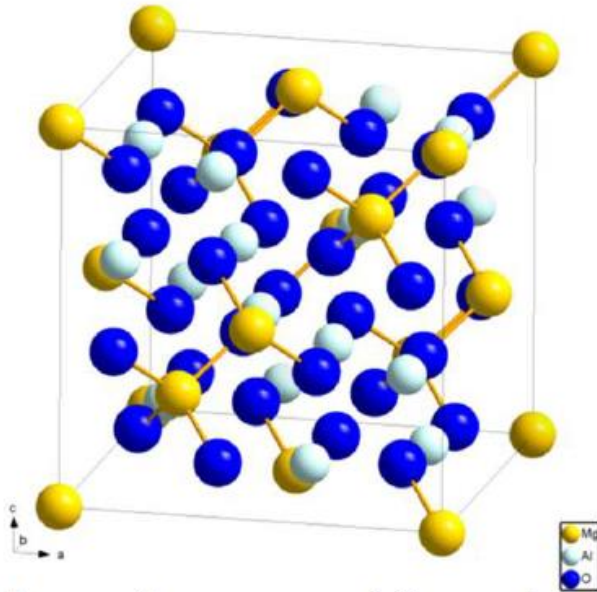
Structure Type – Rhenium Trioxide (ReO_3)



Also called the aluminum fluoride structure, which is adopted by the fluoride of Al, Sc, Fe, Co, Rh, and Pd, the high temperature form of WO_3 , and ReO_3 .

-Corner-sharing network of $[\text{ReO}_6]$ that give a highly symmetrical structure with cubic symmetry.

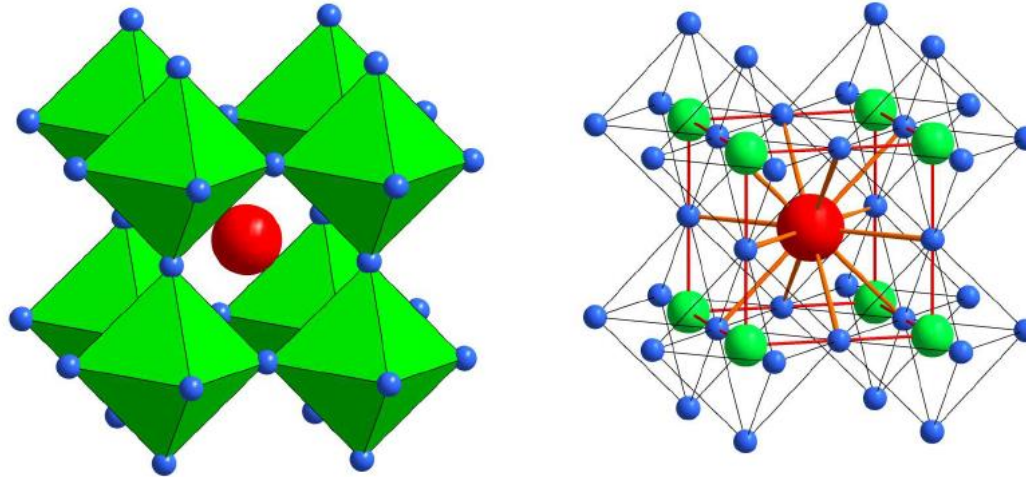
Structure Type – Spinel (MgAl_2O_4)



Spinels have the general formula AB_2O_4 , where generally A is a divalent ion A^{2+} , B is trivalent ion B^{3+} .

The structure can be described as being based on a *cubic close-packed* array of oxide ions, with **A^{2+}** ions occupying tetrahedral holes and **B^{3+}** ions occupying octahedral holes.

Structure Type – Perovskite (CaTiO_3)



The structure is named after the mineral CaTiO_3 .

- It has the same octahedral network as the ReO_3 structure, with the A ion added into the cavity.
- The structure may also be visualized as a *ccp* array of A and X ions with the B ions occupying the octahedral holes. (Compare with NaCl)
- Structures of high-temperature superconductors are based on perovskite.

Solid State Chemistry

Schottky defect

A *Schottky defect* is an example of a *point defect* in a crystal lattice and arises from vacant lattice sites. The stoichiometry and electrical neutrality of the compound are retained. Defects that fall in this category include:

- a vacant atom site in a metal lattice;
- a vacant cation and a vacant anion site in an MX lattice;
- two vacant cations and a vacant anion site in an M_2X lattice;
- a vacant cation and two vacant anion sites in an MX_2 lattice.

Crystal defects

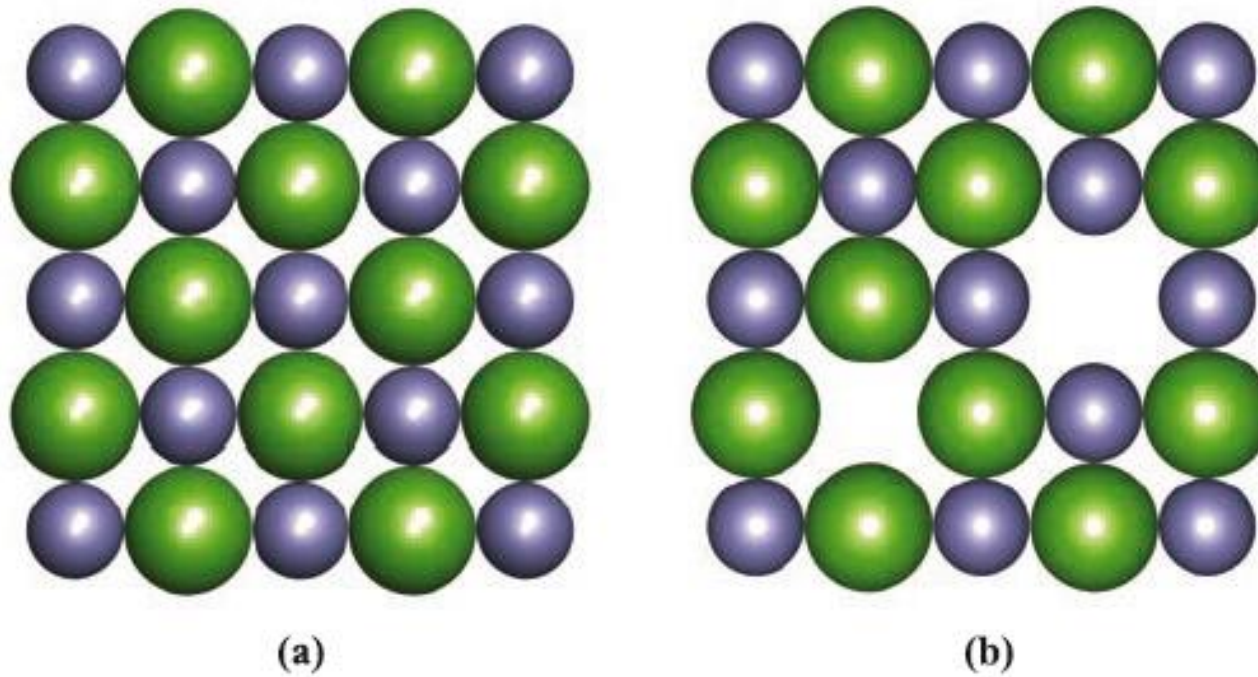


Fig. 6.27 (a) Part of one face of an ideal NaCl structure; compare this with Fig. 6.16. (b) A Schottky defect involves vacant cation and anion sites; equal numbers of cations and anions must be absent to maintain electrical neutrality. Colour code: Na, purple; Cl, green.

Frenkel defect

In ionic lattices in which there is a significant difference in size between the cation and anion, the smaller ion may occupy a site that is vacant in the ideal lattice. This is a *Frenkel defect* (a *point defect*) and does not affect the stoichiometry or electrical neutrality of the compound.

Crystal defects

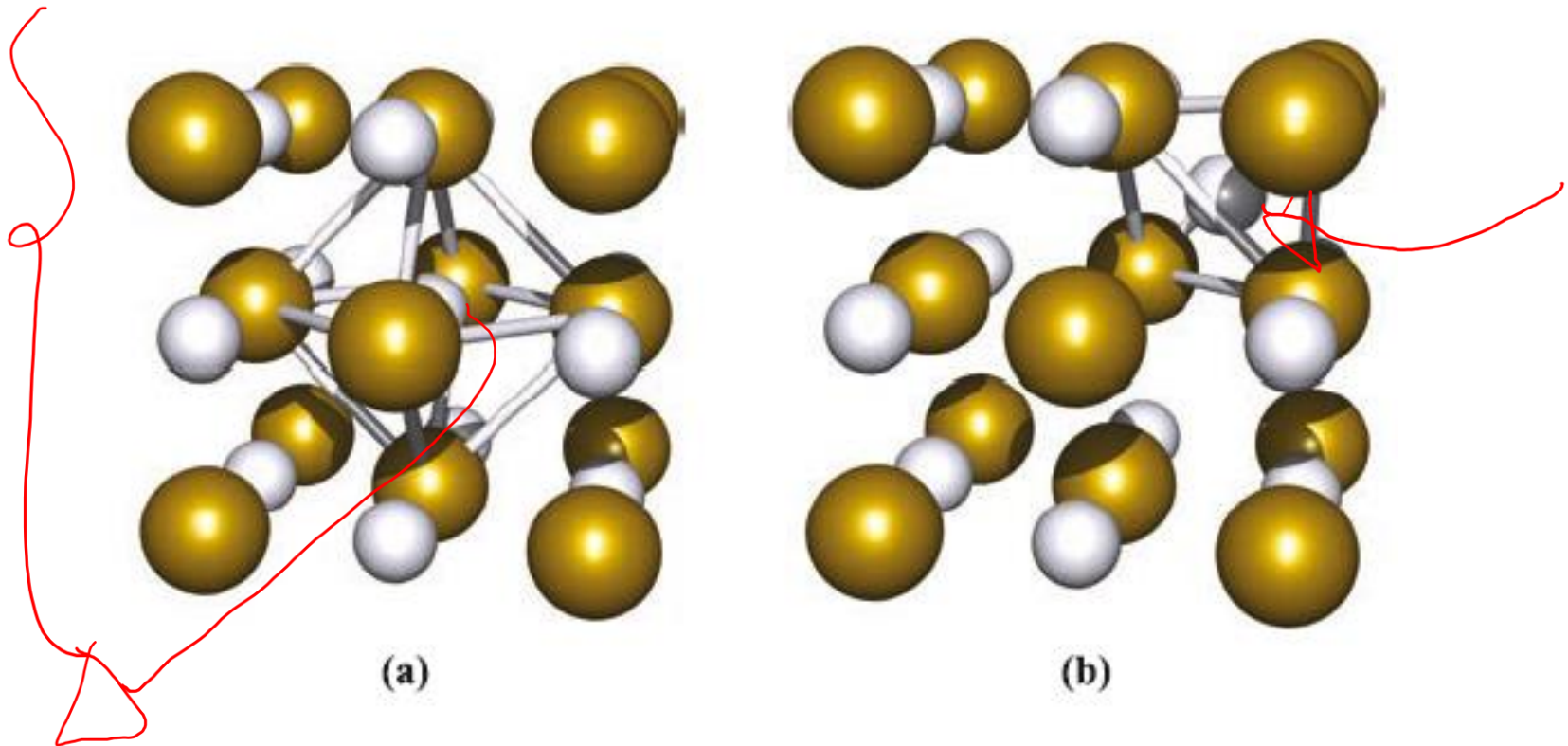
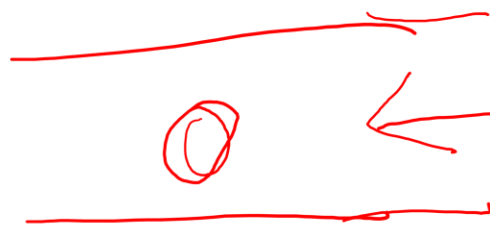
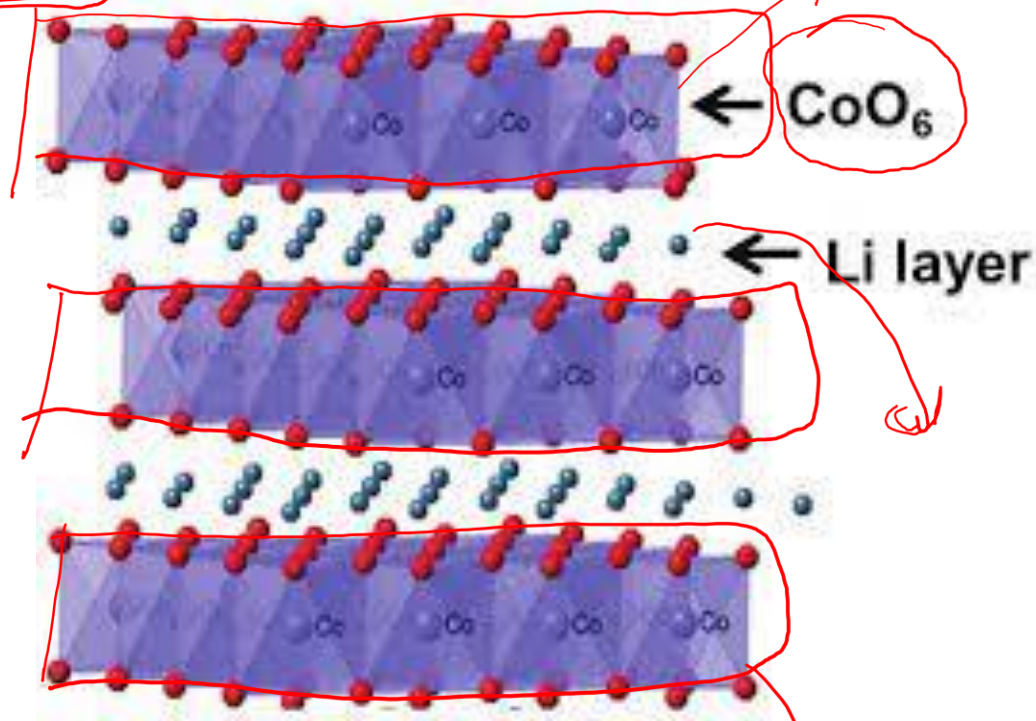
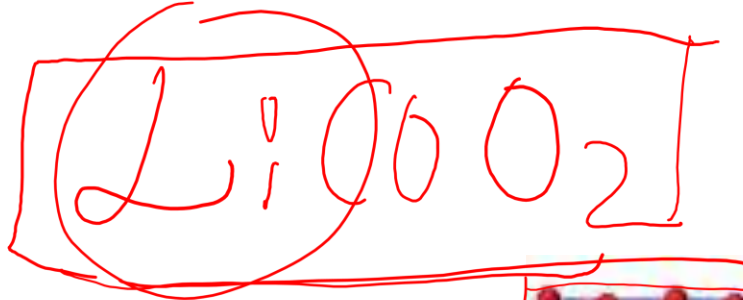


Fig. 6.28 Silver bromide adopts an NaCl structure. (a) An ideal lattice can be described in terms of Ag^+ ions occupying octahedral holes in a cubic close-packed array of bromide ions. (b) A Frenkel defect in AgBr involves the migration of Ag^+ ions into tetrahedral holes; in the diagram, one Ag^+ ion occupies a tetrahedral hole which was originally vacant in (a), leaving the central octahedral hole empty. Colour code: Ag, pale grey; Br, gold.

Structure – Property Relationship

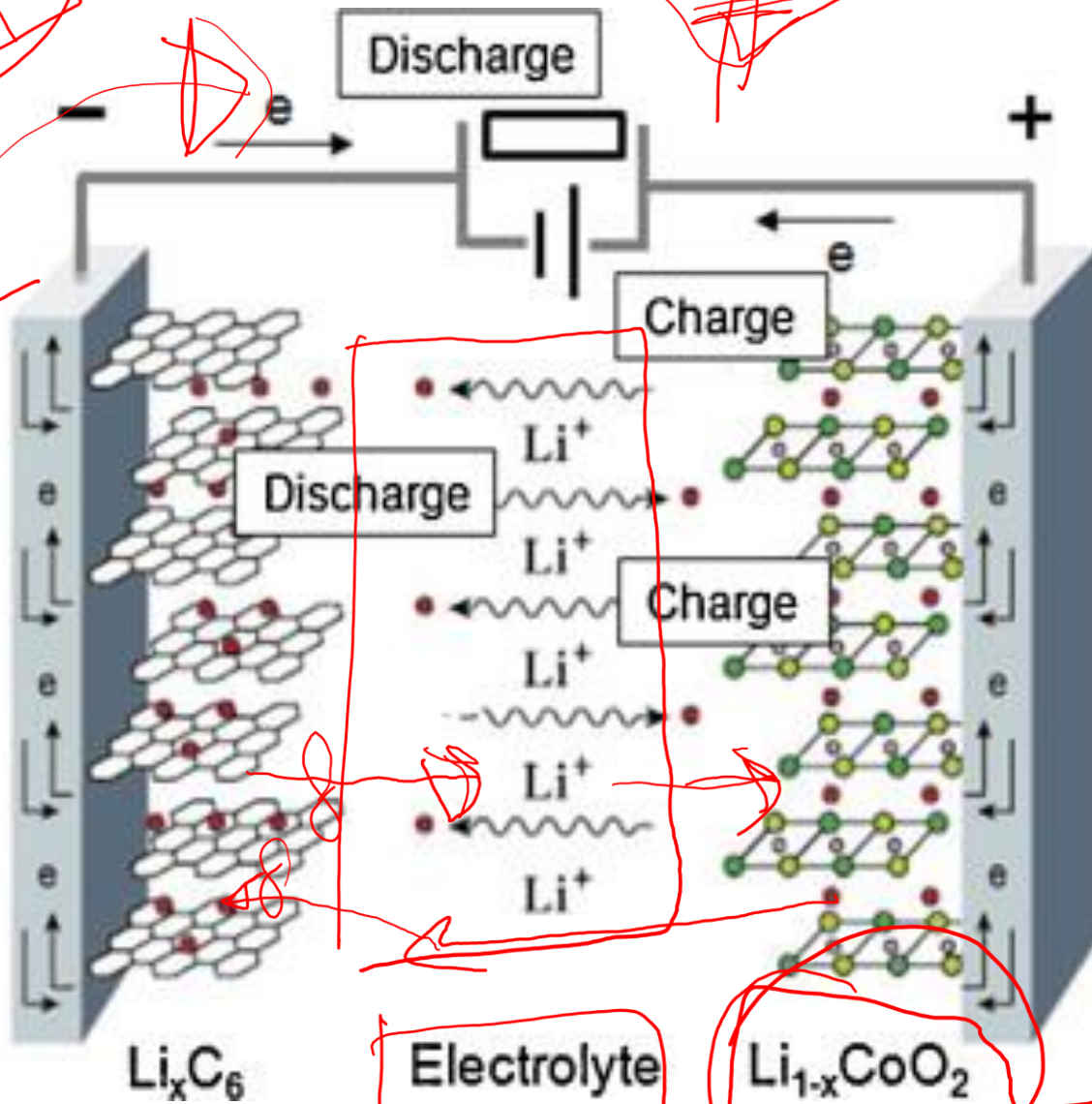




Graphite

Intercalation layer

Secondary Li Ion Battery



Power
LiPF₆

WV
pu

2079

100%

Introduction to Nanomaterials

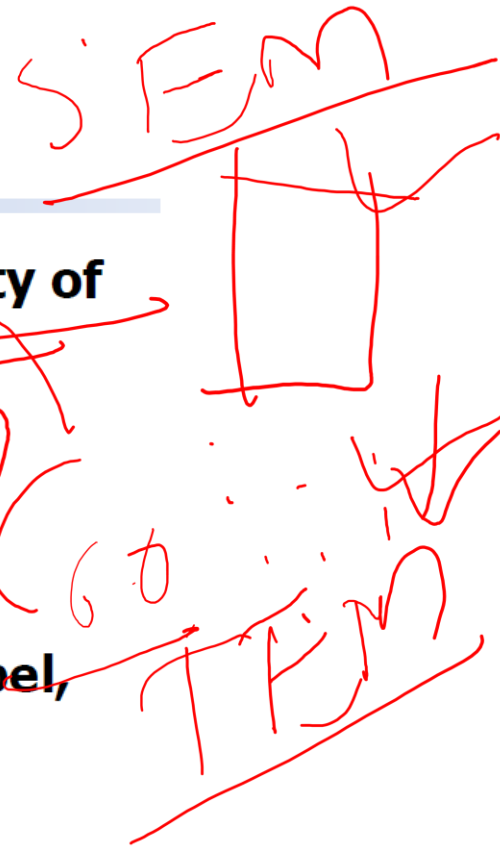


History of nanomaterials

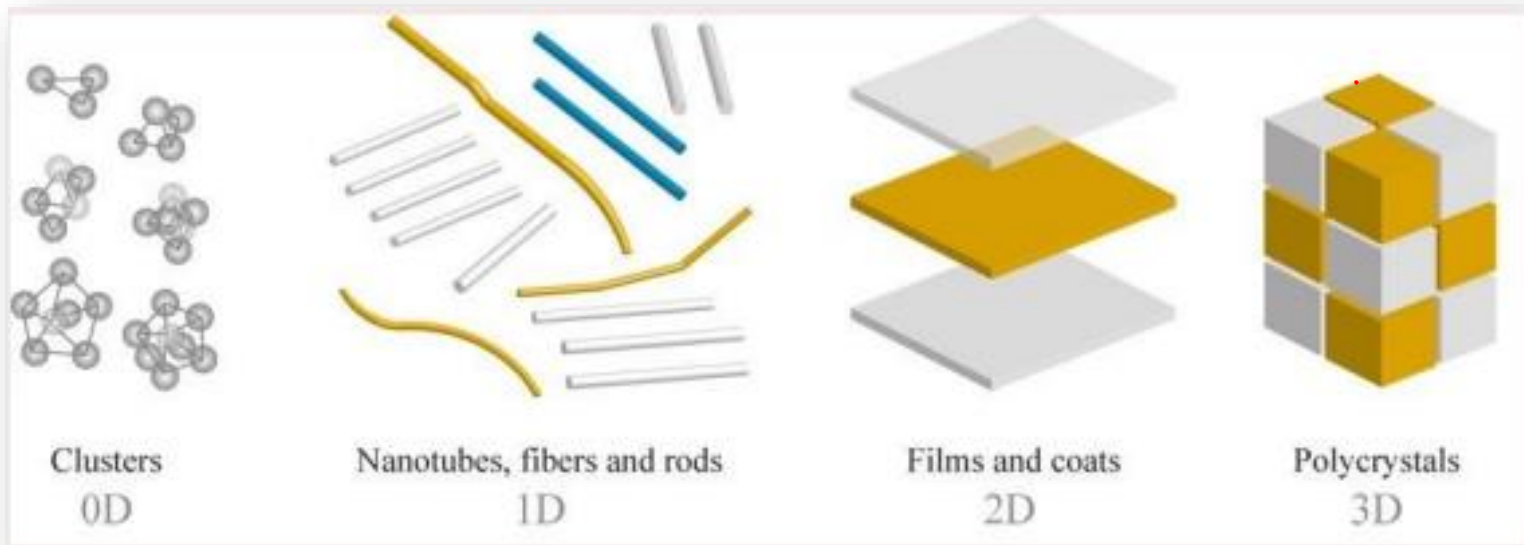
- 1959: Richard Feynman's famed talk. " **There's Plenty of Room at the Bottom** "

- 1981: Binnig and Rohrer created the STM to image individual atoms. (Nobel, Physics 1986)
- 1985: Curl, Kroto, Smalley discovered **fullerene (Nobel, Physics 1996)**
- 1991: Iijima discovered single wall carbon nanotubes.
- 2010 A. Geim and K. Novoselov (**Nobel physics on Graphene**)

2016 Nobel Prize in Chemistry – development of **molecular machines** which can lead to more miniaturization and development of new materials.



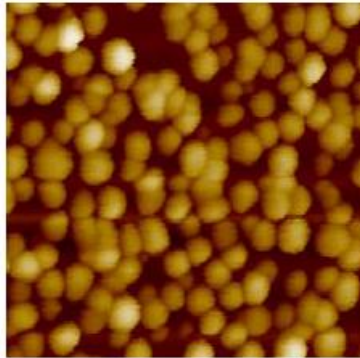
Classification of Nanomaterials: Based on Number of Free Dimensions



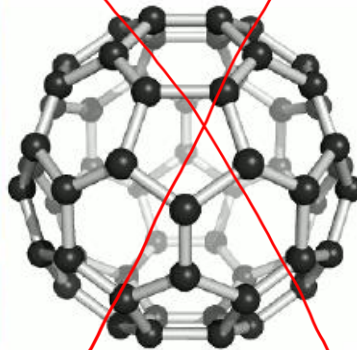
Classification of Nanomaterials: Based on Compositions

- **Carbon-based nanomaterials**
- Inorganic-based nanomaterials
- Organic-based nanomaterials
- Composite-based nanomaterials

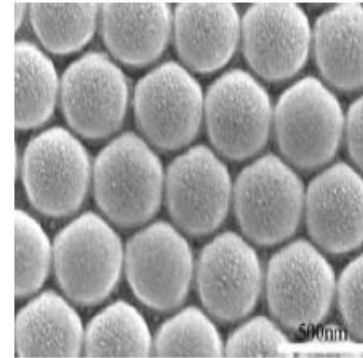
Nanomaterials (gold, carbon, metals, metal oxides and alloys) with variety of morphologies (shapes)



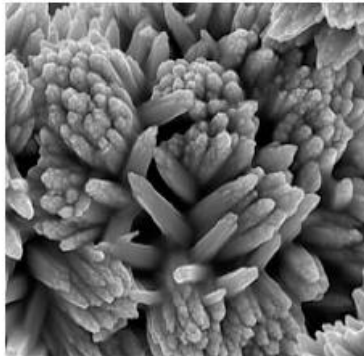
Au nanoparticle



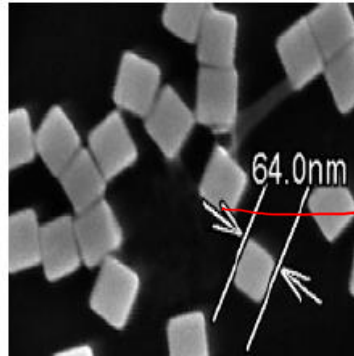
Buckminsterfullerene



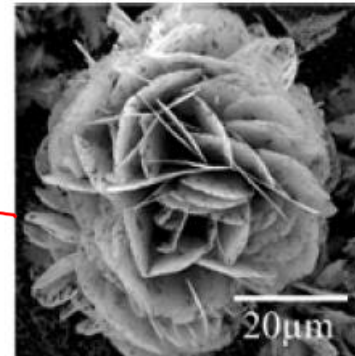
FePt nanosphere



Titanium nanoflower



Silver nanocubes

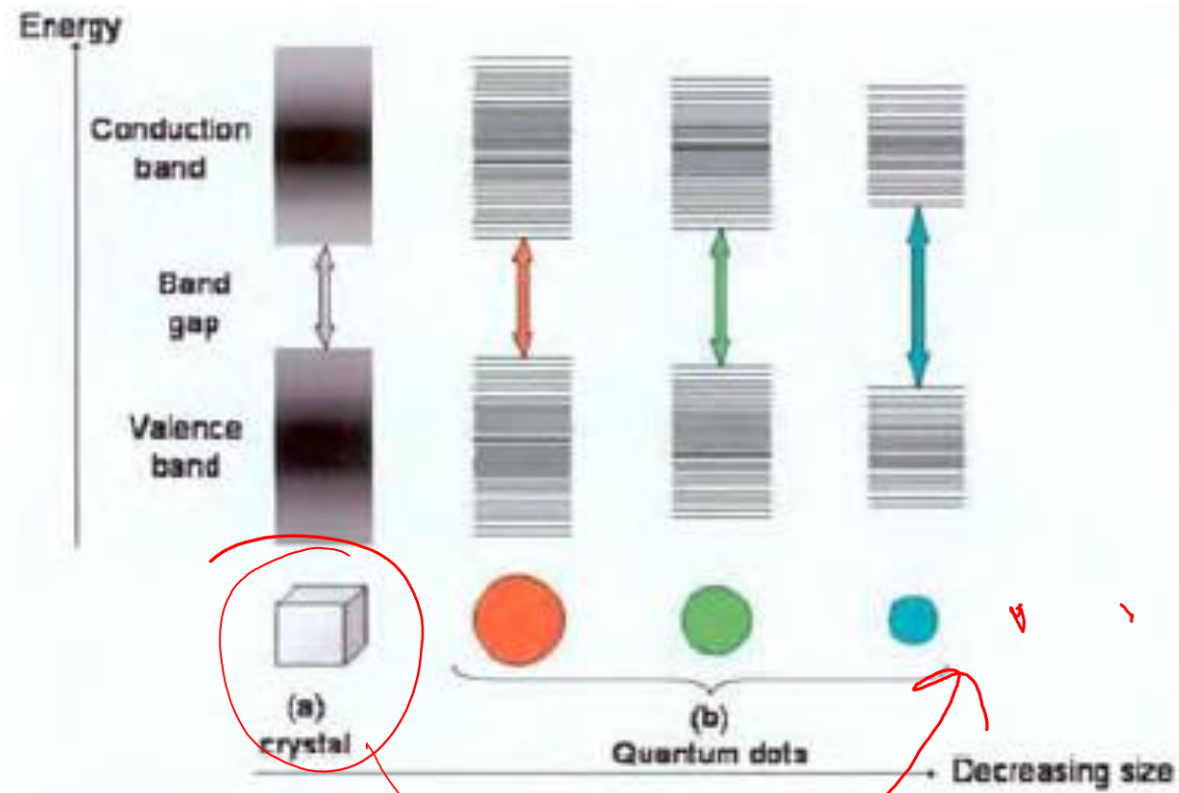


SnO₂ nanoflower

Surface area increase

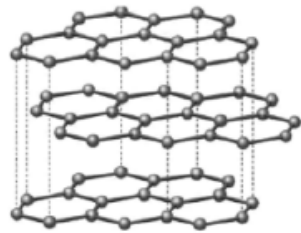
How would the total surface area increase if a cube of 1 m³ were progressively cut into smaller and smaller cubes, until it is composed of 1 nm³ cubes? Table (2) summarizes the results.

Size of cube side	Number of cubes	Collective surface area
1 m	1	6 m ²
0.1 m	1 000	60 m ²
0.01 m = 1 cm	10 ⁶ = 1 million	600 m ²
0.001 m = 1 mm	10 ⁹ = 1 billion	6 000 m ²
10 ⁻⁹ m = 1 nm	10 ²⁷	6 x 10 ⁹ = 6 000 km ²



Carbon-based nanomaterials

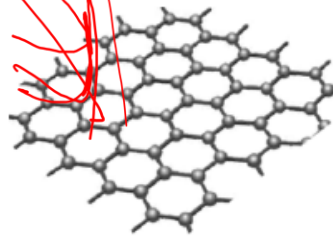
Basic building block



Graphite (3D)

van der Waals stack of graphene
Conductor

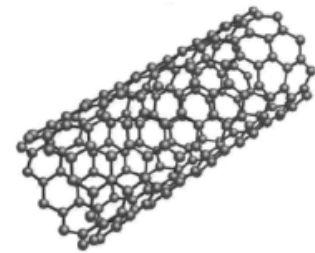
Stack



Graphene (2D)

Single-atom-thick carbon layer
sp² bonding of carbons
Semi-metal

Roll

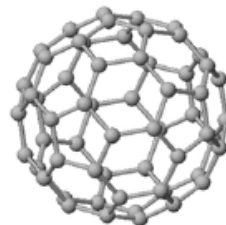


Carbon nanotube (1D)

Rolled graphene
Semiconductor (2/3) or metal (1/3)

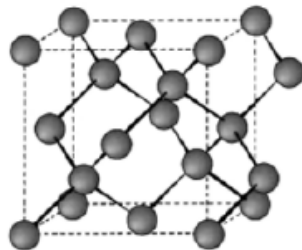


Ball



Fullerene (0D)

Wide band gap (5.5 eV)

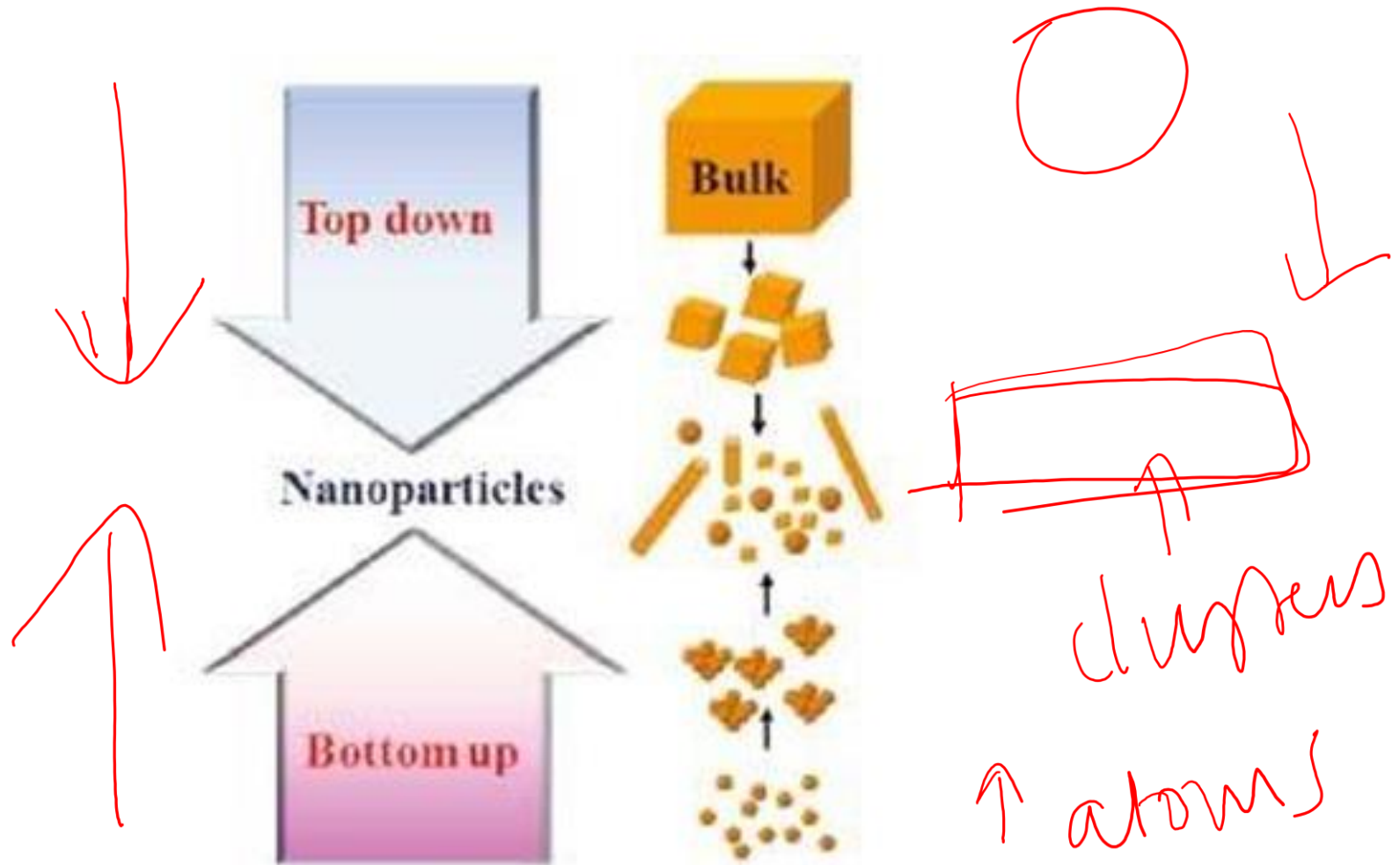


Diamond (3D)

sp³ bonding of carbons
Wide band gap (5.5 eV)

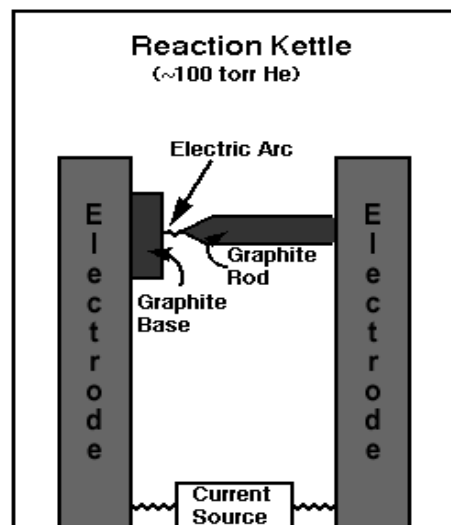
Single element carbon shows
different functionality
depending on its dimension

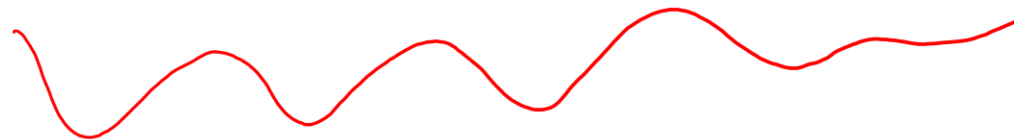
Nanomaterial - synthesis and processing



Fullerenes can be made by vaporizing carbon within a gas medium.

- An electric arc is maintained between two nearly contacting graphite electrodes.
- Most of the power is dissipated in the arc and not in resistive heating of the rod.
- The entire electrode assembly is enclosed in a reaction kettle that is filled with ~ 100 torr pressure of helium.
- Black soot is produced, and extraction with organic solvents yields fullerenes.





C_{60}



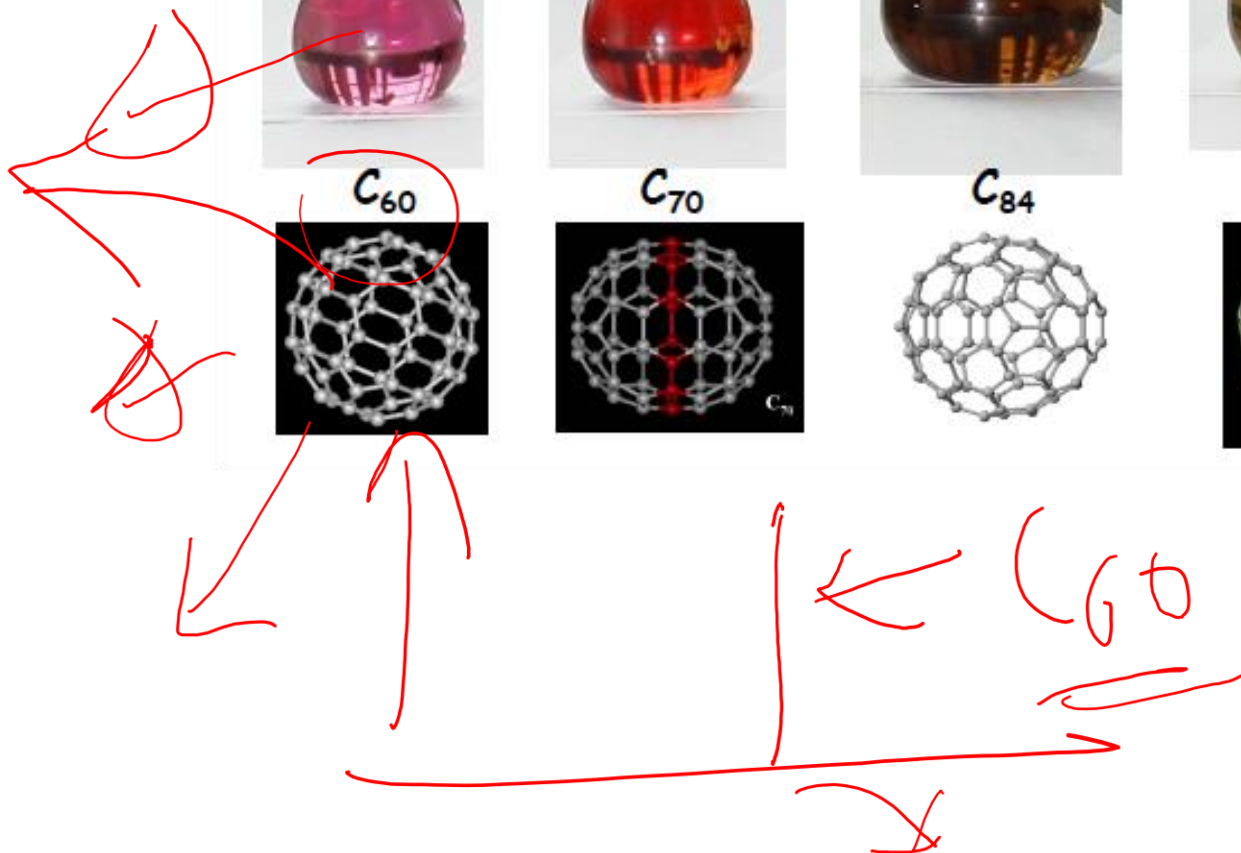
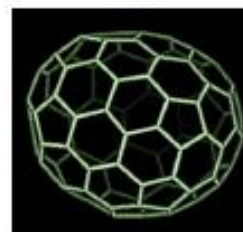
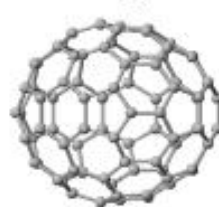
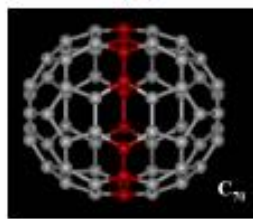
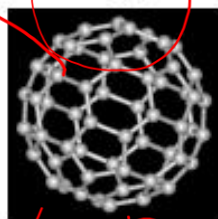
C_{70}



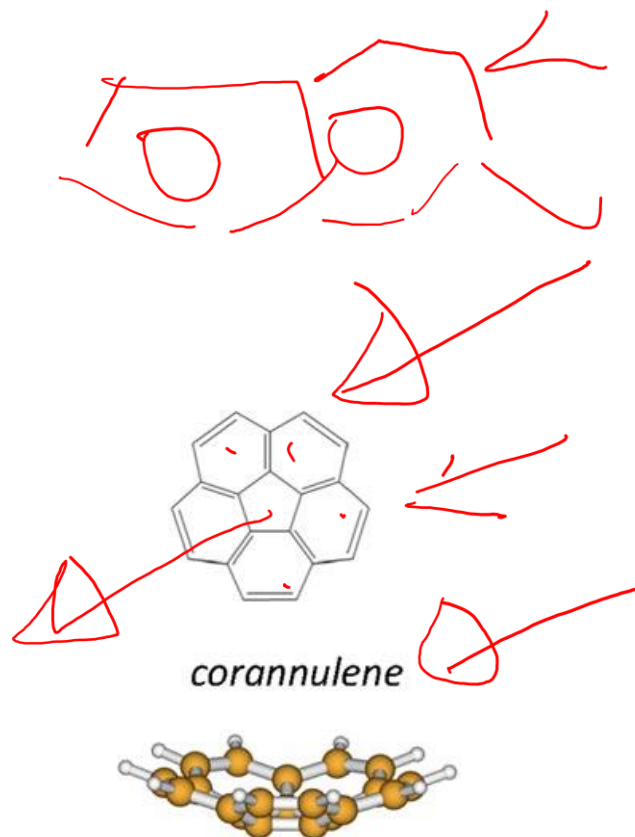
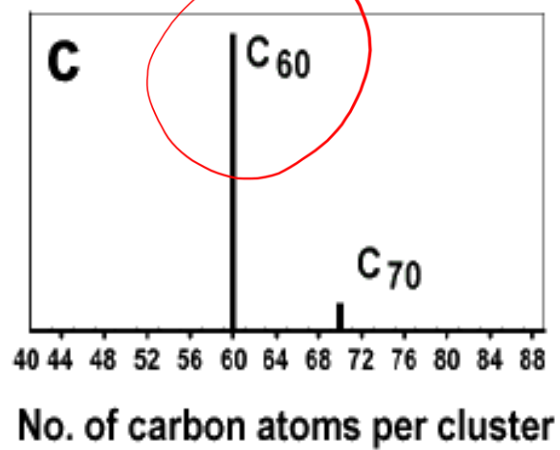
C_{84}

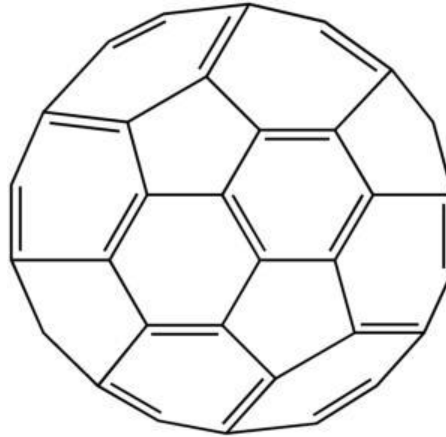


C_{86}



Structure C_{60}

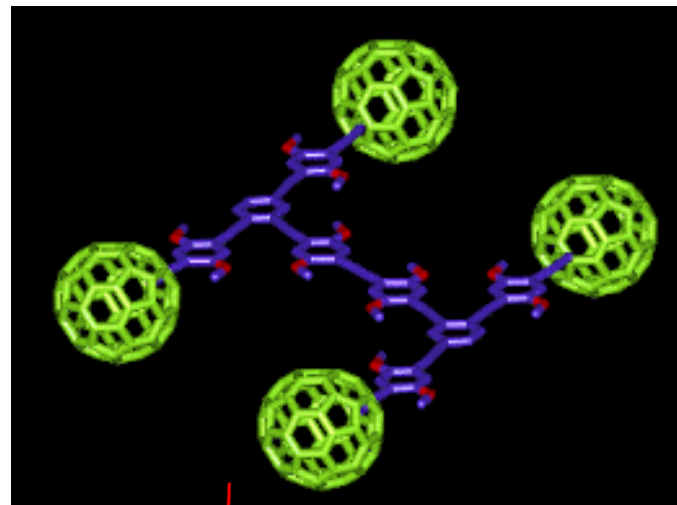
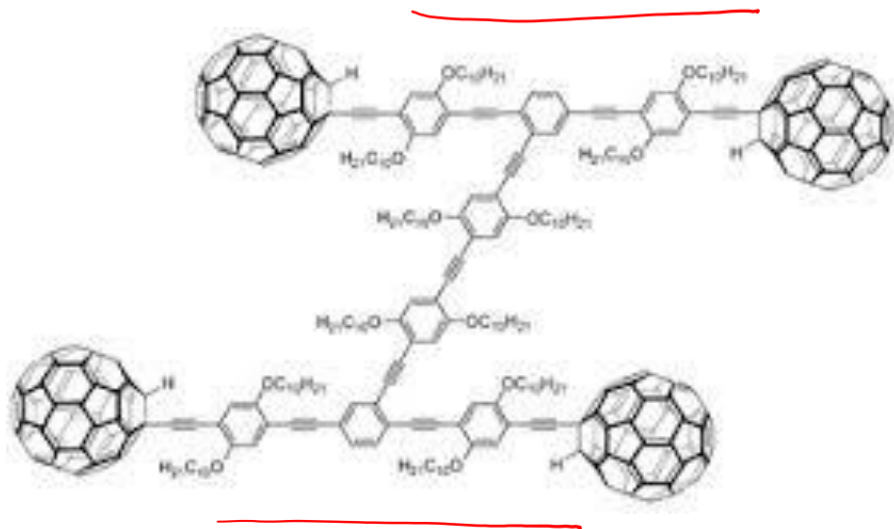




30 [6,6] double bonds connect two hexagons

60 [5,6] bonds connect a hexagon and a pentagon.

The X-ray diffraction bond length values are 135.5 pm for the [6,6] bond and 146.7 pm for the [5,6] bond

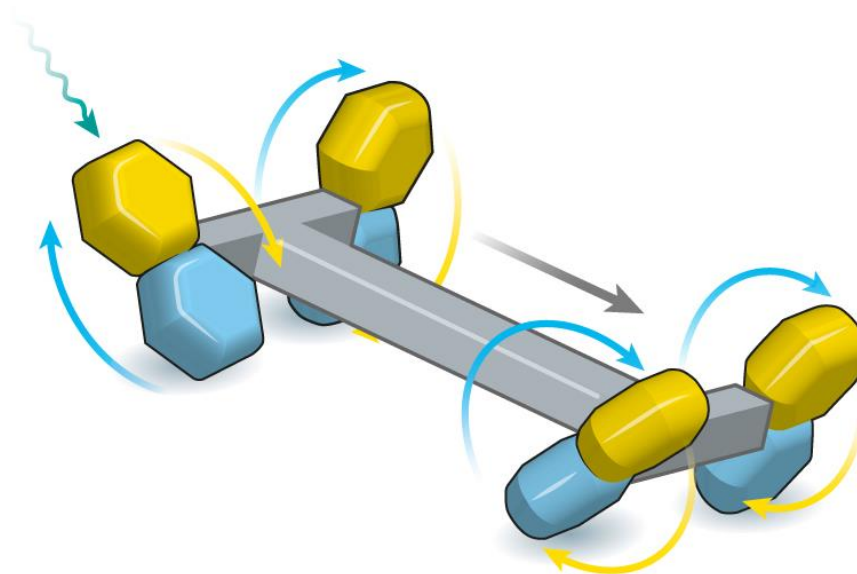


NOT FOR EXAM

"For the greatest benefit to mankind"
Alfred Nobel

2016 NOBEL PRIZE IN CHEMISTRY

Jean-Pierre Sauvage
Sir J. Fraser Stoddart
Bernard L. Feringa



Nanotube

• Discovered 1991, Iijima

• STRIP OF A GRAPHENE SHEET ROLLED INTO A TUBE

